

Introduction

to

**Physics
of Materials**

Efedua E. Eziashi



This text is aimed at giving the undergraduate electrical engineering students an introduction to the basic physical principles needed for the understanding of the properties of electronic materials and devices, which play important role in modern engineering and scientific instrumentation. It has been developed from lecture courses for University Undergraduates at the 300-level. This edition contains additional information on developments and relevant topics aimed at keeping the student abreast with new technologies.

It is important that I recognise and acknowledge all those numerous sources of information, which I invariably consulted as a teacher but forgot with time. Thank you.

metal-oxide is an insulator

Table of Contents

Preface	(ii)
Chapter One : Structure of the Atom	
1.1 Quantum atom model	2
1.2 Band structure of solids	8
1.3 Classification of electrical materials	12
Chapter Two: Semiconductor Materials	16
Intrinsic and extrinsic materials	17
Majority and minority carriers	20
Chapter Three: The Density of Energy States	22
3.1 The uncertainty and exclusion principles	23
3.2 Allowed energies in a crystal	24
3.3 Carrier population in bands	26
Chapter Four: Charge Transport Process	35
4.1 Drift and diffusion flow	36
4.2 Concept of carrier lifetime	41
4.3 Magnetic effect on current carriers	43
Chapter Five: Electron emission from material surfaces	48
5.1 Introduction (image force)	49
5.2 Thermionic emission	53
5.3 Field emission	55
5.4 Electron emission by photoelectric effect	61

STRUCTURE OF THE ATOM

Chapter Six: Characteristics of Junctions	64
6.1 Introduction	65
6.2 Junction between metals	66
6.3 Junction between metal and semiconductor	71
6.4 The p-n junction	79
Chapter Seven: Optoelectronics	87
7.1 Transition in direct and indirect gap materials	88
7.2 Optoelectronic devices	91
Chapter Eight: Dielectrics	103
8.1 Introduction	104
8.2 Applications of dielectrics	109
8.3 Optical fibre transmission medium	112

1.1 Quantum Atom Model	2
Total Energy of the Hydrogen Atom	5
1.2 Band Structure of Solids	8
Covalent and Metallic Bond	9
Energy Band Structure	10
1.3 Classification of Electrical Materials	12
Insulators	13
Conductors	1
Semiconductors	

CHAPTER ONE

STRUCTURE OF THE ATOM

1.1 Quantum Atom Model

The simple atom model consists of a central core (called the nucleus), which is positively charged, with electrons orbiting around it (figure 1.1). For this system to be stable, the centrifugal force on the orbiting electron is counterbalanced by Coulomb's force of electrostatic attraction of the positively charged nucleus. Thus,

$$m\omega^2 r = q^2 / 4\pi\epsilon_0 r^2 \quad \dots\dots\dots (1.1a)$$

Or,

$$\omega^2 r^3 = q^2 / 4\pi\epsilon_0 m \quad \dots\dots\dots (1.1b)$$

where ϵ_0 is the dielectric constant of free space.

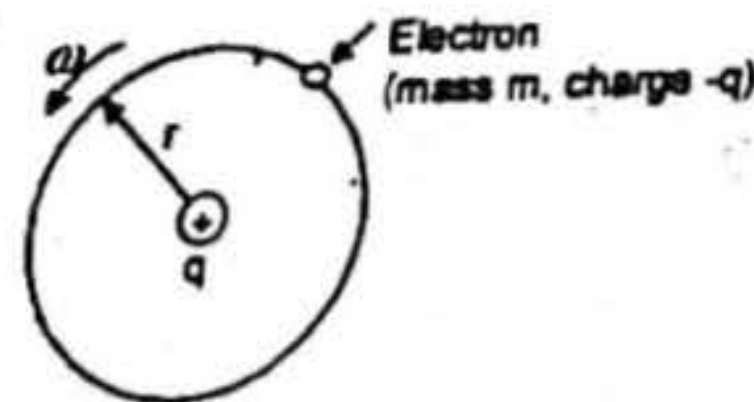


Fig. 1.1 The simple model of the atom

The above model is unsatisfactory. This is because theory shows that an electron, accelerated round its orbit, would radiate energy continuously. An example is the electromagnetic radiation from a radio transmitting antenna, where electrons are accelerated to and fro within the antenna. The continuous loss of energy would result in the continuous reduction in orbital velocity, and hence reduction in orbital radius r . Eventually the electrostatic attraction overcomes the centrifugal force. The result of this is that the electron spirals into the nucleus and the atom becomes unstable.

Plank in 1901 showed that when a source is radiating electromagnetic wave, its energy is quantized (i.e its magnitude is always a multiple of a unit known as a quantum) and so could not change continuously. For electromagnetic radiation, the quantum is called a photon. If the frequency of radiation is f , each photon has energy hf , where h is Plank's constant ($= 6.6 \times 10^{-34}$ J-s). The frequency of electromagnetic radiation covers a very wide range, extending from long radio waves (30 kHz) to X-rays (up to 3×10^{21} Hz). From the relationship $\lambda = c/f$, the corresponding value of the wavelength λ range from 10^4 to 10^{-13} meters, where c is the velocity of electromagnetic radiation (i.e the velocity of light).

In 1924, deBroglie proposed that a moving particle, such as an electron (and indeed all matter in motion), may be considered to have associated wave properties in addition to its particle-like properties. The wave properties are characterised by frequency (f) and wave length (λ), while the particle properties are characterised by energy $E (=hf)$ and momentum $p (=h/\lambda)$. Thus, the wave and particle properties of the electron are linked by Plank's constant.

The lessons of Plank and deBroglie lead to the quantum mechanical theory, which states that 'all particles have associated with them a wave train (or wave packet) which governs their movement. They will only behave as particles when they interact with other matter. The actual position of the particle within the wave train is uncertain and cannot be observed or predicted'. In other words the quantum mechanical theory does not allow us to observe the velocity nor determine the position of an electron precisely. (This is the uncertainty principle).

If we consider the wave train associated with the electron in orbit as reaching right round the atom (figure 1.2), then:

1. It is not now possible any more to consider the electron as a particle orbiting at a steady radius. We can only know that it is somewhere in what was previously the orbital path (shown dotted). That means that orbits are no longer precisely defined, and, in the hydrogen atom for an example, the electron can traverse the whole of the space about the nucleus corresponding to a permitted orbital radius (i.e. the probability of locating the electron is greatest at this distance).
2. The electron-wave stretches right round the nucleus like an elastic band. Consequently, there must be a whole number of wavelengths in the band, other wise there would be discontinuities.

This leads to the equation relating the circumference of the band to the number of wavelengths. Thus,

$$2\pi r = n\lambda \quad \dots\dots\dots (1.2)$$

where n is an integer. Substituting for $\lambda = h/p = h/mr\omega$, we have

$$2\pi r = nh/(mr\omega). \text{ Or,} \\ \omega^2 r^2 = n^2 h^2 / 4\pi^2 m^2 r^2 \quad \dots\dots\dots (1.3)$$

From equations 1.1 and 1.3, we have

$$r = n^2 h^2 e_0 / (q^2 m \pi) \quad \dots\dots\dots (1.4)$$

The quantity n is called the principal quantum number. By giving n integer values, 1, 2, 3, etc, the family of orbital radii r_1, r_2, r_3 , etc is obtained.

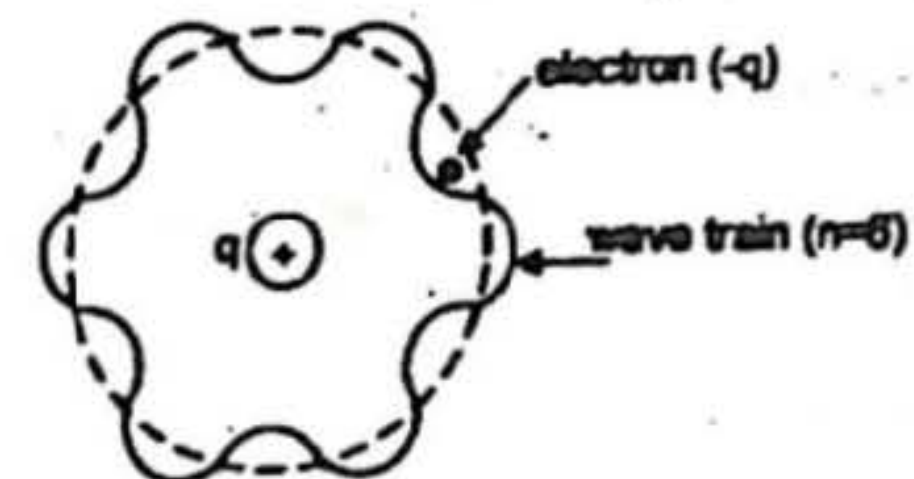


Fig. 1.2 Quantum atom model

Total Energy Of The Hydrogen Atom

The total energy (E) associated with the atom is the sum of the potential energy (V) and kinetic energy (K). In general, the Coulomb force between the nucleus and the electrons is given as

$$F = Zq^+q^-/(4\pi\epsilon r^2)$$

$= Zq^2/(4\pi\epsilon r^2)$ for a material with atomic number Z , where qZ is the total charge on the nucleus. The potential energy (V) at distance r is defined as the work done in moving the electron against this force, from infinity to distance r . Hence,

$$V = \int [Zq^2/(4\pi\epsilon_0 r^2)] dr = -Zq^2/(4\pi\epsilon_0 r) \quad \dots\dots\dots (1.5)$$

For the hydrogen atom, $Z = 1$, and hence

The kinetic energy of the electron is

$$K = \frac{1}{2}mr^2\omega^2$$

Substituting the expression for $r^2\omega^2$ (equation 1.1) gives

$$K = \frac{1}{2}mq^2/(4\pi\epsilon_0mr) = q^2/8\pi\epsilon_0r \quad \dots\dots\dots (1.7)$$

The total energy for the hydrogen atom thus becomes

$$\begin{aligned} E = V + K &= [-q^2/4\pi\epsilon_0r] + [q^2/8\pi\epsilon_0r] \\ &= -q^2/8\pi\epsilon_0r \end{aligned}$$

But from equation 1.4, $r = \epsilon_0 n^2 h^2 / \pi m q^2$. Hence,

$$\begin{aligned} E_n &= -q^4 m / (8 \epsilon_0^2 n^2 h^2) \\ &= -(1/n^2)(q^4 m / 8 h^2 \epsilon_0^2) \quad \dots\dots\dots (1.8a) \end{aligned}$$

$$= -13.58/n^2 \quad (\text{electron-volts}) \quad \dots\dots\dots (1.8b)$$

And,

$$r_n = 0.053 n^2 \quad (\text{nanometer}) \quad \dots\dots\dots (1.9)$$

The values of E_n and r_n for $n = 1, 2, 3$, and 4 are given in Table 1.1, and the potential well representation for the hydrogen atom is given in figure 1.3

The minimum energy, (i.e. -13.58 eV) is obtained for $n = 1$ and represents the energy of a stable or ground state hydrogen atom. The higher permitted levels (i.e. $n > 1$) are unstable or excited states. The electron does not radiate energy while it is in any of the permitted energy states, so its energy remains constant. If the electron makes a transition from the ground state to a higher level, it needs to receive extra energy corresponding to the energy difference between the levels. Thus, a transition from $n = 1$ to $n = 2$ requires extra energy equal to $(13.58 - 3.39) = 10.19$ eV.

Table 1.1 Energy levels and orbital radii for the hydrogen atom

n	E_n (eV)	r_n (nm)
1	-13.58	0.053
2	-3.39	0.212
3	-1.51	0.477
4	-0.85	0.848

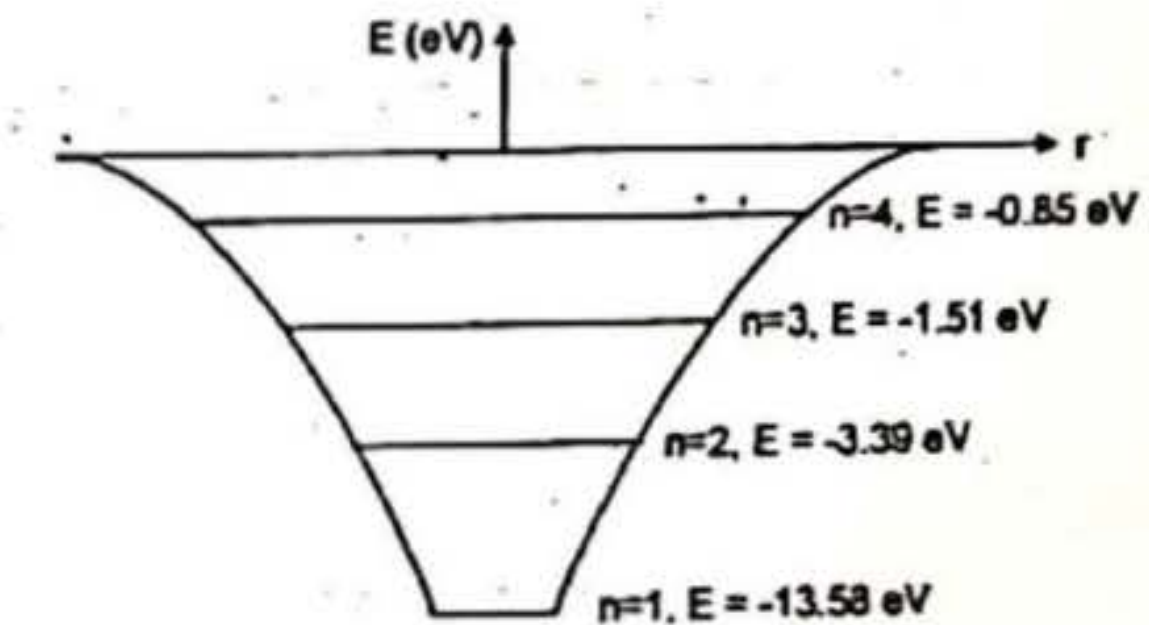


Fig.1.3 Potential well representation of energy levels for hydrogen atom

An electron does not remain in an excited state if a lower energy state is vacant. Thus, the electron in the hydrogen atom spends only a very short time in the excited state and then returns to the ground state, either directly or by way of an intermediate level. When an electron moves from an excited state to one of a lower energy, a quantum of radiation is emitted. The frequency (f) of the radiation is given by

$$hf = E_{n+1} - E_n = \Delta E \quad \dots\dots\dots (1.10)$$

where ΔE is the energy difference between the two levels and h is Plank's constant.

Band Structure Of Solids

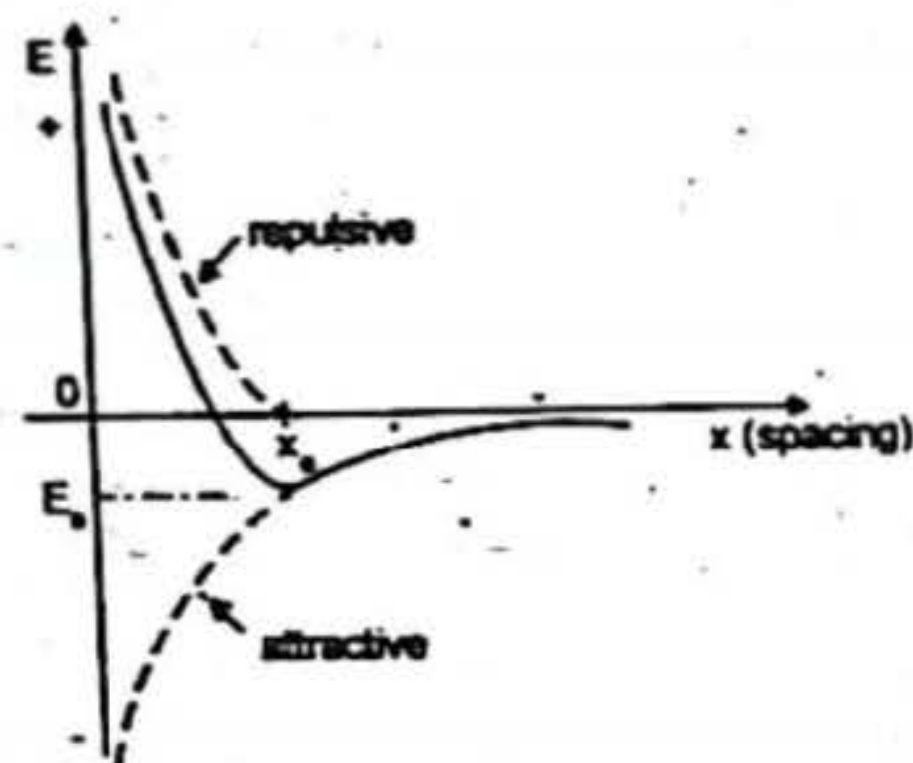


Fig. 1.4 Interatomic energies as a function of spacing

The hydrogen atom we considered in the last section was treated in isolation. In a crystal however, the atoms in the crystal are held in their relative positions by a balance between various types of inter-atomic forces, some repulsive and others attractive. The total energy is the sum of the repulsion and attraction energies (figure 1.4). The corresponding forces are given by the slope dE/dx of the resultant energy curve at a given point. At the normal atomic spacing in the crystal, the total force on the atom is zero. However, there is a bonding or binding energy. Thus, at the equilibrium spacing x_0 in the figure, the total force is zero (since dE/dx is zero) and the binding energy is E_0 .

Covalent And Metallic Bond

There are two important ways in which the binding energy is provided to form a stable structure. These are covalent bond and the metallic bond. In covalent bond, stable structure is attained when atoms share their valence electrons with neighbouring atoms to achieve the inert gas type of structure. For an example, molecular hydrogen is formed when two hydrogen atoms share their electrons so that both electrons are associated with one of the nuclei for part of the time (figure 1.5a). Thus, by this sharing of electrons, a helium type structure is achieved for much of the time.

Covalent bond also occurs in crystals of elements with four valence electrons, where atoms combine to form a tetrahedral structure. Each atom shares its valence electrons with four neighbouring atoms, such that each atom has a share in eight electrons thus achieving the inert gas type of structure (figure 1.5b).

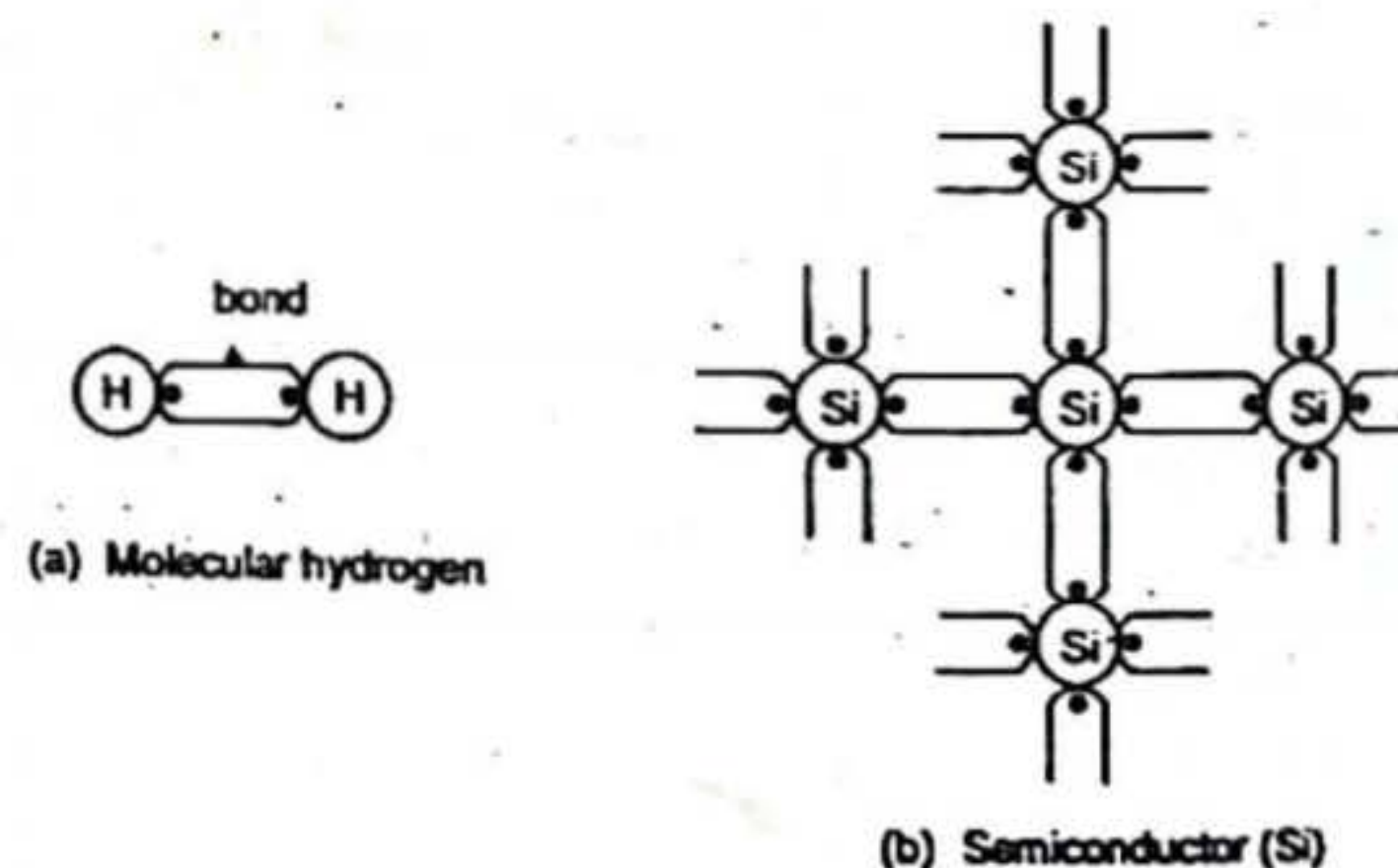


Fig.1.5 Examples of covalent bonding

$$P = 15 \rightarrow 3s^2 3p^3$$

10

$$Ga = 31 - 3d^{10} 4s^2 4p^1$$

$$As = 33 - 3d^{10} 4s^2 4p^3$$

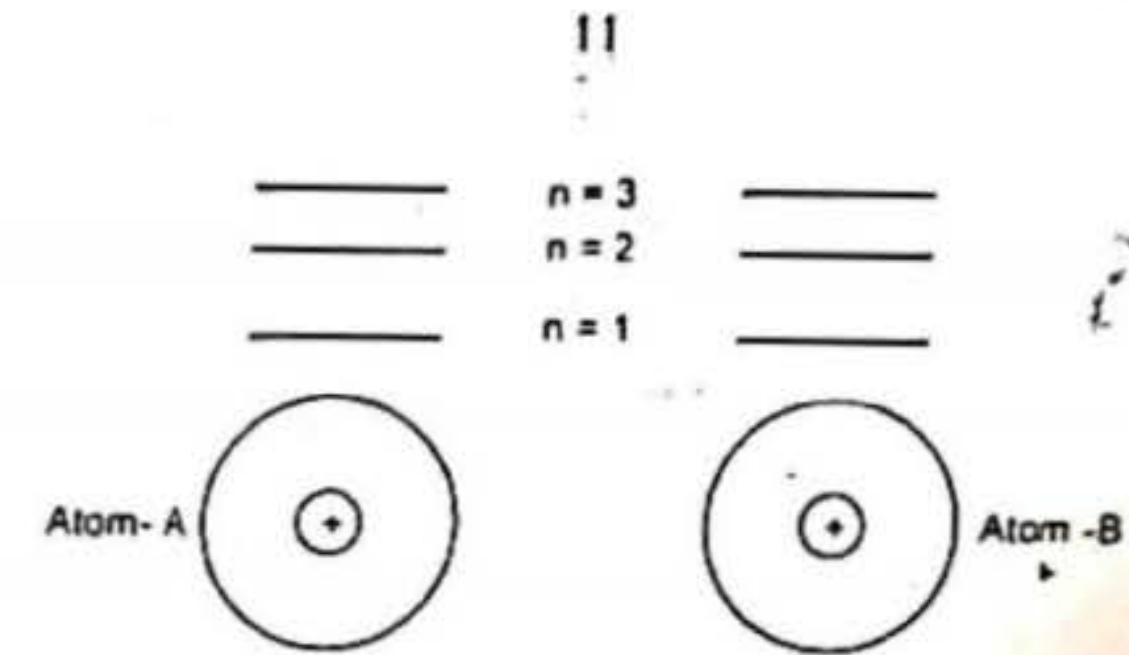
The tetrahedral arrangement of atoms also occurs in compound semiconductors such as gallium arsenide and gallium phosphide, known as III-V compounds. Gallium, with three valence electrons, forms the central atom. It shares its three valence electrons with other neighbouring atoms of Arsenic or Phosphorous, which has five valence electrons.

Metallic bonding occurs in groups I, II, and III elements. A group I element, (such as copper for an example), has one valence electron, which each copper atom releases. The released electron is no longer attached to its parent atom and can wander freely through the crystal. In the copper crystal, each atom releases an electron. The interaction between the cloud of free electrons so released and the nuclei form what is called metallic bond.

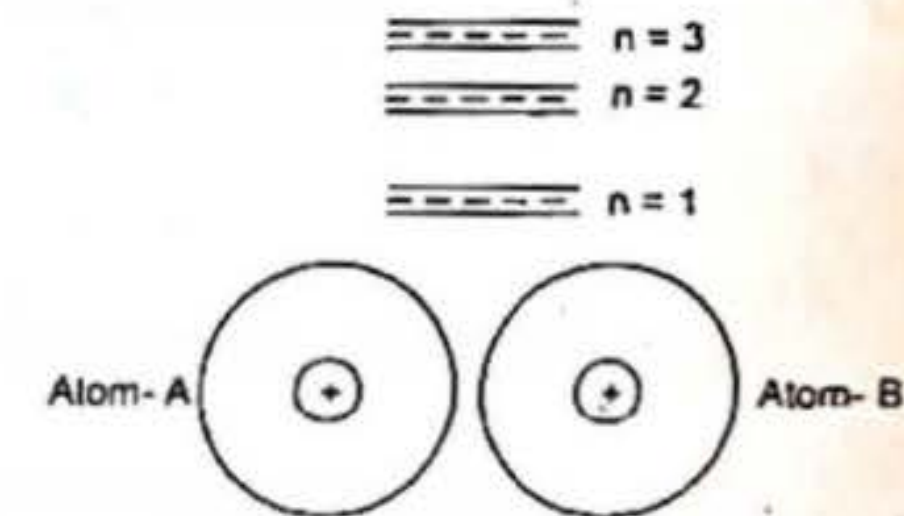
Energy Band Structure

The energy levels calculated for the single isolated hydrogen atom were seen to be discrete levels determined by the quantum number n . If there are two such atoms well separated, both these atoms will have the same energy levels as shown in figure 1.6a. Now, if these atoms are brought closer together, they will begin to interact so that the nucleus and electrons of atom A will exert a force on the electrons of atom B, and vice versa. This interaction will cause the energy levels of the electrons to split so that, in place of each of the single energy levels, the two electrons will have available two energy levels (figure 1.6b) for each value of n .

* In an assembly of N atoms closely packed as in a crystal, each energy state typical of one atom is split into N levels each of which can, by Pauli's



(a) Two hydrogen atoms well separated



(b) Splitting of energy levels

Fig. 1.6 Interaction of atoms leading to energy bands

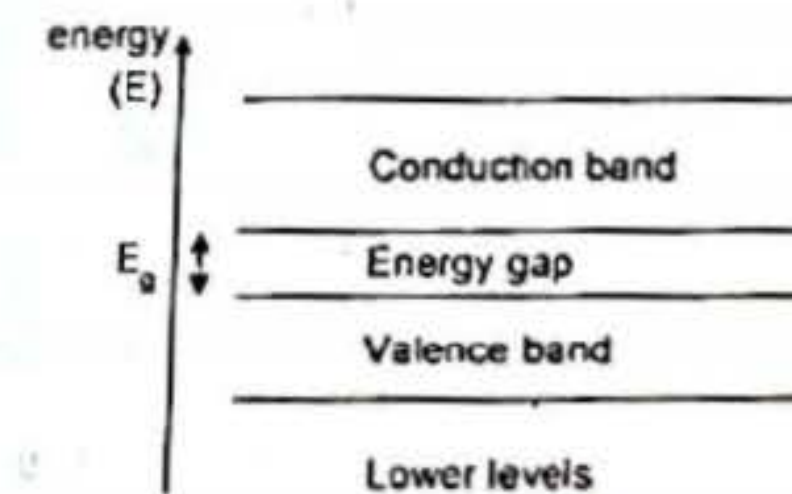


Fig. 1.7 Simplified energy band structure

Exclusion principle, accommodate two electrons of opposite spin.

In an assembly of N atoms therefore, the separation of the N levels are so small that they may be regarded as an almost continuous band. In place of each precisely defined characteristic energy level of an individual atom, the entire crystal possesses an energy band composed of very large number of separate levels closely packed. The simplified band structure is as illustrated in figure 1.7.

In the assembly of atoms comprising a solid, the various energy bands may overlap (as in a metal) in which case its electrons have a continuous distribution of permitted energies; or the energy bands may remain separated (as in a semiconductor and an insulator) and gaps between them represent energies which their electrons cannot possess. Such gaps are called **forbidden energy bands**. The two bands of most interest in considering the electrical properties of materials are the valence band and the conduction band.

Classification Of Electrical Materials

An electron in orbit has a potential energy due to its position. This potential energy increases with increase in orbital radius. Thus, the most distant electrons from the nucleus are at the highest energy level. The electrons at the outermost shell of an atom, called valence electrons, exist in the valence band of energies. The valence band is therefore the highest energy level that an electron can occupy (at zero degree Absolute). Energy bands higher than the valence band are unoccupied at zero degree Absolute and are called the conduction bands.

Materials can be electrically classified as conductors, insulators, and semiconductors according to their electrical resistivity or conductivity. Resistivity and conductivity are dependent on the density (or number) of free electrons (current carriers) in the conduction band, which in turn depends on the energy band structure of the material. The classification of materials may therefore be determined by their energy band structure (figure 1.8).

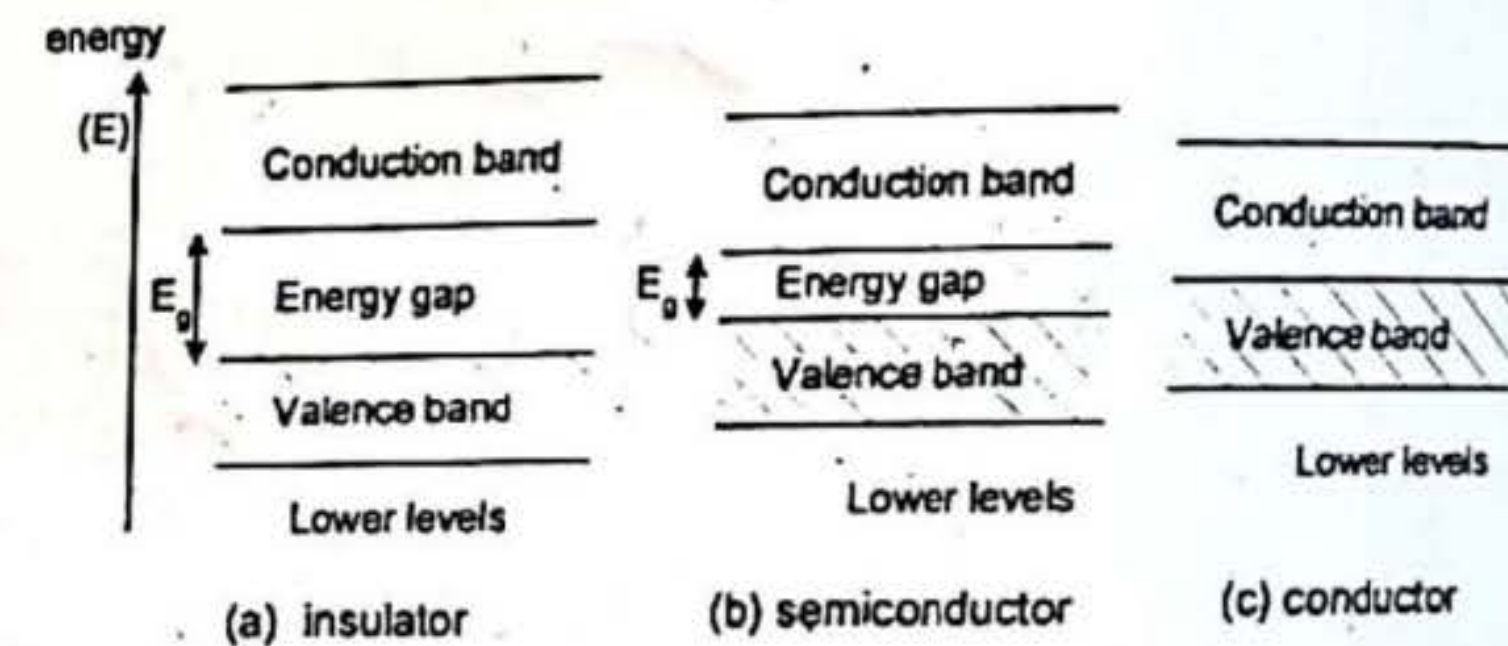


Fig. 1.8 Classification of materials by band structure

Insulators

An insulating material has a very wide energy gap existing between the valence band and the conduction band (figure 1.8a). As a result of the wide gap, the additional energy required by a valence electron to move into the conduction band, (where it can take part in electrical conduction), is rather large. Consequently, the conduction band is virtually empty of free charge carriers since valence electrons remain stuck to their parent atoms. The absence of free electrons in the conduction band results in very high resistivity, or zero conductivity.

Insulating materials are therefore used as dielectrics, which have the property of blocking the passage of electricity. Examples of insulating materials are glass, quartz, mica, distilled water, hard rubber, dry paper and dry wood, Teflon, and silicon dioxide (SiO_2).

Conductors

The energy band structure of materials, which are considered to be good electrical conductors, is such that the valence band and the conduction band overlap (figure 1.8c). Thus, valence electrons require little or no extra energy to move into the conduction band. Good conductors, (which are mostly metals), have very large number of free electrons (of the order of $10^{28}/\text{m}^3$) available in the conduction band. The most used conductors are copper, aluminium, silver and gold, in that order. These metals are used as wires and conductors in electrical and electronic circuits and systems. The more resistive alloys of nickel and chromium are used in heating elements. Tungsten filaments are used in electric lamps, and carbon compounds are used in precision resistors.

Semiconductors

The energy gap in the semiconductor material is much narrower than for the insulator (figure 1.8b). At low temperatures, well below room temperature, the semiconductor behaves as an insulator (because the extra thermal energy gained by the valence electrons is not sufficient to bridge the energy gap).

At room and higher temperatures, valence electrons acquire extra thermal energy sufficient to bridge the energy gap and move into the conduction band.

It follows therefore that the semiconductor material is neither a good conductor nor a good insulator. Its conductivity is very much dependent on temperature. The density of charge carriers in semiconductor materials vary from about $10^{19}/\text{m}^3$ to $10^{25}/\text{m}^3$, depending on temperature and doping concentration (soon to be explained). The usefulness of the semiconducting material to the electronic Engineer lies in the fact that the number of free charge carriers can be controlled (by doping, or by directing electromagnetic radiation on them, or by subjecting them to high electric fields, etc). In fact, the operation of all semiconductor devices is governed by the behaviour of charge carriers in the semiconductor under thermal equilibrium conditions and under the influence of external fields (See Chapter Two for further discussion). The densities of charge carriers and the energy gap for some materials is given in Table 1.2.

Table 1.2 Electron density (n) and energy gap (E_g) at 300°K

Conductors			Semiconductors			Insulators		
	E_g (eV)	n (m^{-3})		E_g (eV)	n (m^{-3})		E_g (eV)	n (m^{-3})
copper	0	10^{29}	germanium	0.7	10^{19}	diamond	5.5	0
aluminium	0	10^{29}	silicon	1.1	10^{16}			
silver	0	10^{29}	GaAs	1.4	10^{23}	SiO_2	8.0	0
gold	0	10^{29}	GaP	2.3	10^8			

CHAPTER TWO

SEMICONDUCTOR MATERIALS

	Page
Intrinsic and Extrinsic Materials	17
• P-Type material	19
N-Type material	19
Majority and minority Current Carriers	20
Improved Conductivity Of Extrinsic Material	20

CHAPTER TWO

SEMICONDUCTOR MATERIALS

2.1 Intrinsic And Extrinsic Materials

Semiconductor materials, such as Germanium and Silicon, have four valence electrons in their outer shells. In the pure (i.e. intrinsic state), they form covalent bonds by each atom sharing its valence electrons with four neighbouring atoms, each neighbouring atom providing one electron as illustrated in figure 1.5b.

In the pure state and at zero Absolute temperature, the covalent bonds remain intact and hence no free electrons are available for conduction. The material therefore behaves as an insulator at zero degree Absolute. As the temperature is increased to room temperature (about 300°K), covalent bonds get broken, (as electrons gain thermal (heat) energy sufficient to bridge (overcome) the energy gap and move into the conduction band). Thus, when a bond is broken, an electron is released and moves into the conduction band. For a bond to be broken, the electron must acquire an extra energy of at least (the gap energy).

When an electron moves into the conduction band, the positively charged nucleus left behind in the valence band is called a hole. A hole is therefore considered as an electron vacancy. We note therefore that for each bond that is broken, an electron-hole pair is formed.

Both the electron and the hole can take part in electrical conduction. The electron constitutes a negative charge carrier, while the hole is a positive charge carrier. Electron conduction takes place in the conduction band, while hole conduction takes place in the valence band.

The concept of hole conduction may be explained as follows: Since a hole is an electron vacancy, this vacancy can be filled by an electron transferring from a neighbouring atom, thus leaving behind a hole. To an observer outside, the hole has moved from the first position to the second. A succession of such electron transfers is conveniently described as motion of positive holes. We should realise that the hole motion is apparent, since the hole cannot physically change its position.

An extrinsic semiconductor is formed when a precisely controlled quantity of another material (which is considered an impurity) is added to the pure semiconductor material. The deliberate addition of 'impurity' is called **doping**. There are two types of controlled doping:

- (a) The addition of a trivalent material (e.g. Boron, Aluminium, or Gallium) to produce p-type material, and
- (b) The addition of a pentavalent material (e.g. Arsenic, Phosphorous, or Antimony) to produce n-type material.

The concentration of impurity atoms in the semiconductor crystal is of the order of one atom of impurity to 10^7 atoms of the semiconductor material. Thus, the major bulk of the material is intrinsic. This level of impurity is however sufficient to alter the characteristics of the semiconductor material.

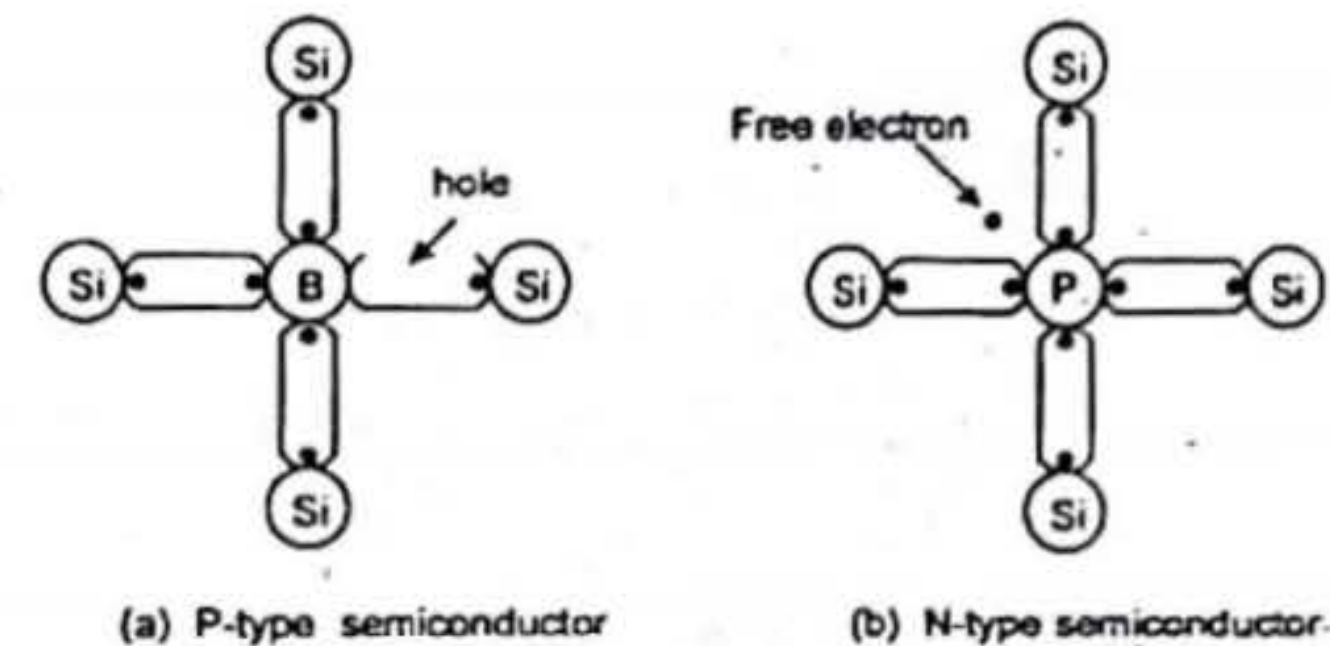


Fig.2.1 Bonding in semiconductor materials

P-Type Material

The bonding of one atom of Boron (a trivalent material) with neighbouring Silicon atoms is illustrated in figure 2.1a. The three Boron electrons are shared, leaving an electron vacancy or hole. The hole can attract electrons from a neighbouring silicon atom to fill it. The dopant boron is called an **acceptor** and the material so formed is called a **p-type** semiconductor material. Electrical conduction in a p-type material is mainly by hole or positive charge carriers, and takes place in the valence band.

N-Type Material

Doped with a pentavalent impurity (e.g. Phosphorous), we observe in figure 2.1(b) that the fifth electron from the Phosphorous atom is not bonded or attached to any atom and hence is free to move into the conduction band.

The material so produced is called a **n-type** material, and the dopant material (Phosphorous in this case) is called a **donour**. Conduction in the n-type semiconductor is mainly by negative charge carriers (electrons) and takes place in the conduction band.

Majority And Minority Current Carriers

At temperatures well below room temperature, all of the donour electrons or acceptor holes are ionised (in the extrinsic material) and can take part in electrical conduction. As temperature rises to room temperature and above, covalent bonds are broken and electron-hole pairs are released. Therefore, both electrons and holes are present and constitute the current carriers in any semiconductor material. In the intrinsic material, there are equal number of electrons and holes. In an extrinsic material however, we do have majority and minority current carriers. The majority current carriers in the n-type material are electrons, while holes constitute minority carriers. In the p-type material, holes form the majority current carriers and electrons are minority carriers. It should be observed that the generation of minority current carriers in each type of material is due to thermal effects, which give rise to the breaking of covalent bonds and the consequent release of electron-hole pairs. (As we will see in chapter seven, photoelectric effects can also be used to release minority current carriers)

Improved Conductivity Of Extrinsic Material

The simplified energy band diagrams for intrinsic and extrinsic materials are illustrated in figure 2.2.

The impurity energy levels are shown dotted and lie within the forbidden gap. The donour level E_d lies just below the bottom of the conduction band (figure 2.2b), and the acceptor level E_a just above the top of the valence band (figure 2.2c). In the n-type material therefore, it is easy to see that a donour electron needs only a little extra energy (much less than E_g) to make a transition into the conduction band and take part in electrical conduction.

In the p-type material, electrons in the valence band require only little extra energy to move up and fill the acceptor holes, leaving behind an equal number of holes in the valence band which can take part in hole conduction.

Consequently, all impurity atoms are ionised at near zero temperature, thus releasing current carriers even at that temperature. At higher temperatures, covalent bonds get broken, as in the intrinsic material. The concentration (density) of charge carriers is therefore higher in the extrinsic material than in the intrinsic material.

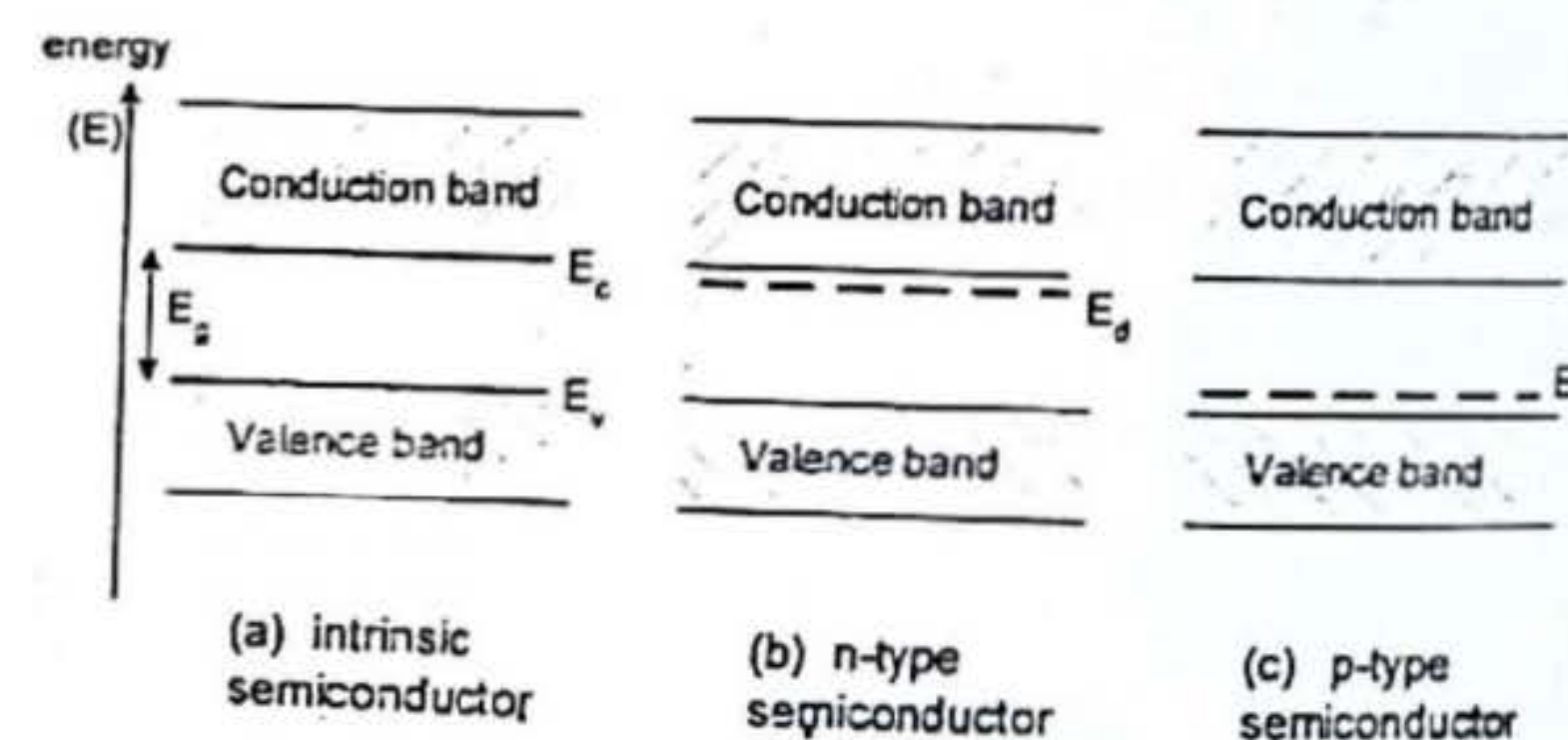


Fig. 2.2 Simplified energy level diagrams for semiconductor materials

CHAPTER THREE

THE DENSITY OF ENERGY STATES

	Page
3.1 The Uncertainty and Exclusion Principles	23
3.2 Allowed Energies in a Crystal	24
3.3 Carrier Population in Bands	26
The Fermi Energy and Probability Function	26
Carrier Densities in a Semiconductor	29
Temperature Effects	32
Examples	33

CHAPTER THREE

THE DENSITY OF ENERGY STATES

The density of energy states examines how many energy values an electron may take up in a given range of energies from E to $E + dE$, when the electron is confined in a crystal. The derivation of the density of states expression requires the use to two physical principles; the **Principle of Indeterminacy** (or the **Uncertainty Principle**) and the **Pauli's Exclusion Principle**.

3.1 The Uncertainty And Exclusion Principles

The quantum mechanical theory does not allow us to observe the velocity (u) of an electron nor determine the position (x) precisely. If the uncertainty (or the error) in determining the value of u is Δu , and in x is Δx , then the uncertainty principle states that

$$m \Delta u \Delta x \geq h \quad \dots\dots\dots (3.1a)$$

Or

$$\Delta p \Delta x \geq h \quad \dots\dots\dots (3.1b)$$

where $p (= m \Delta u)$ is the uncertainty in determining the momentum of the electron.

We observe from this expression that, if the velocity u is more accurately known (i.e Δu is small), then the greater will be the uncertainty in it position x (i.e Δx is large), and of course vice versa.

Re-writing equation 3.1a, as $\Delta u \Delta x \geq h/m$, we see that for particles of very low mass such as the electron, the error involved can be very large indeed. For an example, if the position of the electron is determined to within 2×10^{-9} , the error in its speed is

$$\begin{aligned}\Delta u &= [6.6 \times 10^{-34}] / [9.1 \times 10^{-31} \times 2 \times 10^{-9}] \\ &= 3.638 \times 10^5 \text{ m/s}\end{aligned}$$

The Exclusion principle states that no more than two electrons in an atom can have the same energy. This arises from the fact that an electron can spin about its axis in two ways. Thus each orbit can accommodate two electrons, each spinning in the opposite direction to the other.

3.2 Allowed Energies In A Crystal

Assume for the moment that an electron, mass m , is constrained to move in a crystal only along a line, such that the uncertainty in its position is Δl . The uncertainty in its velocity (Δu) is obtained from equation 3.1 as

$$\Delta u \geq h/m\Delta l \quad \dots\dots\dots (3.2)$$

Consider velocities lying between u_1 and u_2 ; also along a straight line (figure 3.1a). The number of separately distinguishable velocities lying between u_1 and u_2 is given as

$$(u_2 - u_1) / \Delta u = m\Delta l (u_2 - u_1) / h \quad \dots\dots\dots (3.3)$$

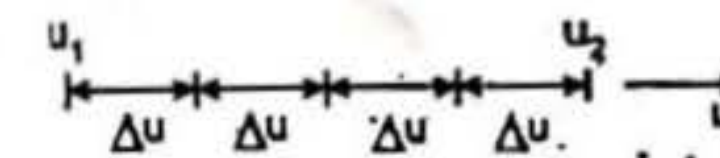
If the electron is now allowed to move in three dimension, all possible velocities must now lie in spherical shell of inner radius u_1 and outer radius u_2 (figure 3.1b). The volume of the shell in 'velocity space' is $(4\pi/3)(u_2^3 - u_1^3)$. Thus, from equation 3.2, each velocity must be separated from its nearest neighbour by a distance $\Delta u = h/(m\Delta l)$.

Each velocity requires a volume in velocity space of $(h/m\Delta l)^3$.

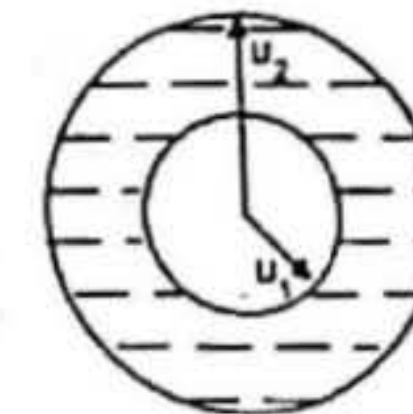
Suppose dU is the number of allowed velocities between u_1 and u_2 .

Then,

$$\begin{aligned}dU &= (\text{volume of shell}) / (\text{volume required by each velocity}) \\ &= (4\pi/3)(u_2^3 - u_1^3) / [h/m\Delta l]^3 \\ &= (4\pi/3) [m\Delta l/h]^3 (u_2^3 - u_1^3) \quad \dots\dots\dots (3.4)\end{aligned}$$



(a) Distinguishable velocities in one dimensional crystal



(b) Distinguishable velocities (dashed lines) in a sphere

Fig.3.1 Distinguishable velocities

In a cube of unit length, ($\Delta l = 1$), the number of allowed velocities is given by

$$dU = (4\pi/3) [m/h]^3 (u_2^3 - u_1^3) \quad \dots\dots\dots (3.5)$$

An electron in orbit has velocity and hence kinetic energy. Let E_1 and E_2 be the energies associated with the velocities u_1 and u_2 respectively. Then,

$$E_1 = \frac{1}{2} m u_1^2 \text{ and } E_2 = \frac{1}{2} m u_2^2$$

In terms of energies E_1 and E_2 , the number of electrons (and hence the number of energy states) per unit volume of crystal is given as

$$dN = (8\pi/3) \sqrt{2} [(m/h)^3] (E_2^{3/2} - E_1^{3/2}) \quad (3.6)$$

For $E_1 = E$ and $E_2 = E + dE$, then

$$\begin{aligned} E_2^{3/2} - E_1^{3/2} &= (E + dE)^{3/2} - E^{3/2} \\ &\approx E^{3/2} + \sqrt{E} dE - E^{3/2}, \quad (\text{for } dE \text{ small}) \\ &= \sqrt{E} dE \end{aligned}$$

Thus,

$$dN = (8\pi m^3/3h^3) \sqrt{2mE} dE = N(E) dE \quad (3.7)$$

This is the density of energy states expression. $N(E) = (8\pi m^3/3h^3) \sqrt{2mE}$ is the density of available energy states in energy range dE that may be occupied by an electron. We note however that two electrons, spinning in opposite directions, may be allowed to share each energy state as allowed by the Exclusion principle.

3.3 Carrier Population In Bands

The carrier population (or density) in a band of energies can be determined if we note that the density of occupied energy states, (and hence the number of electrons per unit volume in energy range dE) is the product of the density of states $N(E)dE$ and the probability of occupancy $f(E)$

The Fermi Probability (or Distribution) Function And Fermi Energy

The probability that an electron in the crystal, at a temperature $T^\circ K$, will occupy a particular energy state E is given by the Fermi distribution function $f(E)$ expressed as

$$f(E) = 1/(1 + \exp[(E - E_f)/kT]) \quad (3.8)$$

where k is Boltzmann's constant, and E_f is Fermi energy. At zero degree Absolute, the function $f(E)$ may take two distinct values. Thus, for

$$E < E_f, \quad f(E) = 1/[1 + e^{-\infty}] = 1$$

And for

$$E > E_f, \quad f(E) = 1/[1 + e^{\infty}] = 0$$

The probability of 1 means that all allowed energies between 0 and E_f are occupied (or filled), and the probability of 0 means that all energies above E_f are unoccupied (or empty). From the above, the Fermi energy E_f can then be defined as the highest energy level (or state) that any electron can occupy at zero degree Absolute. The energy E_f is a reference energy level, which may not correspond to any allowed energy state. The two values of the Fermi function, at zero degree, are plotted (rectangular curve) as shown in figure 3.2 (full line).

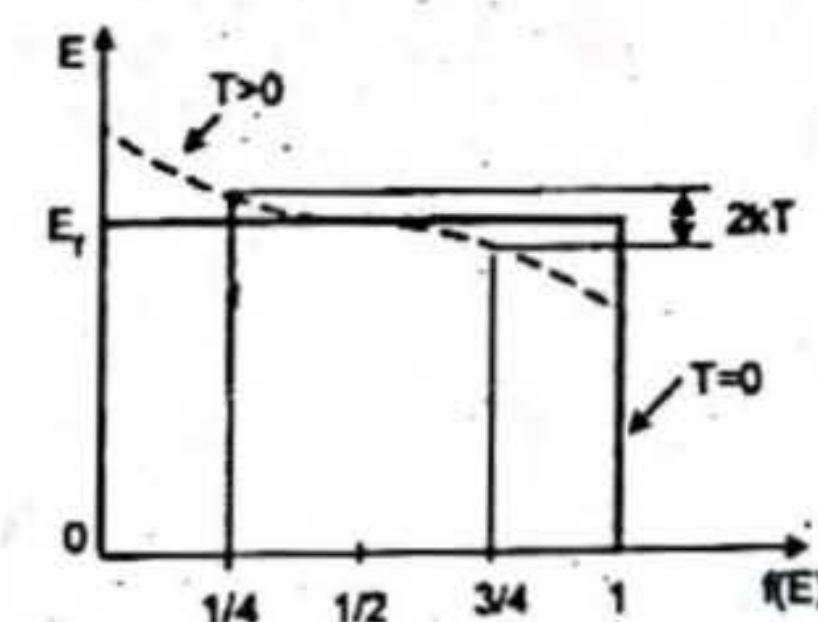


Fig. 3.2 Probability function plots

At a finite temperature $T^\circ\text{K}$, we note that at the energy

$$E = (E_f + kT), \quad f(E) = 1/(1 + e^1) = 1/2$$

At $E = E_f, \quad f(E) = 1/(1 + e^0) = 1/2$

And at

$$E = (E_f - kT), \quad f(E) = 1/(1 + e^{-1}) = 1/2$$

These values are also plotted (dashed line) in figure 3.2. The probability of $1/2$ means that an electron is likely to occupy any energy level above the Fermi energy as below it at the finite temperature $T^\circ\text{K}$.

The magnitude of the Fermi energy, for a given material, can be calculated. To do this, we note that if we place n electrons in a unit volume of crystal lattice at zero degree absolute temperature, the electrons will fill up available energy slots from 0 to the energy E_f according to equation 3.7. (Observe that n in this case is the density of electrons in the crystal). The value of E_f may be obtained by writing

$$n = \int_0^{E_f} dN f(E) \quad \dots \dots \dots (3.9)$$

Since at zero degree $f(E) = 1$, all energies below E_f are occupied. Then equation 3.9 becomes

$$n = \int_0^{E_f} dN = \int_0^{E_f} (8\pi m/h^3) \sqrt{2mE} dE \quad \dots \dots \dots (3.10)$$

This integrates to give,

$$n = (8\pi/3)(2m/h^2)^{3/2} E_f^{3/2}$$

From which,

$$\begin{aligned} E_f &= n^{2/3} [h^2/2m \times (3/8\pi)^{2/3}] \text{ joules} \\ &= n^{2/3} [(h^2/2m) \times (3/8\pi)^{2/3}] \times [1/1.6 \times 10^{-19}] \text{ eV} \end{aligned} \quad \dots \dots \dots (3.11)$$

It is observed from equation 3.11 that, under thermal equilibrium, a variation in electron density (n) (or a redistribution of electrons) affects the Fermi energy. From the above analysis therefore, the Fermi energy for a given metal can be found if the density (n) of electrons in that metal is known.

Carrier Densities In A Semiconductor

In this section, we will discuss the density of charge carriers (electrons and holes) in bands. In practice, electron conduction takes place in the conduction band, while hole conduction takes place in the valence band. From the Fermi function $f(E)$, defined as the probability of occupancy of an available energy state at E , we note that if there is no available state (as in the energy gap) there is no probability of finding an electron there.

The density of conduction band electrons (n_0), can be written as

$$n_0 = \int_{E_c}^{\infty} f(E) N(E) dE \quad \dots \dots \dots (3.12)$$

where the energy at the top of the conduction band can be considered, to a good approximation, to be infinite. The subscript n_0 indicates equilibrium condition. The quantity $N(E) = (8\pi m/h^3) \sqrt{2mE}$ is the density of states in the energy range dE . The probability of occupancy of energy E_c is given as

$$f(E_c) = 1/[1 + \exp(E_c - E_f)/kT]$$

For a semiconductor $\exp(E_c - E_f)/kT$ is greater than 1, and hence $f(E_c)$ can be approximated to

$$f(E_c) = \exp -(E_c - E_f)/kT$$

Thus, equation 3.12 becomes

$$\begin{aligned} n_0 &= \int_{E_c}^{\infty} [(8\pi m_0/h^3) \sqrt{2m_0E}] e^{-(E_c - E_f)/kT} dE \\ &= 2[(2\pi m_0 kT/h^2)^{3/2}] e^{-(E_c - E_f)/kT} \\ &= N_c e^{-(E_c - E_f)/kT} = N_c f(E_c) \end{aligned} \quad \dots \dots \dots (3.13)$$

And for the intrinsic material, the intrinsic density is

$$n_i = N_c e^{-(E_c - E_f)/kT} \quad \dots\dots\dots (3.14)$$

where m_n is the mass of an electron, and E_f is the Fermi energy for the intrinsic material. The quantity $N_c = 2[2\pi m_n kT/h^2]^{3/2}$ is considered the 'effective density of states' located at conduction band edge E_c .

Re-writing equation 3.13, we have

$$(E_c - E_f) = kT \ln(N_c/n_o) \quad \dots\dots\dots (3.15)$$

Equations 3.13 and 3.15 predict that, as electron concentration in the conduction band increases from n_i to higher values, the Fermi energy E_f moves closer to the conduction band.

For hole concentration in the valence band, we note that a hole is the absence of an electron. Therefore the probability of an empty state containing no electron at energy range E is

$$\begin{aligned} 1 - f(E) &= 1/[1 + e^{(E_f - E)/kT}] \\ &= e^{-(E_f - E)/kT} \quad (\text{for } E_f > E) \quad \dots\dots\dots (3.16) \end{aligned}$$

This is the probability of hole occupancy. Using the analytical procedure for electron concentration, the hole concentration in the valence band can be shown to be given as

$$p_o = N_v e^{-(E_f - E_v)/kT} \quad (\text{for extrinsic material}) \quad \dots\dots\dots (3.17a)$$

And,

$$p_i = N_v e^{-(E_i - E_v)/kT} \quad (\text{for intrinsic material}) \quad \dots\dots\dots (3.17b)$$

where $N_v = 2[(2\pi m_p kT/h^2)^{3/2}]$ is the 'effective density of state' located at the valence band edge E_v , and m_p is the mass of a hole. Equation 3.17 can also be re-written as

$$(E_f - E_v) = kT \ln(N_v/p_o) \quad \dots\dots\dots (3.18)$$

Equation 3.17 and 3.18 also predict that as hole concentration increases from p_i to higher values, E_f moves closer to the valence band. These predictions are valid for both intrinsic and doped semiconductors provided thermal equilibrium is maintained.

At a given temperature, the generation and recombination mechanism precisely balance, so that the equilibrium hole and electron concentration satisfy the relationship

$$n_o p_o = n_i^2 \quad \dots\dots\dots (3.19)$$

This is the 'mass action law' where n_o and p_o are equilibrium electron and hole concentrations, and n_i is intrinsic carrier concentration. In the intrinsic material, $n_o = p_o = n_i$. In the extrinsic material, n_o is not equal to p_o , but it can be shown that $n_o p_o = n_i^2$. The value of n_i depends on temperature and may be given as

$$n_i p_i = n_i^2 = N_c N_v e^{-(E_c - E_v)/kT} = A_o T^3 e^{-(E_{go}/kT)}$$

And,

$$n_i = (A_o T^3)^{1/2} e^{-(E_{go}/2kT)} \quad \dots\dots\dots (3.20)$$

where $E_{go} = (E_c - E_v)$ is the zero temperature energy gap, and A_o is a constant, both of which are properties of the specific semiconductor.

A second constraint on the value of n_o and p_o comes from the general requirement that charge neutrality must be maintained every where in the crystal. That is, there must be equal densities of positive and negative charges in the material. The charge neutrality condition is stated as,

$$n_o + N_A = p_o + N_D \quad \dots\dots\dots (3.21)$$

where N_A and N_D are atomic densities of acceptor and donor impurities respectively. Equations 3.19 and 3.21 must be solved simultaneously in order to find the hole and electron concentrations in doped semiconductor.

Temperature Effects

Carrier concentration is temperature dependent. This is because both n_i and E_f are temperature dependent. For the temperature dependence of n_i , we take logarithm of equation 3.20. This gives,

$$\ln(n_i) = (\ln(A_0)^{1/2} + \ln T^{3/2}) - (E_g/2k)(1/T) \dots \dots \dots (3.22)$$

For temperatures above about 200°K, the term containing $1/T$ varies more rapidly with temperature. Consequently, $\ln(n_i)$ increases almost linearly as $1/T$ decreases. The temperature dependence of electron concentration in an extrinsic material is illustrated in figure 3.3.

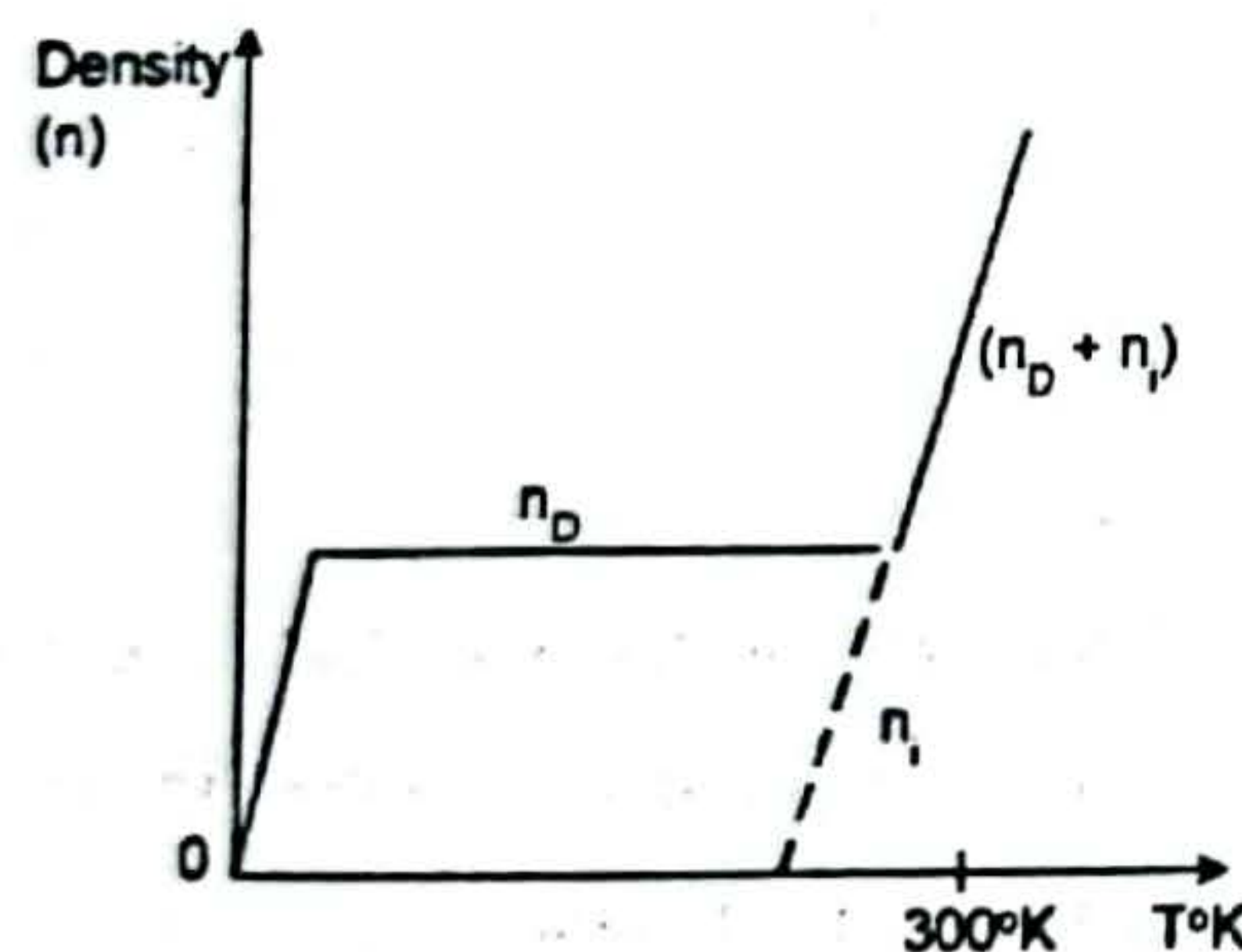


Fig. 3.3 Temperature effects on electron density

For temperatures slightly higher than 0°K, all the donor atoms are ionised and remains essentially constant at n_D . For temperatures above about 200°K, $n = n_D + n_i$ where $n_D = N_D$ (approximately).

Examples

E3.1

A piece of Silicon is uniformly doped with 10^{17} Arsenic atoms per cm^3 . If n_i for Silicon is $1.5 \times 10^{10} \text{ cm}^{-3}$, calculate the equilibrium hole concentration at 300°K. Show that $(E_f - E_i) = 0.4$.

Solution

Since $N_D > n_i$, we can approximate $n_o = N_D$. Hence,

$$p_o = n_i^2/n_o = 2.25 \times 10^{20}/10^{17} = 2.25 \times 10^3 \text{ cm}^{-3}$$

From the expression, $n_o = n_i e^{(E_f - E_i)/kT}$ we obtain

$$\begin{aligned} (E_f - E_i) &= kT \ln(n_o/n_i) = 0.025 \ln(10^{17}/1.5 \times 10^{10}) \\ &= 0.407 \text{ eV} \end{aligned}$$

E3.2

Silicon has intrinsic carrier concentration $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$. It is doped with Phosphorous atoms to a density of $N_D = 10^{11} \text{ cm}^{-3}$. Calculate the equilibrium hole and electron densities p_o and n_o respectively in the resulting extrinsic material.

Solution

From the expression $n_o p_o = n_i^2$, we have $p_o = n_i^2/n_o \dots \dots \dots (a)$

From charge neutrality expression and noting that $N_A = 0$, we have

$$n_o = p_o + N_D = n_i^2/n_o + N_D$$

From which,

$$n_o^2 - n_o N_D - n_i^2 = 0.$$

This solves to give,

$$n_o = N_D/2 \pm \frac{1}{2}[N_D^2 + 4n_i^2]^{1/2}$$

Taking the positive sign gives, $n_o = 1.02 \times 10^{11} \text{ cm}^{-3}$.

(Note that this value of n_o is approximately equal to N_D . Hence, for $N_D \gg n_i$, n_o can be approximated to N_D).

Substituting for n_o in equation (a), gives $p_o = 2.25 \times 10^9 \text{ cm}^{-3}$.

CHAPTER FOUR

CHARGE TRANSPORT PROCESS

	Page
4.1 Drift and Diffusion flow	36
Drift Flow	36
Diffusion Flow	39
4.2 Concept of Carrier Lifetime	41
4.3 Magnetic effects on Current Carriers (The Hall effect)	43
4.4 Examples	46

CHAPTER FOUR

CHARGE TRANSPORT PROCESS

4.1 Drift And Diffusion flow

In general, there are two mechanisms of current flow; drift and diffusion flow.

Drift flow

In a metal such as copper, the density of atoms is of the order of 10^{29} atoms per meter cubed. If each atom produces one valence electron, there are 10^{29} free electrons/ m^3 . At normal temperature, these free electrons are in ceaseless random motion due to their thermal or heat energy. There is however no net flow of electrons (and hence no net flow of charges) in any one given direction. There is therefore no current flow. The random zig-zag motion of the electrons is due to

- (i) collision with the crystal atoms, which themselves are vibrating about their mean position in the lattice, and
- (ii) abrupt deflection of electron path, when the electron passes near an impurity ion.

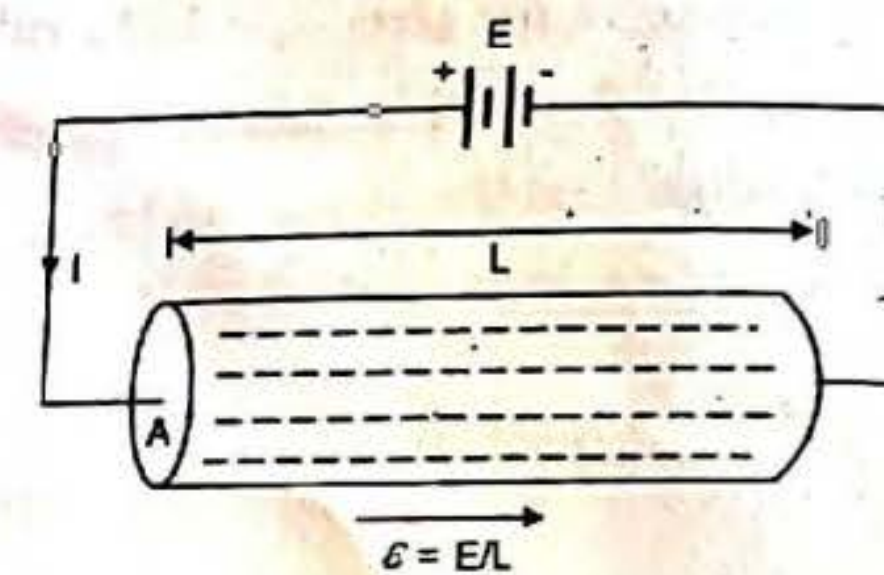


Fig. 4.1 Drift current in a conductor with applied emf

If an electric field is applied, the random motion of the electrons continue, but the electrons tend to drift in a direction parallel to the field. Hence, in a given time interval, there is a net flow of electrons across a given plane and consequently there is current flow. This is drift current, which is due to the influence of an electric field.

Consider the conducting solid (figure 4.1), of length L and cross-sectional area A , to which is connected a voltage source E . The electric field E ($= E/L$) causes electrons to drift in the solid towards the positive terminal of the voltage source. A continuous flow of charges thus circulates through the closed loop. Note that the direction of conventional current flow (i.e the direction in which a positive charge flows) is opposite to the direction of electron flow.

Let the conductor be considered to have N electrons distributed through it. If an electron travels the length L in time T seconds, due to the influence of the electric field force acting on it, then the average drift velocity $u = L/T$. The current I , defined as time rate of flow of charge, is given by

$$I = qN/T = qNu/L \text{ Amps} \quad \dots\dots\dots (4.1)$$

where q is charge on the electron.

The current density J (defined as current per unit area) in the conducting medium is

$$J = I/A = qNu/AL \text{ A/m}^2 \quad \dots\dots\dots (4.2)$$

But AL is the volume of the conducting medium containing N electrons. Thus, electron density $n = N/AL$. Substituting in 4.2, gives the current density

$$J = qnu \text{ A/m}^2 \quad \dots\dots\dots (4.3)$$

The quantity qn is the volume charge density in Coulombs/m³.

The drift velocity u is proportional to the applied electric field ⁽¹⁾ (and hence to the applied voltage). Thus,

$$u = \mu \epsilon \quad \dots\dots\dots (4.4)$$

where μ is the constant of proportionality and is called the mobility. It is defined as the average drift velocity per unit electric field. Substituting for u in the expression for current density, gives $J = (qn)u = qn\mu\epsilon$.

Since $\epsilon = E/L$, then $J = q\mu nE/L$, and $I = q\mu nAE/L$.

From Ohm's law, $I = E/R$. Therefore, $R = (\rho L/A) = L/q\mu nA$ and hence $\rho = 1/q\mu n$ is the resistivity of the conducting medium. The expression for conductivity then becomes $\sigma = (1/\rho) = q\mu n \text{ (Ohm-m)}^{-1}$

Considering that there are two types of current carriers in a semiconductor, (i.e. electrons and holes) we can talk of electron mobility (μ_n) and hole mobility (μ_p). The current density expression is therefore modified to read;

(1) Note that if τ is the mean-free-time between collisions, then $u = a\tau = (q\epsilon/m)\tau$, where a is acceleration, m is mass of electron, and $q\epsilon$ is force on an electron in an electric field.

$J = q\epsilon (n\mu_n + p\mu_p)$, and conductivity $\sigma = q(n\mu_n + p\mu_p)$. For the intrinsic semiconductor material, $n = p = n_i$. Hence,

$$\sigma = qn_i(\mu_n + \mu_p) \quad \dots\dots\dots (4.5)$$

Diffusion Flow

Diffusion mechanism of charge transport arises due to non-uniform charge concentration. Suppose, for an example, a large number of electrons are injected instantaneously into the centre of a piece of semiconductor crystal (figure 4.2). The resulting non-uniform charge distribution leads to a charge gradient (dQ/dx) being set up. A plot of the electron density (or concentration) versus distance x from $x = 0$ along the crystal, with time, reveals an exponential behaviour, which is symmetrical in both directions. Note that this is a two-dimensional model of a three-dimensional problem. This electron distribution involves charge movement through the material and therefore we can talk of a current flow, the diffusion current flow. The charge flow is from a region of high concentration to a region of low concentration. This diffusion process tends to keep the densities of free electrons and holes constant throughout the material (when equilibrium is attained). The diffusion process is very important in the operation of such semiconductor devices as the transistor.

Diffusion current density J is proportional to the charge gradient. It is given by $J = -D (dQ/dx)$, where D is the diffusion constant. The electron current density becomes

$$J_n = D_n q (dn/dx) \text{ A/m}^2 \quad \dots\dots\dots (4.6)$$

And the hole current density

$$J_p = -D_p q (dp/dx) \text{ A/m}^2 \quad \dots\dots\dots (4.7)$$

where D_n and D_p are diffusion constants for electron and hole diffusion respectively. Diffusion constant D and mobility μ are related by Einstein diffusion equation given as

$$D_n/\mu_n = D_p/\mu_p = kT/q \quad \dots\dots\dots (4.8)$$

The total current in the semiconductor crystal is the sum of drift plus diffusion flow. Hence, total current density for electron conduction is

$$J_n = [qn\mu_n E + D_n q (dn/dx)] \text{ A/m}^2 \quad \dots\dots\dots (4.9)$$

And for hole conduction,

$$J_p = [qp\mu_p E - D_p q (dp/dx)] \text{ A/m}^2 \quad \dots\dots\dots (4.10)$$

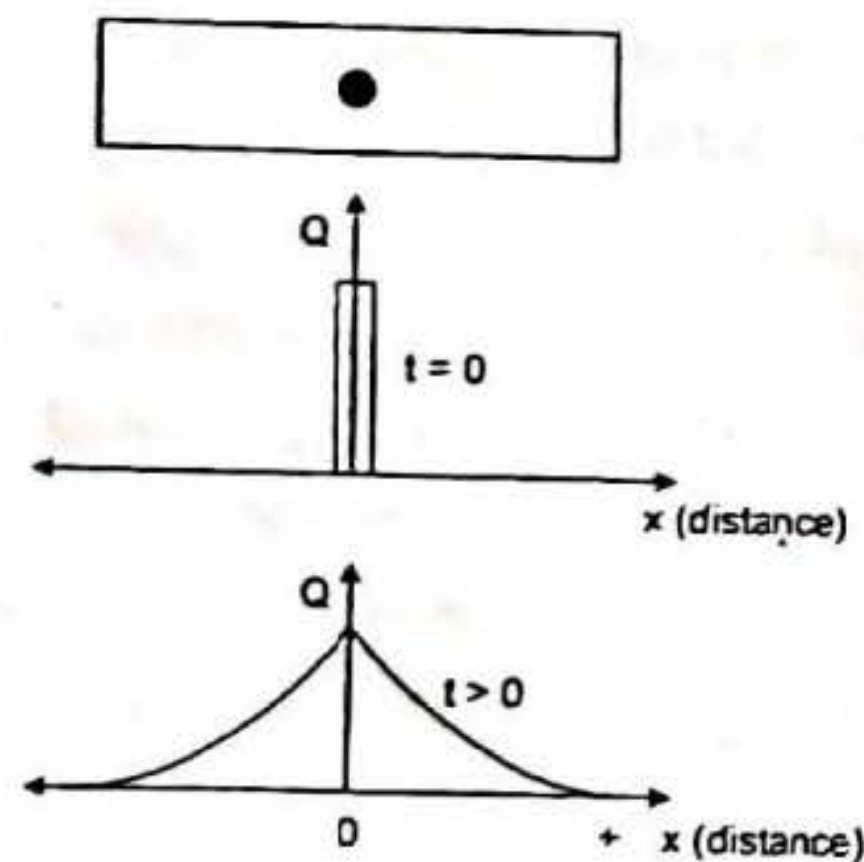


Fig. 4.2 Diffusion of charge carriers in a crystal

4.2 Concept Of Carrier Lifetime

The thermal vibration of the constituent atoms in a semiconductor constantly excites electrons up into the conduction band, leaving behind an equal number of free holes in the valence band. Existing free electrons also fall back into the valence band, neutralizing the free holes. As a result of this constant and equal process of generation and recombination, the population of free current carriers in a semiconductor is kept constant. Recombination centres (or traps) in the semiconductor materials include imperfections such as ionised foreign impurities and broken covalent bonds.

A finite time elapses between the generation and recombination of a charge carrier, during which the charge is free to carry current. The mean value of this period for all carriers of the same type is known as the lifetime (τ). If the generation rate is g (electrons and holes per cubic metre per second) and the lifetime of electron-hole pair is τ , then a stable population density n (free carriers per cubic metre) is obtained, where $g = n/\tau$. In a semiconductor, we can talk of the lifetime for electrons and lifetime for holes.

Carrier lifetime cannot be directly computed. We note however that at a given temperature, a state of thermal equilibrium is attained in which the rate of generation equals the rate of recombination. If an external source of energy (e.g. light or photon radiation) is used to excite electrons, the thermal equilibrium condition is upset. The rate at which the electron-hole pairs are generated increases, and so also does the rate of recombination. Thus, a new set of equilibrium condition will be attained if the irradiation remains constant.

The increase in electron-hole pair density (Δn) causes the conductivity of the material to increase by an amount $\Delta\sigma$. If now the source of energy is removed, the conductivity of the material will be observed to decrease exponentially with time, until its value reduces to what it was before the irradiation. This process is similar to the discharge of a capacitor through a resistance.

For negative charge carriers, the electron density n expressed as a function of time is given as

$$n(t) = n_0 + \Delta n \exp(-t/\tau)$$

where τ is the time constant of the rate of decay which is the same as the lifetime. The quantity n_0 is the carrier density prior to excitation by the external energy. The time rate of change

$$\begin{aligned} \frac{dn}{dt} &= (-\Delta n/\tau) \exp(-t/\tau) = (-\Delta n/\tau)(n - n_0)/\Delta n \\ &= (n_0/\tau) - (n/\tau) \end{aligned} \quad \dots\dots\dots (4.11)$$

The term n_0/τ represents the rate of thermal generation of the charge carriers, and the ratio n/τ represents the rate of recombination. Thus, at equilibrium, $n_0/\tau = n/\tau$ and $dn/dt = 0$.

The lifetime of a carrier is a reflection of crystal imperfection and hence is a measure of the purity of the crystal. Values of τ range typically from 10 ns to 10 μ s. We note that in transistors, large values of τ result in high transistor current gain, while high transistor switching speed requires that τ must be low valued.

Associated with lifetime is diffusion length (L), defined as the average net displacement of a carrier in the crystal before it is trapped or neutralized by a recombination centre. Thus, the average carrier will travel one diffusion length in time τ seconds. Typical values of L range from 10^{-6} m to 10^{-3} m.

4.3 Magnetic Effects On Current Carriers (The Hall Effect)

The effect of a magnetic field on current carriers is illustrated by the Hall effect. We recall that if a current carrying conductor (length l) is placed in a magnetic field (of flux density B), the conductor experiences a Lorentz force $F = Bil$, where i is current. Similarly, if an electric field applied to a bulk semiconductor produces an average drift velocity (u) in the x -direction (say), and a magnetic field (B) is applied in the y -direction (say), then the charge carriers in the semiconductor slab will experience a Lorentz force deflecting them sideways (in the z -direction) as they travel through the semiconductor. Thus, carriers of the same type will crowd on the same side of the slab.

Consider a uniformly doped n -type Silicon slab with dimensions a , b , and L , as shown in figure 4.3. Let an emf E bias the slab as shown. The majority current carriers (electrons) drift in the z -direction with a velocity determined by the electric field $\epsilon_z (= E/L)$. Thus,

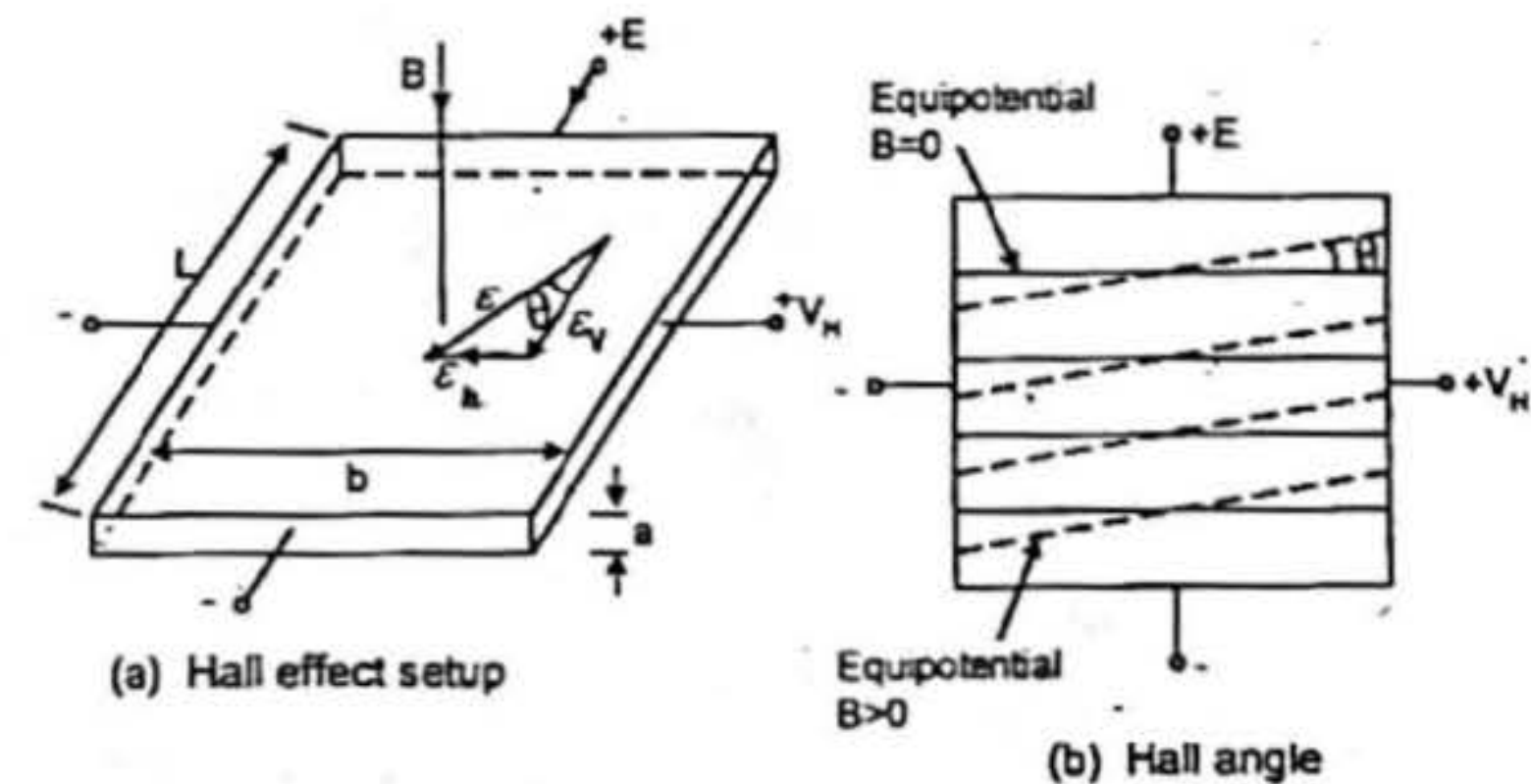


Fig. 4.3 The Hall effect

$$u = \mu_n E_v = \mu_n E/L \quad \dots\dots\dots (4.12)$$

If a transverse magnetic field of flux density B is applied in the y -direction as indicated, the force exerted on an electron due to this field is

$$F_B = qBu = qB\mu_n E/L \quad \dots\dots\dots (4.13)$$

This force tends to deflect the electrons sideways in the minus x -direction. Therefore electrons accumulate on the left side of the slab, thus developing a transverse voltage V_H called the Hall voltage. The presence of V_H creates a transverse electric field $\epsilon_H (= V_H/b)$. The force $F_H (= q\epsilon_H)$, developed by the electric field ϵ_H , tends to oppose the effect of F_B . When equilibrium is attained, $F_H = F_B$, and hence,

$$qV_H/b = qB\mu_n E/L$$

$$\text{Or } \epsilon_H / E_v = \mu_n B \quad \dots\dots\dots (4.14a)$$

From which,

$$\mu_n = (L/bB)(V_H/E) \quad \dots\dots\dots (4.14b)$$

If I is the current in the slab, due to the voltage E , then $E/I = L/\sigma A$, where $\sigma (=qn\mu_n)$ is the conductivity of the slab, n is electron density, and $A (=ab)$ is cross-sectional area of the slab. Thus, $E = IL/(nq\mu_n ab)$

Substituting into equation 4.14b, we have that

$$1/nq = (V_H/b)(ab/I)(1/B) = R_H \quad \dots\dots\dots (4.15a)$$

where R_H is called the Hall coefficient.

If we set the current density (I/ab) and the magnetic flux density (B) to equal unity, then equation 4.15a becomes

$$1/nq = V_H/b = \epsilon_H \quad \dots\dots\dots (4.15b)$$

The quantity ϵ_H can be measured (although it requires sensitive instrument with high input impedance) and hence the electron density of the material can be found.

A parameter called Hall angle (θ) is defined as the angle through which the equipotential surfaces are rotated when magnetic field (B) is applied (figure 4.3b). We note that the total electric field within the semiconductor material is the vector sum of ϵ_v and ϵ_H . The angle θ is the angle the resultant field ϵ makes with ϵ_v . Thus,

$$\tan \theta = \epsilon_H / \epsilon_v = (V_H/b)(L/E) \quad \dots\dots\dots (4.16a)$$

Hence, from 4.14a,

$$\tan \theta = \mu_n B \quad \dots\dots\dots (4.16b)$$

Since ϵ_H is small compared with ϵ_v , the angle θ is small and hence $\tan \theta = \theta$.

Therefore,

$$\theta = \mu_n B \quad \dots\dots\dots (4.17)$$

Since the Hall angle can be measured, mobility μ_n can be computed.

From the above, it is established that the Hall effect transducer can be used to measure the density (n or p) and mobility (μ_n or μ_p) of charge carriers. The predominant majority current carriers will depend on whether the semiconductor is n -type or p -type. The type of current carrier is identified by the sign of V_H (negative for electrons and positive for holes). We note also from equation 4.15a, that $V_H (= IB/qna)$ is proportional to the product of current (I) and magnetic flux density (B). This property is useful as the basis for using the Hall probe (or Hall plate) as a multiplying device in analogue computation. The main application of the Hall effect transducer is as a gaussmeter, for the measurement of magnetic flux.



Example E.4**E.4.1**

A bar of n-type Silicon, 3mm long with rectangular cross-section ($50 \mu\text{m} \times 100 \mu\text{m}$) has donor concentration of $5 \times 10^{14} \text{ cm}^{-3}$ at 300°K . This impurity concentration is 1-in- 10^8 atoms of Silicon. An emf drives a steady current of $1 \mu\text{A}$ through the bar. If the intrinsic density n_i is 1.45×10^{10} , determine

- the atomic density for silicon,
- the electron and hole densities,
- the conductivity and resistance of the bar, given that $\mu_n = 0.15$,
- the source voltage.

Solution

- Atomic density of Silicon = $10^8 \times 5 \times 10^{14} = 5 \times 10^{22} \text{ atoms cm}^{-3}$
- For $N_D \gg n_i$, $n_o = N_D = 5 \times 10^{14} \text{ cm}^{-3}$, and $p_o = n_i^2/n_o = 4.2 \times 10^5 \text{ cm}^{-3}$.
- Since $n_o \gg p_o$, conductivity can be calculated on the bases of n_o only.

Given that $\mu_n = 0.15$, then, conductivity

$$\sigma = n_o q \mu_n = 5 \times 10^{14} \times 1.6 \times 10^{-19} \times 0.15 = 1.2 \times 10^{-5}$$

- Source voltage,

$$E = IR = I\rho l/A = I/\sigma A = (1 \times 10^{-6} \times 3 \times 10^{-3}) / (1.2 \times 10^{-5} \times 50 \times 100 \times 10^{-12}) \\ = 0.5 \times 10^5 \text{ V}$$

The slab resistance is given as

$$R = (3 \times 10^{-3}) / (1.2 \times 10^{-5} \times 50 \times 100 \times 10^{-12}) = 0.5 \times 10^{11} \Omega$$

E.4.2

From the free electron theory, the electrons that effectively contribute to electrical conductivity are those with energies near the Fermi energy E_f . If u_f is the average velocity of the electrons, then $\frac{1}{2}mu^2 = E_f$ and the mean-free-path between collisions (L) may be written as $L = u_f \tau$, where τ is the mean-free-time between collisions. Given that conductivity (σ) for copper is $6 \times 10^7 \text{ mho/m}$, electron density $n = 8 \times 10^{28} \text{ m}^{-3}$, Fermi energy $E_f = 7 \text{ eV}$, and $m = 9.1 \times 10^{-31} \text{ kg}$, calculate the mean-free-path for electrons in copper.

$$(\text{Answer} = 4.18 \times 10^{-8} \text{ m})$$

CHAPTER FIVE

ELECTRON EMISSION FROM SURFACE OF MATERIALS

	Page
5.1 Introduction (Image Force)	49
Work-Function	52
5.2 Thermionic Emission	53
Emission Current	54
5.3 Field Emission	55
Motion of an Electron in Electric Field	55
Field Effect on Emission Current	57
5.4 Electron Emission by Photoelectric Effect	61
Examples	63

CHAPTER FIVE

ELECTRON EMISSION FROM SURFACES OF MATERIALS

5.1 Introduction

Electron emission is the escape of electrons from the surface of a material (metal or semiconductor). Electrons are normally held in the confine of a solid by the energy barrier existing at the surface of the solid. For an electron to escape and hence become free of the solid, it must acquire a minimum extra energy.

An important group of devices used for display and other purposes and which rely on electron emission for their operation, include the cathode-ray tube and the electron microscope. These devices operate in very low-pressure vacuum enclosure so that the motion of the electron is not interrupted by collision with air particles. In its simplest form, the device consists of a solid plate (called the cathode) from which electrons are emitted, and a second plate (called the anode), which collects the emitted electrons. The anode and the cathode plates are known as electrodes.

Image Force

An electron inside a solid crystal moves about, under thermal agitation, with complete freedom since it is in a region of electrical neutrality. The forces exerted upon it are balanced due to the symmetrically arranged neighbouring atoms.

Its motion is restricted near the surface by a force of attraction directed towards the interior, normal to the surface. Near the surface of the material, the electron may have a velocity component, which will take it away from the surface. However, as soon as it leaves the crystal, the positive charge left behind exerts a force, which acts to prevent its escape. If we assume that the surface of the material is plane and continuous, the force acting on the electron may be computed from ordinary electrostatics.

In figure 5.1a, the electric field lines from an electron (charge $-q$), situated at point x outside the surface of the material, are normal to the surface. These field lines (dashed lines in the figure) would seem to originate from a positive point charge ($+q$) appearing as an image charge at a distance $-x$ inside the crystal. The force of attraction, between the charges, is given by Coulomb's law as,

$$F_1(x) = -q^2/4\pi\epsilon_0(2x)^2 = -q^2/(16\pi\epsilon_0x^2) \quad \dots\dots\dots (5.1)$$

The quantity $\epsilon_0 (= 8.85 \times 10^{-12} \text{ F/m})$ is the dielectric constant of free space. A plot of the F_1/x curve, figure 5.1b (dotted line), suggests that when $x = 0$ (i.e. electron just on the surface), F_1 is infinite. We recall however (Section 1.2) that the force on an electron inside a crystal is zero. Therefore, the 'image' force theory does not apply at near the surface of the material. The practical F/x curve, figure 5.1b (full line), differs from the image curve at near the surface of the material. The force increases from zero inside the crystal and reaches a maximum just outside the surface when all the surface atoms contribute to it. The image value is seen to be reached at about the distance $x = 1 \text{ nm}$ from the surface.

The potential energy (E), associated with this force, constitutes the potential barrier preventing the emission of the electron from the surface of the material. It is given as

$$E = \int F(x)dx = \int (-q^2/16\pi\epsilon_0x^2)dx = -q^2/16\pi\epsilon_0x \quad \dots\dots\dots (5.2)$$

Equation 5.2 is similar to equation 1.8, noting that the distance of the electron from its 'image' is $2x$. Note that equation 5.2 is in error near the surface..

Computed on the assumption that $\dot{E} = 0$ at $x = \infty$, the $E(x)$ graph is sketched in figure 5.2a. By shifting the x -axis down to the edge of the conduction band, the potential barrier curve can be redrawn as in figure 5.2b.

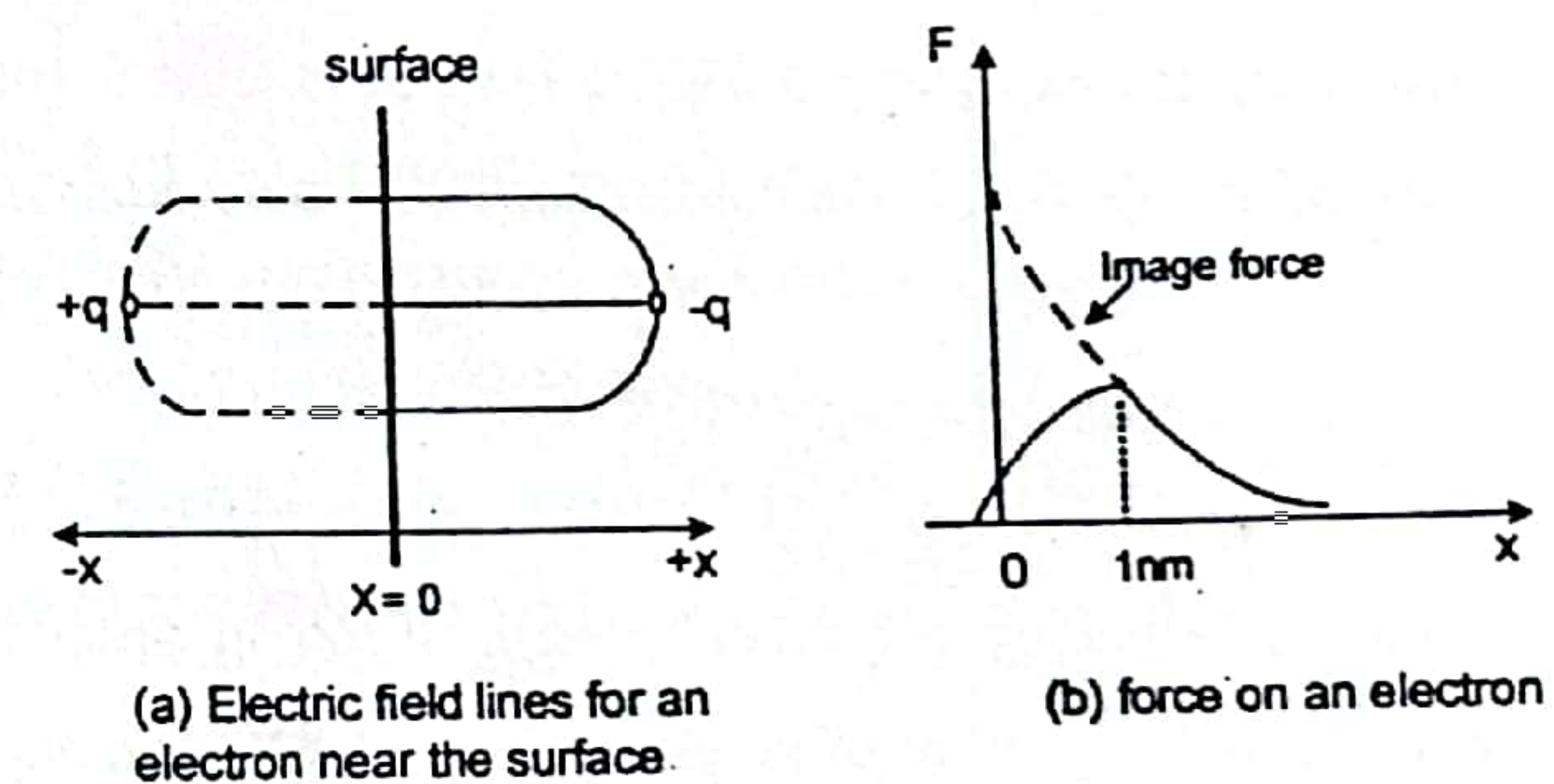


Fig. 5.1 The concept of Image force

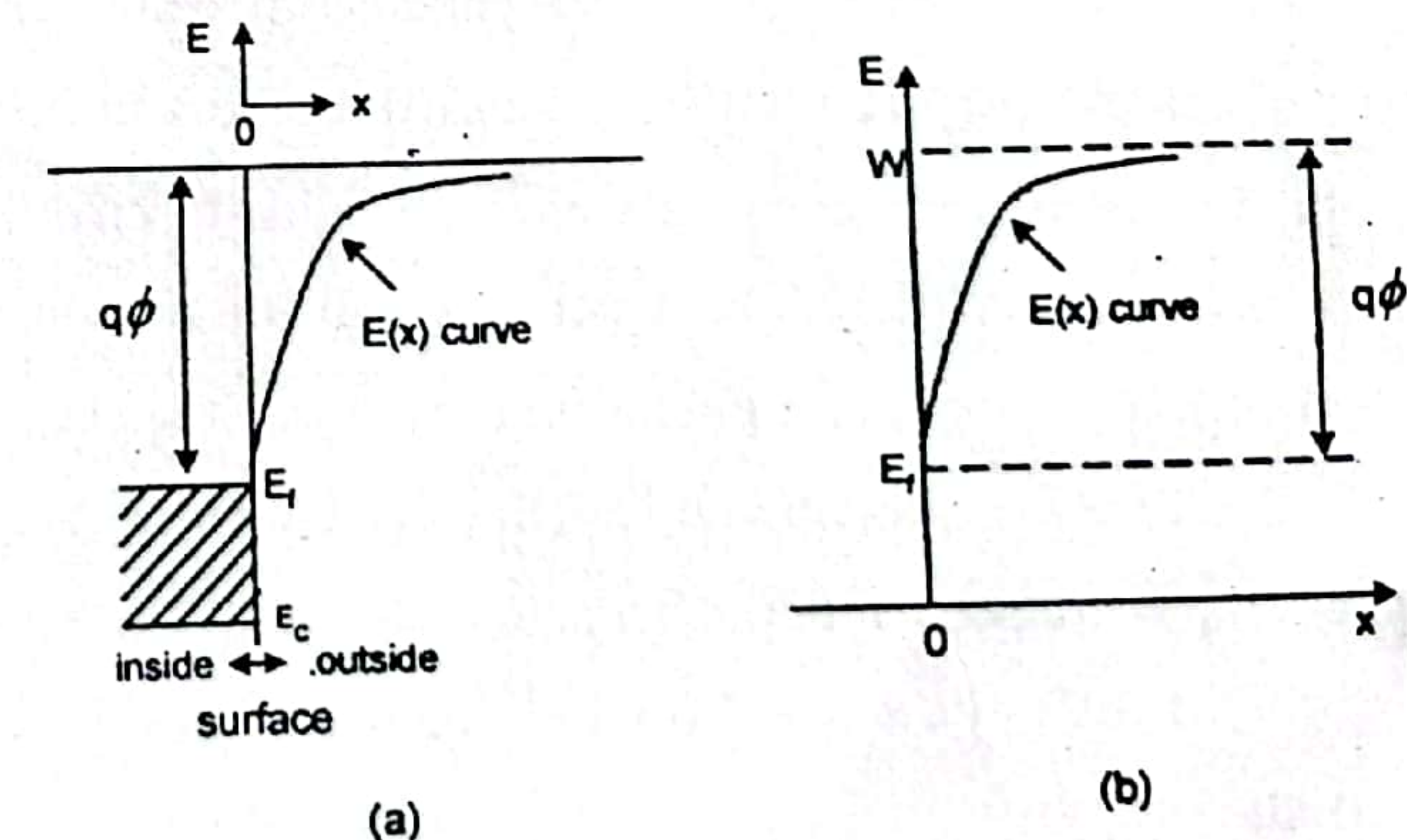


Fig. 5.2 Potential energy of electron as a function of its distance x from the surface of a material

The curve approaches a constant value $W (= E_f + q\phi)$ asymptotically, where W is the height of the potential well that an electron must climb before it can escape. In practice, the energy W is reached at a distance of $x = 10$ nm from the surface of the material. For an electron to escape from the surface of the material, its total energy therefore must be at least W (otherwise it could travel beyond the surface, but must fall back into the material).

Work function

The energy acquired by an electron in order to free itself from the material is in the form of kinetic energy. Its kinetic energy must therefore be at least W in order for the electron to escape from the surface. The energy E_f (in figure 5.2) is the Fermi energy. It represents the maximum energy an electron can possess at the temperature of zero degree Absolute. The difference energy $q\phi = (W - E_f)$ is called the work function of the material. It represents the minimum extra energy needed to take the electron from the conduction band out of the material at zero Absolute temperature. The work function ($q\phi$) is expressed in the unit of joules. It can be expressed in electron-volt (eV). One electron volt is defined as the energy acquired by an electron when it is accelerated through a potential of one volt. Thus,

$$1 \text{ (eV)} = q \times 1 = 1.6 \times 10^{-19} \text{ joules}$$

And,

$$\phi \text{ (eV)} = q\phi = \phi \times 1.6 \times 10^{-19} \text{ joules}$$

The work function of some commonly used emitter materials are given in Table 5.1.

The extra energy needed for an electron to escape from the surface of a material can be acquired from the following effects; **thermal effect** (thermionic emission), **field effect** (field emission), **photoelectric effect** (photoelectric emission), and **secondary emission** (in which other electrons are released when the surface of a solid is bombarded with a beam of high energy, high velocity electrons called primaries).

5.2 Thermionic Emission

Thermionic emission is the emission of electrons by a hot metal or semiconductor. The cathode materials are heated to temperatures between 1000°K and 2400°K . As indicated in the last section, the energy barrier W determines escape condition of the electron from the surface of the cathode. Since the energy acquired by an electron in order to escape from the surface is in the form of kinetic energy, (i.e. $\frac{1}{2}mu^2$), the escape condition is given as

$$\frac{1}{2}mu^2 \geq W, \text{ or } u = (2W/m)^{1/2} \quad \dots\dots\dots (5.3)$$

where u is the velocity component of the electron in a direction normal (i.e. perpendicular) to the surface of the cathode material, and m is the mass of an electron. In terms of momentum (p), the escape condition becomes

$$p_x (= mu) \geq (2mW)^{1/2} \geq [2m(E_f + \phi)]^{1/2} \quad \dots\dots\dots (5.4a)$$

where p_x is the component of p perpendicular to the surface and positive. If we denote the critical or minimum momentum by p_{xc} , then

$$p_{xc} = [2m(E_f + \phi)]^{1/2} \quad \dots\dots\dots (5.4b)$$

Thus, for an electron residing at the bottom of the conduction band to escape, a necessary but not sufficient condition is that its energy must be greater than $(E_f + \phi)$. This condition is not sufficient since the momentum of the electron must

not be directed normal to the surface. If the momentum is not normal to the surface (figure 5.3), the electron energy will need to be much greater than $(E_f + \phi)$, otherwise it will be turned back from the surface. The necessary and sufficient condition is therefore given by equation 5.4. In figure 5.3, electron B (with $p_B > p_{xc}$) is emitted, while electron A (with $p_A < p_{xc}$) is turned back.

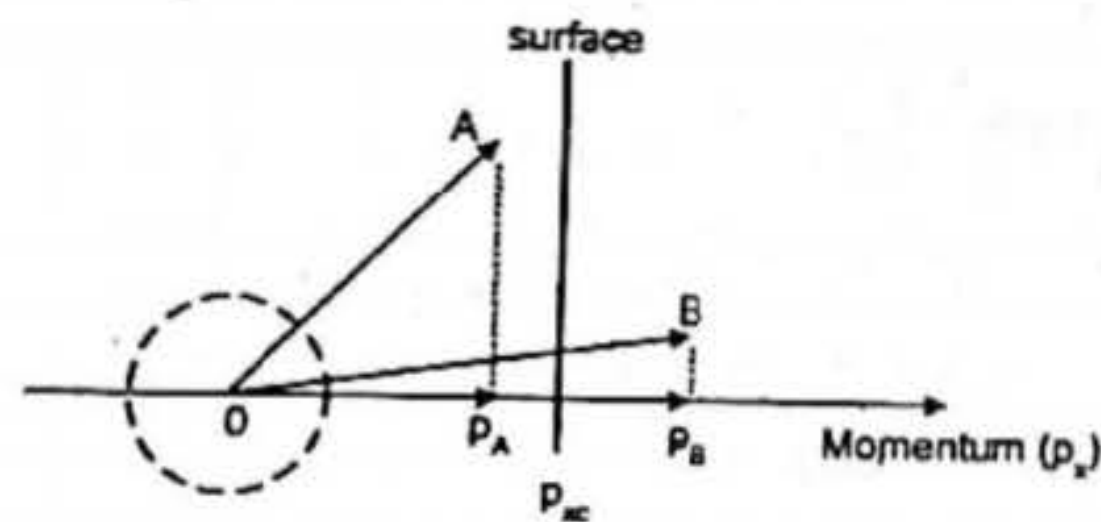


Fig. 5.3 Momentum-space plot (condition for electron emission)

Emission Current

The probability of an electron escaping from the surface of a cathode material is the probability of an electron having energy W . This is given by Fermi function as

$$f(W) = 1/[1 + \exp(W - E_f)/kT]$$

This can be approximated to

$$f(W) = 1/\exp[(W - E_f)/kT] = \exp(-q\phi/kT) \quad \dots\dots\dots (5.5)$$

where ϕ (eV) = $(W - E_f) = q\phi$ joules. Thus, the probability of an electron escaping from the surface of a cathode material increases as the temperature of the material is raised.

The number of the emitted electrons, and hence the current density J_s , depends on $f(W)$. J_s is the saturated or temperature limited current density. The relationship between J_s and $f(W)$ is given by Richardson-Dushman equation

$$J_s = AT^2 \exp(-q\phi/kT) = AT^2 \exp(-b/T) \quad \dots\dots\dots (5.6)$$

where $b = q\phi/k$ is a constant (which depends on the workfunction), A is a universal constant, and k is Boltzmann constant ($= 1.38 \times 10^{-23}$ J/K). The exponential term has much greater effect on J_s than AT^2 term.

The emission current at a temperature T , as indicated by the current density equation 5.6, cannot be sustained for long. This is because, as electrons leave the surface of the metal, the metal becomes positively charged.

① This would have the effect of making it difficult for further electrons to leave the surface. Secondly, the emitted electrons form electron cloud ^{or space charge} in the immediate vicinity of the emitting cathode, which discourages further electrons emission.

These two effects are removed by connecting a dc voltage source (emf) between the anode and the emitting cathode. If the anode potential is positive with respect to the cathode (figure 5.4), the electric field produced helps to sweep the electrons to the anode, thus eliminating the electron cloud. The presence of the emf also means that for every electron that is emitted from the cathode, another is supplied by the dc source to replace it. The metal does not now get positively charged but maintains its charge neutrality. As noted earlier, the anode-cathode system is often enclosed in an evacuated glass envelope. This is done to prevent scattering of the electrons by air molecules.

5.3 Field Emission

Motion of an Electron in Electric Field

Let us consider what happens to an electron after it has escaped from the surface of the cathode material and is under the influence of an electric field. The study of the motion of an electron in an electric field is important in understanding how electric field strength influences emission current.

Take two parallel electrodes (the cathode and anode plates in figure 5.4) separated by a distance d , with a potential difference V which maintains the anode positive with respect to the cathode. The electric field between the electrodes is given by

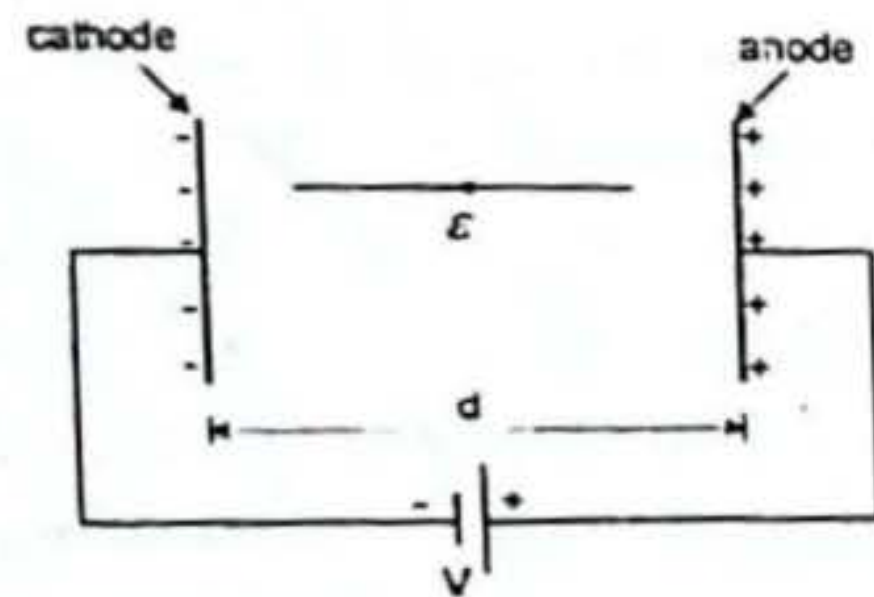


Fig. 5.4. Electron in uniform electric field

$$E = dV/dx = V/d \text{ (in the x-direction)} \quad \dots\dots\dots (5.7)$$

If the free electron finds itself in this field, it experiences a force $F = qE$ which accelerates it towards the positive anode. In the process, the electron gains velocity $u (= dx/dt)$. From the expression force $(F) = (\text{mass} \times \text{acceleration}) = mdu/dt$, we have that

$$\begin{aligned} \text{Energy (E)} &= \left(\int F dx \right) = \int (mdu/dt) dx \\ &= \int m u du = \frac{1}{2} m u^2 \quad \dots\dots\dots (5.8a) \end{aligned}$$

The electron has thus gained kinetic energy whilst in the electric field

We note also that the work done by the electric field force (qE), in moving the electron away from the cathode, is given from equation 5.7 as

$$E_c = \int_0^x F dx = \int_0^x qE dx = \int qdV$$

Hence,

$$E_c = qEx = qV \quad \dots\dots\dots (5.8b)$$

For E constant, the E/x curve is a straight line. Equating 5.7 and 5.8b, we have

$$\frac{1}{2} m u^2 = qV. \text{ From which,}$$

$$u = [2qV/m]^{\frac{1}{2}} \quad \dots\dots\dots (5.9)$$

This is the expression for the terminal velocity of the electron as it arrives at the anode.

Field Effect on Emission Current

Comparing the forces acting on the electron emitted into the electric field, we note that the Coulomb (or image) force acts to decelerate the electron,

while the electric field force accelerates the electron. Thus while the Coulomb force produces a potential energy barrier, the electron gains kinetic energy in the electric field. The 'straight line' energy curve (b), due to the electric field force (figure 5.5), is plotted superimposed on the image force contribution to the energy barrier (curve a). Considering the resultant energy curve (c), we see from the definition of work-function, that the resulting work-function ϕ_c is less than ϕ_a .

Considering the total potential energy curve (c); the total electric field force is zero at the maximum height of the curve (since at that position the gradient of the potential energy dE/dx is zero). The position x_0 of this maximum is found by equating the total force to zero (or by setting $dE/dx = 0$).

Thus, $-q^2/16\pi\epsilon_0 x_0^2 + q\epsilon = 0$. From which,

$$x_0 = [q/16\pi\epsilon_0\epsilon]^{\frac{1}{2}} \quad \text{.....} \quad (5.10)$$

Any electron, which can reach this maximum point of the total potential energy barrier, has sufficient energy to escape from the surface.

With electric field applied, the maximum energy barrier is given as

$E = (E_i - E_e)$ at $x = x_0$. Substituting equations 5.2 and 5.8 respectively yields

$$E = [-q^2/(16\pi\epsilon_0 x_0) - q\epsilon x_0]. \quad \text{.....} \quad (5.11)$$

Substituting for x_0 from equation 5.10 yields the expression for the maximum energy barrier at x_0 as

$$E = -[q^3\epsilon/4\pi\epsilon_0]^{\frac{1}{2}} \quad \text{.....} \quad (5.12)$$

We note that in the absence of an electric field, the potential energy barrier at $x = \infty$ is zero (Figure 5.2a). When an electric field is applied, the reduction in the total potential energy barrier (and hence the decrease $\Delta\phi$ in the effective work-function) is numerically equal to the total potential energy at $x = x_0$. Hence, the reduction $\Delta\phi$ is given by equation 5.12 as,

$$q\Delta\phi = -2q^{\frac{3}{2}}\epsilon^{\frac{1}{2}}/(16\pi\epsilon_0)^{\frac{1}{2}} = -[q^3\epsilon/4\pi\epsilon_0]^{\frac{1}{2}} \quad \text{joules.....} \quad (5.13a)$$

Or

$$\Delta\phi = -[q\epsilon/4\pi\epsilon_0]^{\frac{1}{2}} \quad \text{eV.} \quad \text{.....} \quad (5.13b)$$

From the graphs in figure 5.5, the effective work-function $\phi_e = (\phi_a - \Delta\phi)$. The resulting emission current density becomes

$$J = AT^2 \exp(-q\phi_e/kT) = AT^2 \exp[(\phi_a - \Delta\phi)/kT] \quad \text{.....} \quad (5.14)$$

The effect of an electric field on the work function (ϕ) of the material, and hence on the current density (J), is known as **Schottky effect**. As the electric field strength (ϵ) increases, the slope of the energy curve (b) increases and ϕ_e decreases (figure 5.5b). For very strong electric fields (of the order of 10^{10} V/m) ϕ_e is very small and the energy barrier virtually reduces to zero. Consequently, electrons are emitted freely and quantum mechanical tunnelling can occur. This is called **field emission**, and it is practically independent of temperature. The emission properties of some cathode materials are tabulated in Table 5.1.

Electron Emission By Photoelectric Effect

Photoelectric emission is the emission of electrons from the surface of a material that is exposed to light (or generally to electromagnetic radiation). Light is accepted to be composed of discrete lumps or bundles of pure energy called photon. The photon is thought of as a packet of waves whose energy (E) and frequency (f) are related by the expression $E = hf$ (joules), where h is Plank's constant. It travels with the speed of light ($c = 3 \times 10^8$ m/s). The wavelength (λ) associated with the wave-packet is defined as the distance travelled in period time T at the speed of light. Hence, $\lambda = cT = c/f$. The expression for energy then becomes

$$E = hf = hc/\lambda \quad \dots\dots\dots (5.11)$$

This equation shows that photons, having different wavelengths, have different energy.

☞ Photoelectric effect relies on the principle that when a photon of light is absorbed in a semiconductor, an electron gains the photon energy. On the other hand, when an electron loses energy, it releases photons. This photon-electron interaction is essential in understanding photoelectric devices, which convert light energy into electrical energy or vice versa.

For an electron to be emitted from the surface of a material, it must absorb photon energy equal to or greater than the magnitude of the work-function ($q\phi$) of the material. Thus, at zero degree absolute, photoelectric emission will occur if, $E (=hf) \geq q\phi$, or $f \geq q\phi/h$.

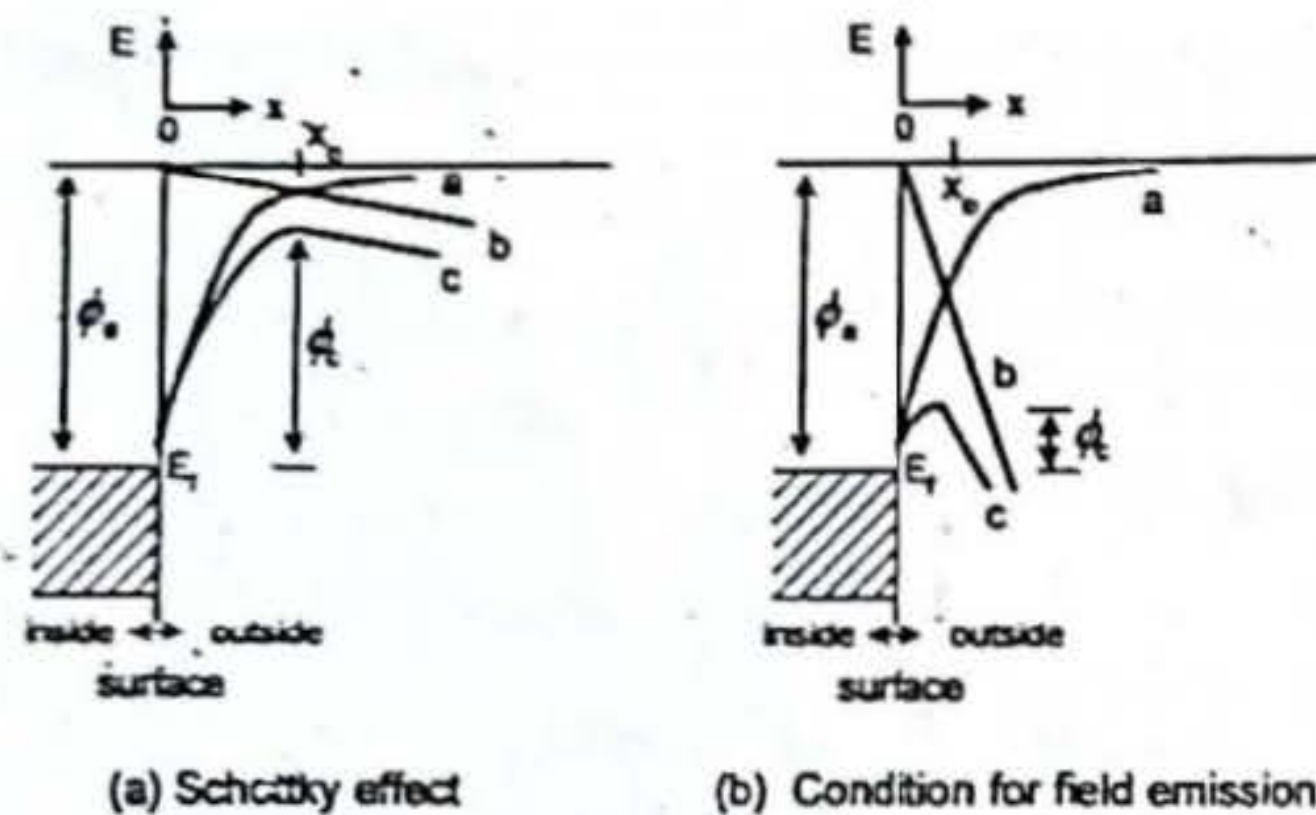


Fig. 5.5 Effect of electric field on workfunction

Table 5.1 Emission properties of some cathode materials

metal	workfunction (ϕ eV)	constant (A)	Operating temp. (°K)
caesium	1.81	16×10^4	150
tungsten	4.52	60×10^4	2400
Thoriodal- tungsten	2.6	3×10^4	1800
Barium strontium oxide	1.3	0.01×10^4	1000

Therefore, a certain minimum frequency (and hence maximum wavelength) exists for each material below which photoelectric emission cannot occur. This threshold frequency f_0 is given by

$$f_0 = q\phi/h = 2.42 \times 10^{14} \phi \quad \dots\dots\dots (5.12a)$$

Or,

$$\lambda_0 = c/f_0 = hc/q\phi \quad \dots\dots\dots (5.12b)$$

where c (the velocity of light) is in meters/second and ϕ is in electron-volt. The wavelength is often measured in Angstrom unit ($\text{\AA}$) instead of meters, where $1 \text{\AA} = 10^{-10} \text{ m}$. Thus, threshold wavelength $\lambda_0 = 1.24 \times 10^4 / \phi$ ($\text{\AA}$).

If the wavelength of the electromagnetic radiation is such that the photon energy ($hf = hc/\lambda$) exceeds the work-function ($q\phi$) of the surface, the surplus energy is transferred to the emitted electron as kinetic energy. Thus, we have the relationship

$$hf = (hc/\lambda) = q\phi + \frac{1}{2}mu^2 \quad \dots\dots\dots (5.13)$$

where u is the velocity of the electron as it leaves the surface. The kinetic energy that an electron possesses on emission depends on the frequency (and hence on the wavelength) of the incident radiation and not on the light intensity. However, the number of electrons emitted depends on the light intensity. The relationship between emission current density and light intensity (lumens) is found to be linear over wide range of light intensities.

Example E5.1

- Calculate the threshold frequency f_0 and wavelength λ_0 of a cathode material coated with cesium (having $\phi = 1.18 \text{ eV}$).
- Determine the escape velocity of an electron, if radiation of wavelength $5000 \text{ \AA}$ bombards the cathode material.

Solution

- From the energy relationship $hf_0 = q\phi$, where hf is in joules,

$$f_0 = q\phi/h = 2.41 \times 10^{14} \phi = 4.37 \times 10^{14} \text{ Hz}$$

$$\lambda_0 = c/f_0 = 6.88 \times 10^{-7} \text{ m}$$

- For $\lambda = 5000 \text{ \AA} = 5 \times 10^{-7} \text{ m}$, $f = c/\lambda = 3 \times 10^8 / 5 \times 10^{-7}$

$$\text{Energy absorbed by electron } hf = 3.98 \times 10^{-19} \text{ J}$$

$$\text{Excess energy converted to K.E} = 3.98 \times 10^{-19} - q\phi$$

$$= 3.98 \times 10^{-19} - 1.6 \times 10^{-19} \times 1.18 = 1.09 \times 10^{-19} \text{ J}$$

Therefore, $\frac{1}{2}mu^2 = 1.09 \times 10^{-19} \text{ J}$. Substituting for m ($= 9.1 \times 10^{-31} \text{ kg}$) gives,

$$u = 4.89 \times 10^5 \text{ m/s}$$

CHAPTER SIX

CHARACTERISTICS OF JUNCTIONS

	Page
6.1 Introduction	65
6.2 Junction Between Metals	66
Effect of Insulating Layer	68
6.3 Junction Between Metal and Semiconductor	71
Metal-(n-type)Semiconductor Junction	71
Band Bending in Semiconductor Material	73
Width of Ionisation at Semiconductor Junction	74
Metal- (n+-type)Semiconductor Junction	75
Metal-Oxide-Semiconductor Junction	75
6.4 The P-N Junction	79
The Tunnel Diode	84

CHAPTER SIX

CHARACTERISTICS OF JUNCTIONS

6.1 Introduction

A junction is the contact between two materials. Contacts of interest in our study of junctions are the contacts between two metals, or a metal and semiconductor, or between two semiconductor materials. The study of these contacts helps us to understand the characteristics of junction devices. In particular, we want to know whether the junction is Ohmic or Rectifying. A contact is Ohmic if electrons have equal freedom of motion in both directions, as in a conductor. In a rectifying contact, electrons can move freely in one direction only, but are restricted in the other (as in a rectifying diode). Both Ohmic and rectifying contacts find useful practical applications, e.g. in integrated circuits and multiple junction devices.

To help us understand this subject, let us review what we have studied in the previous chapters. Thus,

- Electrons can occupy only allowed energy states.
- An electron cannot remain in an excited state (i.e. higher energy level) when a stable lower energy level is vacant.

If we place 'n' electrons in a unit volume of crystal at zero degree absolute temperature, the electrons (in line with the Exclusion principle) will occupy allowed energy states starting from the lowest energy level to an energy state E_F called the Fermi energy.

If the number 'n' of electrons increases, the highest occupied energy state is higher. It is lower if 'n' decreases. Thus, we conclude that a redistribution of electrons or charges affects the Fermi energy. When two materials are brought into contact, and there is a redistribution of charges, the Fermi levels change. When equilibrium condition is reached, the Fermi levels of the two materials must line up at the same level.

The nature of the contact (i.e. whether ohmic or rectifying) will depend on the work-function of the materials, and on the charges present. The work-function of a material is generally defined as the energy level difference between the Fermi energy and the vacuum level. It represents the minimum extra energy needed to take an electron from the conduction band out of the material at zero Absolute temperature.

6.2 Junction Between Two Metals

In a metal, the valence and conduction bands overlap and hence the Fermi level lies in the conduction band. Consider metals A and B with different work functions ϕ_A and ϕ_B , where ϕ_B is greater than ϕ_A (figure 6.1). This means that the Fermi energy E_{FA} is higher than E_{FB} , and hence electrons in metal A have higher energies than the electrons in metal B. When the two metals are brought together into contact, the system is initially unstable. The higher energy electrons in metal A move across the junction to occupy the lower energy levels in metal B.

This movement or redistribution of charges has the effect of leaving metal A positively charged and metal B negatively charged. Consequently, an electric field is set up across the junction with a polarity that tends to discourage the flow of electrons from metal A to metal B. Since the Fermi levels lie within the conduction band and there is no energy gap, electrons can still wander across the junction in either direction. When the system attains equilibrium, the net flow of electrons across the junction is however zero.

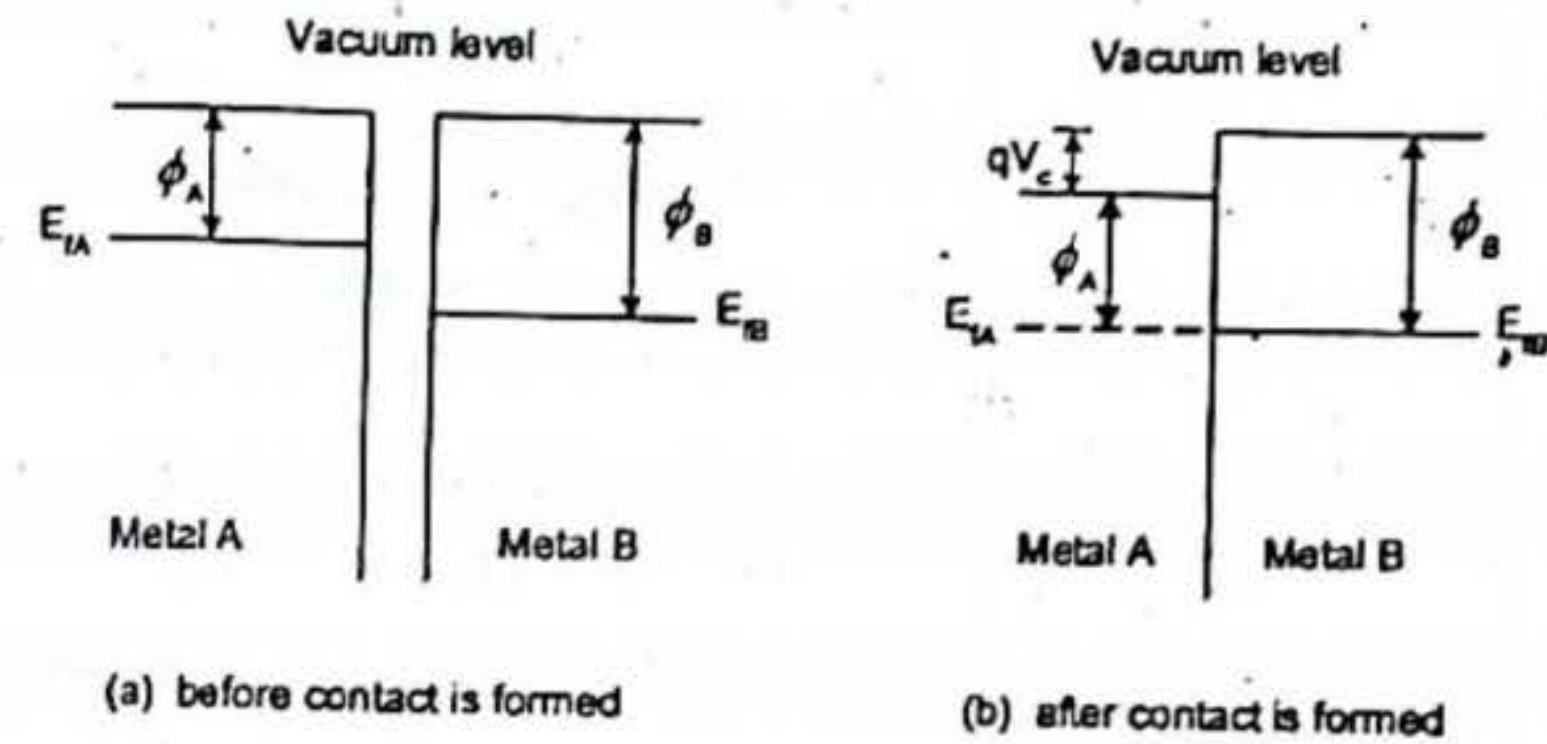


Fig. 6.1 Energy diagrams for junction between two metals

The energy diagrams before and after the junction is formed are illustrated in figure 6.1. When contact is formed and the system is in equilibrium (figure 6.1b), the Fermi levels line up. As a result of the charge transfer between the two metals, there will be a difference in their potentials. This potential difference is given as

$$V_c = (\phi_B - \phi_A)/q \quad \dots\dots\dots (6.1)$$

The quantity V_c is called the **contact potential** and it is the difference in the Fermi levels of the two materials.

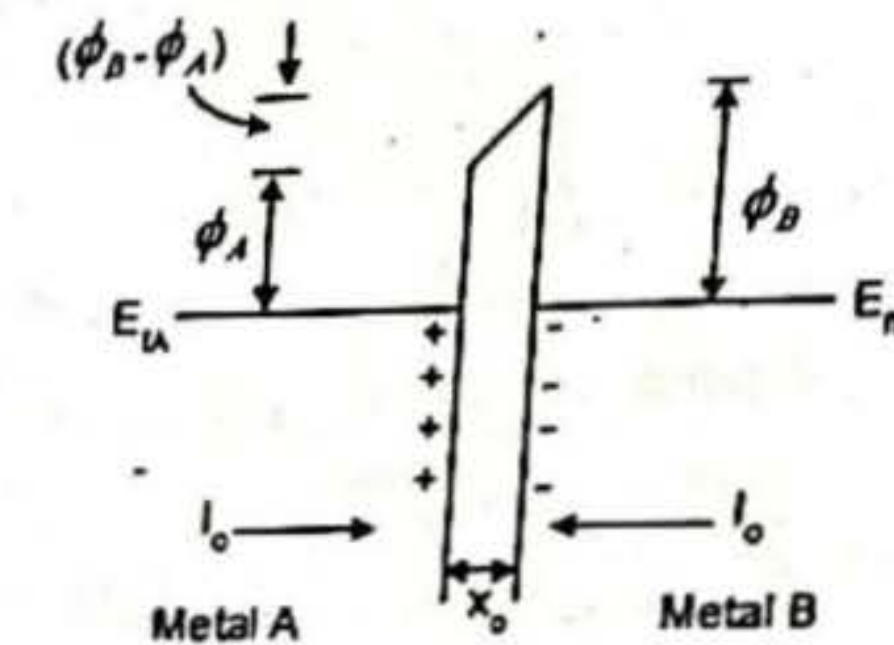
We note that in metal, the Fermi energy lies within the conduction band. Hence, there is no barrier potential to overcome in order to cross the junction from one side to the other. Thus, if an external voltage source is connected across the junction between the metals, a net flow of electrons results in one direction or the other depending on the polarity of the applied voltage (as obtains in a conductor). Thus the metal-to-metal contact is generally ohmic.

Effect Of Insulating Layer At Junction Between Two Metals

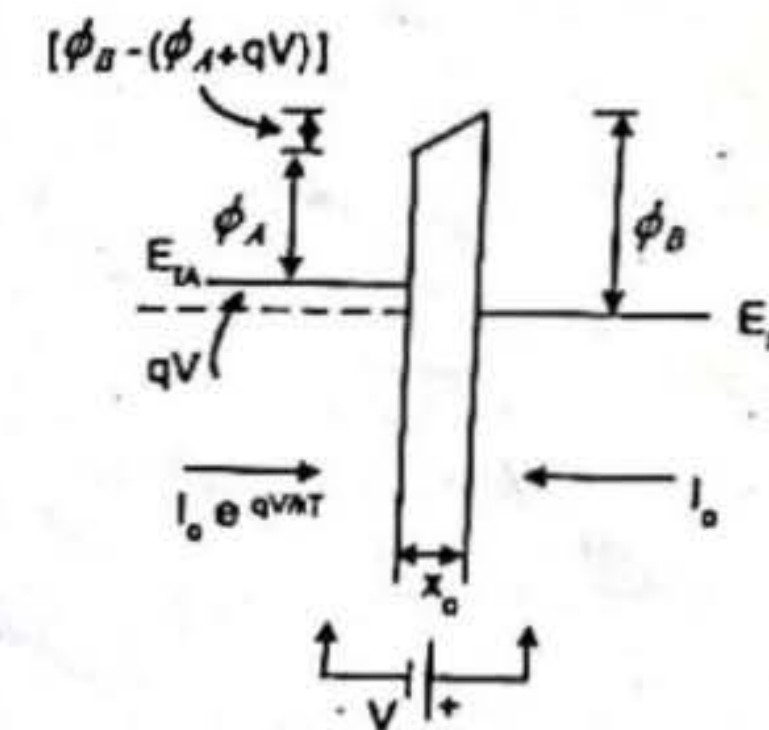
Now suppose a thin film of insulating layer, (i.e. a dielectric material), of width x_0 , separates the metals at the junction. The resulting energy diagrams are sketched in figure 6.2. With zero bias (figure 6.2a) movement of electrons across the junction is now restricted. To move from metal A to metal B, (and vice versa) an electron needs to acquire an extra energy equal to ϕ_0 , in order to achieve the conduction band energy of the dielectric. At equilibrium, the net flow of electrons across the junction is zero. Thus, the same current I_0 would flow from A to B as from B to A. If we assume that the extra energy is acquire thermionically, then by Richardson's equation

$$I_0 = AT^2 \exp(-\phi_B/kT) \quad \dots\dots\dots (6.2)$$

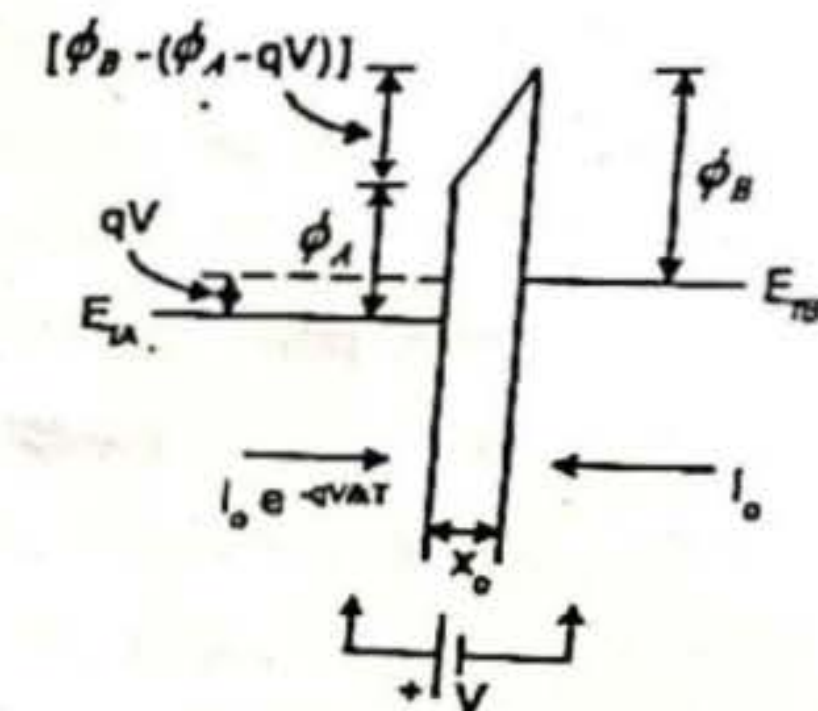
Now let a potential V be applied (figure 6.2b) such that metal A is made negative with respect to metal B. The effect of such a negative potential will be to raise the energy levels in metal A (relative to energy levels in metal B) by an amount qV (joules).



(a) zero external bias condition



(b) forward bias



(c) reverse bias

Fig. 6.2 Rectification at a metal-to-metal junction with thin insulating material

This in effect raises the Fermi level by qV thereby decreasing the height of the potential barrier for electrons in metal A to $(\phi_B - qV)$. However, electrons from metal B must still overcome the work function ϕ_B , in order to cross to metal A.

Thus, current I_0 continues to flow from metal B to metal A, while the electron current flow from A to B is increased to a value given by

$$I_{AB} = AT^2 \exp[-(\phi_B - qV)/kT] = AT^2 e^{-\phi_B/kT} e^{+qV/kT} \\ = I_0 e^{(qV/kT)} \quad \dots\dots\dots (6.3)$$

The total current flow due to applied voltage (V) then becomes

$$I_T = I_0 e^{(qV/kT)} - I_0 = I_0 [\exp(qV/kT) - 1] \quad \dots\dots\dots (6.4)$$

This is the forward current.

If the applied voltage V is reversed in polarity (as in figure 6.2c), the Fermi level in metal A drops by an energy qV joules. Thus, while electrons in metal B still see a work function ϕ_B , electrons in metal A see $(\phi_B + qV)$. Thus, the electron current from metal A to metal B becomes, $I_0 \exp(-qV/kT)$. The total reverse current flowing in this case is

$$I_r = I_0 [1 - \exp(-qV/kT)] \approx I_0 \quad (\text{approx.}) \quad \dots\dots\dots (6.5)$$

The metal-to-metal contact, where the junction contains a film of insulating material, is therefore a rectifying junction. The rectifying characteristic is sketched in figure 6.3.

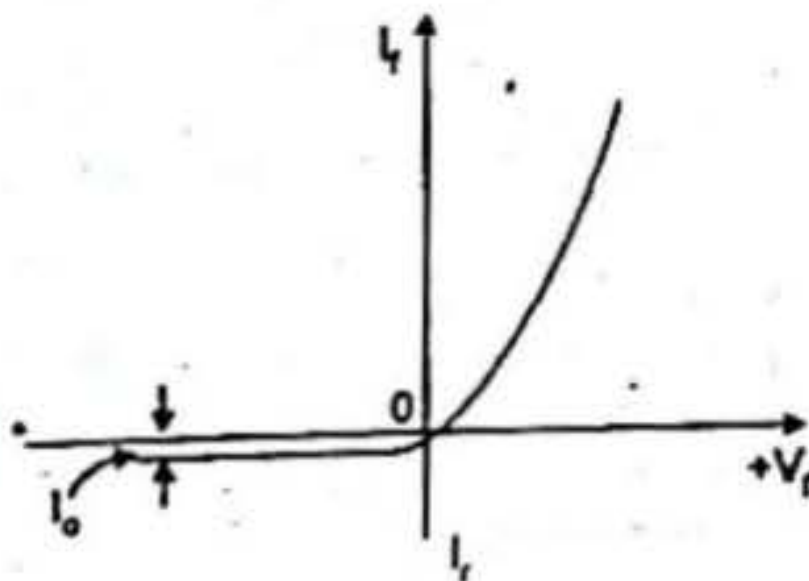


Fig. 6.3 Rectifying characteristic of metal-metal contact
(Due to film of insulating material at the junction)

6.3 Junction Between Metal and Semiconductor

The behaviour of a metal-semiconductor junction will depend on,

- (i) whether the semiconductor is n-type or p-type, and
- (ii) whether the work function of the metal is greater or less than that of the semiconductor.

Let ϕ_m and ϕ_s be the work function of the metal and semiconductor respectively. In general, if $\phi_m < \phi_s$, the contact is ohmic for n-type material and rectifying for p-type material. When $\phi_m > \phi_s$, the contact is rectifying for n-type material and ohmic for p-type material. Note that this prediction is not always true in practice.

Metal-(n-type) Semiconductor Junction

We will consider the arrangement where $\phi_m > \phi_s$ and the Fermi levels are E_{fm} and E_{fs} respectively, as illustrated in figure 6.4. Figure 6.4a shows the energy level diagram as the materials are brought into contact and prior to equilibrium being established. The Fermi level of the semiconductor is above the Fermi level of the metal by an amount of $(\phi_m - \phi_s)$. Hence, the higher energy electrons from the semiconductor climb over into the metal, leaving the donor atoms positively ionised. This has the effect of lowering the Fermi level in the semiconductor and raising the Fermi level in the metal. Consequently, when equilibrium has been established, the two Fermi levels line up as in figure 6.4b. The ionised donor atoms constitute distributed or space charges, and hence a depletion region (width x) exists near the junction in the semiconductor.

Since the depletion region has no charge carriers, it functions as an insulating region. Thus, an insulating material separates the metal and the conducting region of the semiconductor. This gives rise to an insulating potential barrier given by $h = qV_c = (\phi_m - \phi_s)$, where V_c is the contact potential (volts) for the metal-semiconductor contact. The junction can therefore act as a rectifying contact, in the same manner as for the metal-insulator-metal junction of figure 6.2.

By a similar analysis, the forward and reverse currents are given respectively as,

$$I_f = I_0 [\exp(qV/kT) - 1] \quad \text{..... (6.6a)}$$

And

$$I_r = I_0 [1 - \exp(-qV/kT)] \quad \text{..... (6.6b)}$$

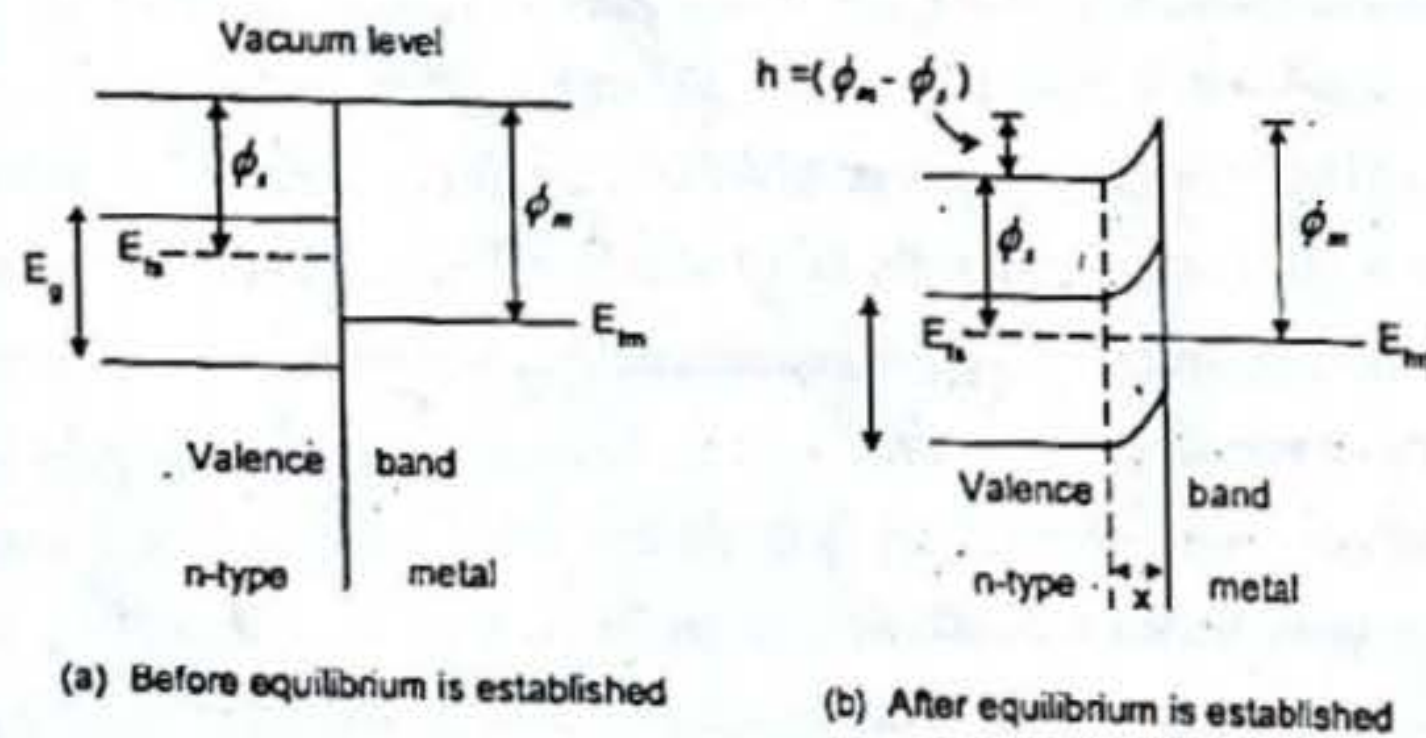


Fig. 6.4 Energy level diagram for metal-(n-type) semiconductor junction

Band Bending in Semiconductor Material

In figure 6.4, we observe that the bands bend at near the junction on the semiconductor side. This is explained as follows: The Fermi level in an intrinsic semiconductor is located in the middle of the energy gap. If the material acquires more electrons, the Fermi level moves up nearer to the bottom of the conduction band. On the other hand, if the electron density reduces or hole density is increased, the Fermi level moves down nearer to the top of the valence band. The position of the Fermi level in a material therefore depends on the density of charge carriers in the material. Thus, if the electron density (n) in an n-type bar varies with distance as in figure 6.5, its energy band diagram takes the form of figure 6.5c.

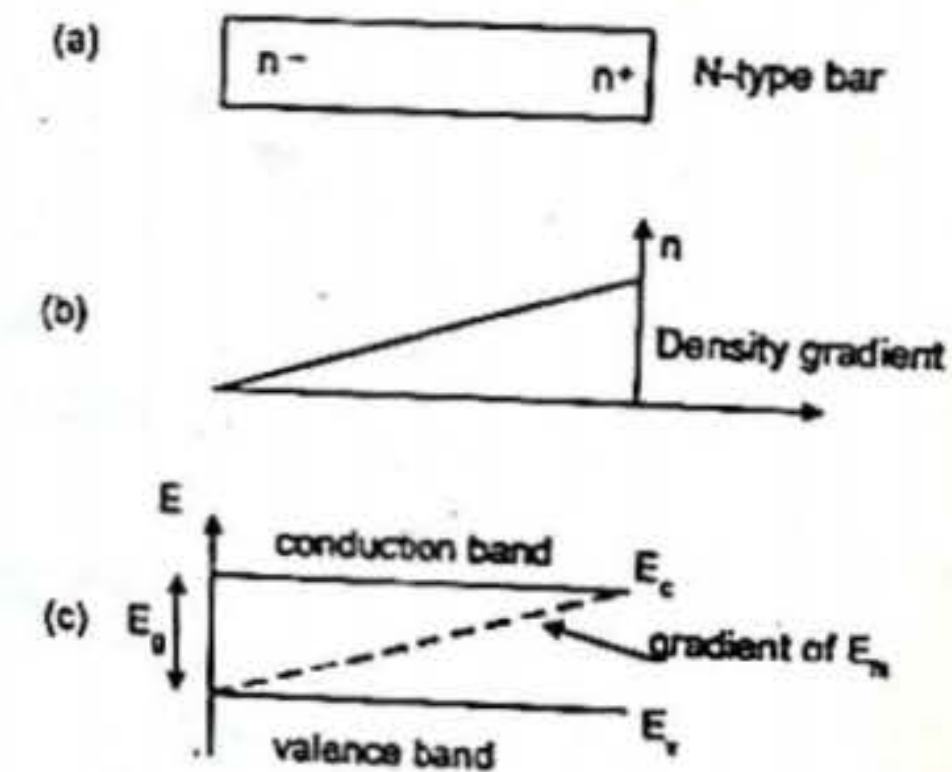


Fig. 6.5 Variation of Fermi energy with electron density (N-type material)

The variation in electron density may result from a redistribution of electrons (or charge carriers) in the semiconductor material, as happens when a junction is formed. In the metal-(n-type) semiconductor junction in equilibrium (figure 6.4b), we note that the electron density in the semiconductor region reduces as we approach the junction. Consequently, the distance of the Fermi level from the bottom of the conduction band increases (from its normal position in an n-type material, as the semiconductor becomes less n-type. The energy diagram therefore bends upwards in the depletion region near the junction, as in figure 6.4b. Band bending therefore occurs due to the non-uniform charge density along the length of the semiconductor material.

Width of Ionisation at Semiconductor Junction

Poisson's equation relating the density (ρ) of charge in a material to the potential (V) in its neighbourhood is given as

$$d^2V/dx^2 = \rho/\epsilon \quad \dots\dots\dots (6.7)$$

where ϵ is absolute permittivity. Integrating twice gives the expression

$$V = \rho x^2/2\epsilon. \text{ From which}$$

$$x = \sqrt{[2\epsilon V/\rho]} \quad \dots\dots\dots (6.8a)$$

Equation 6.8a allows us to calculate the distance x_0 (figure 6.4b) over which donor atoms are ionised before the Fermi levels equalize. For an n-type semiconductor material, the charge density (ρ) in the depletion region equals qN_D , where N_D is the donor atom density. The associated potential (V) is the contact potential (V_c). Equation 6.8a can therefore be written in the form

$$x_0 = [2\epsilon V_c/qN_D]^{1/2} \quad \dots\dots\dots (6.8b)$$

The distance x_0 is seen to be inversely proportional to N_D , and hence decreases with increase in doping concentration.

Equation 6.8b can be written in the form

$$x_0 = [(2\epsilon/qN_D)(V_c/x_0)] = (2\epsilon/qN_D)E \quad \dots\dots\dots (6.9)$$

where $E = V_c/x_0$ is the electric field due to contact potential V_c .

Metal - (n⁺-type) Semiconductor Junction

Suppose the n-type material in figure 6.4 is very heavily doped. Its Fermi level would move into the conduction band. The semiconductor material is then said to have become **degenerate** (or near metallic). Given this condition, the characteristic of the metal - (n⁺-type) semiconductor contact would be similar to the contact between two metals (Section 6.2), and is therefore **ohmic** even though $\phi_m > \phi_s$. Such a contact finds practical applications in making connections to integrated circuit, where aluminium (with $\phi_m = 4.24$ eV) and n⁺-type semiconductor (with $\phi_s = 3.1$ eV) form ohmic contact.

Metal-Oxide-Semiconductor Junction

In this section, we will consider surface effects, which arise in the metal-oxide-semiconductor (MOS) structure, which is essentially a capacitor. The simplified band diagrams, for the ideal MOS structure in equilibrium, are shown in figure 6.6. In the diagram, ϕ_m is the modified work function for metal-oxide interface (measured from E_{an} to the conduction band of the oxide).

Similarly, the energy ϕ_s is the modified work function at the semiconductor-oxide interface. In this idealized case, we assume that $\phi_m = \phi_s$ and that the semiconductor is a p-type material. The energy difference $\phi_f = (E_i - E_b)$ is a measure of the position of the Fermi level E_b below the intrinsic level E_i for the semiconductor, and indicates how strongly p-type the semiconductor is.

The effects of external bias voltage are illustrated in figure 6.6(b&c). In figure 6.6b, the metal voltage is made negative with respect to the semiconductor and hence negative charges are deposited on the metal. Consequently, positive charges (holes) accumulate on the semiconductor-oxide interface. This raises the electron energies in the metal relative to the semiconductor and E_{fm} rises above the equilibrium level. Conversely E_b now lies below the equilibrium level by qV , where V is the applied voltage.

Since ϕ_m and ϕ_s do not change with applied voltage, moving E_{fm} up in energy relative to E_b causes a tilt in the oxide conduction band. Also, since hole concentration can be given by

$$p_o = n_i e^{(E_i - E_b)/kT} = n_i e^{(q\phi_f/kT)},$$

the increase in the concentration of holes at the oxide-semiconductor interface means that, at near this interface, $(E_i - E_b)$ increases as E_b moves closer to the valence band energy E_v . However, hole concentration remains normal inside the semiconductor crystal and hence the relative positions of E_b , E_i , E_c , and E_v are maintained as for zero bias. The result is that the energy bands of the semiconductor bend at near the interface to accommodate the accumulation of holes.

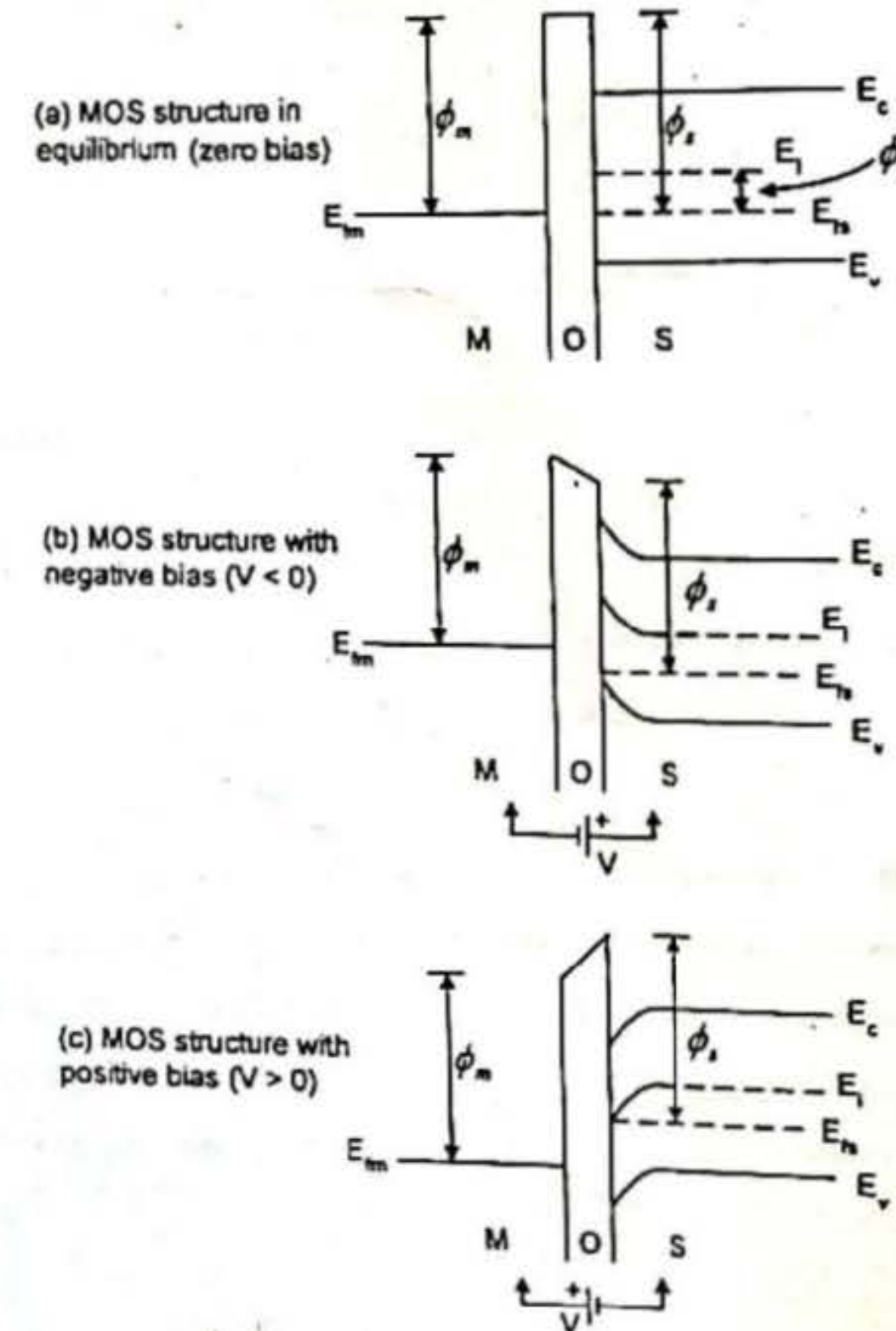


Fig. 6.6 Band diagram for the ideal MOS structure

When an external positive potential is applied to the metal relative to the semiconductor, positive charges are deposited at the surface of the metal and correspondingly negative charges appear at the oxide-semiconductor interface. Electron concentration in the conduction band of the metal is reduced and hence E_{Fn} drops below its equilibrium level. The electrons on the semiconductor interface have the effect of depleting the hole concentration at that interface. Thus, at near the interface, the Fermi level E_F moves further away from E_v and $\phi_r = (E_i - E_F)$ decreases. The down bending of bands near the interface on the semiconductor side is illustrated in figure 6.6c.

Further increase in the applied positive potential will bend the bands more strongly down at the semiconductor interface. In fact, if a sufficiently large voltage is applied, the intrinsic energy level E_i can bend below E_F (figure 6.7).

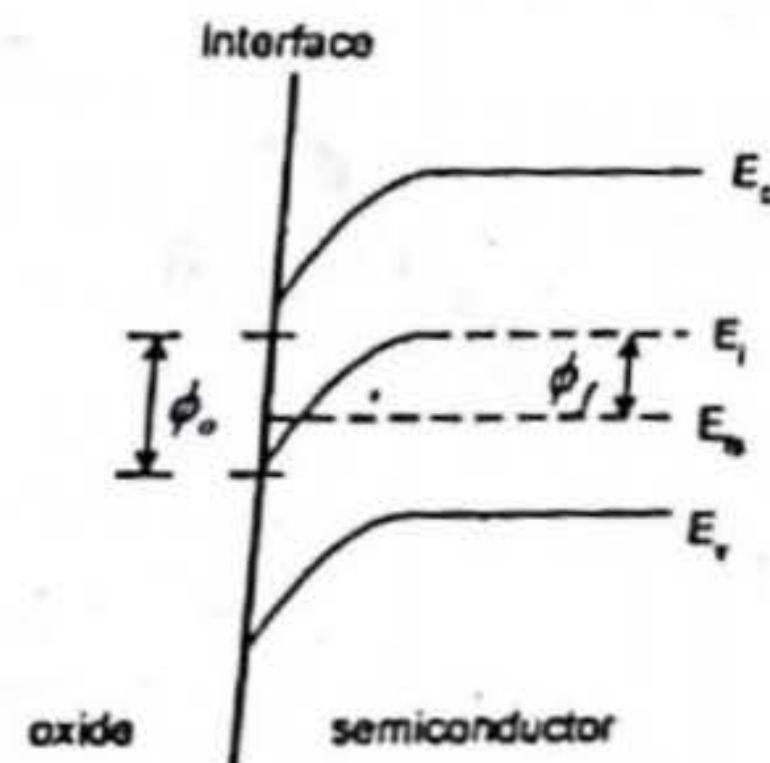


Fig. 6.7 Illustrating the inversion process in MOS

The condition $E_F \gg E_i$ implies a large electron concentration in the conduction band, (typical of n-type material). Thus, the region near the semiconductor interface in this case has conducting properties typical of n-type material. This n-type material is formed, not by doping, but instead by a process called **INVERSION** of the original p-type material resulting from the applied voltage. This inversion layer, separated from the underlying p-type material by a depletion region, is the key to MOS transistor operation.

In figure 6.7, let ϕ_0 be the extent of band bending at the interface. In figure 6.6a, we note that $\phi_0 = 0$. This is referred to as the **flat band** condition for the ideal MOS case. For $\phi_0 < 0$, the bands bend up at the interface as in figure 6.6b and we have hole accumulation. When $\phi_0 > 0$, we have hole depletion (figure 6.6c). Finally, when $\phi_0 > \phi_f$, inversion is obtained. For a true n-type conducting channel to exist, E_i should lie as far below E_F at the interface as it is above E_F further into the semiconductor crystal. This condition occurs when

$$\phi_0 (\text{inversion}) = 2\phi_f = (2kT/q) \ln(N_A/n_i) \quad \dots\dots\dots (6.10)$$

where N_A is the density of acceptor impurities, and n_i is intrinsic density.

6.4 The P-N Junction

We will now consider the contact between two semiconductor materials. The particular junction of interest is the p-n junction, where the contact is between p-type and n-type semiconductors of the same material (e.g. Silicon). In this case, the energy gap $E_g (= E_c - E_v)$ for both materials is of the same magnitude. The energy band diagrams, prior to the formation of a junction, are illustrated in figure 6.8a.

Since the Fermi energy level E_{fn} is higher than E_{fp} , the electrons in the conduction band of the n-type material are at higher energies. Thus, when the junction is formed and prior to equilibrium, the higher energy donor electrons spill over the junction into the p-type side to fill acceptor holes which are at lower energy levels. This movement of donor electrons leaves behind fixed ionised positive donor charges near the junction on the n-side. The acquisition of electrons by acceptor holes produces fixed negative acceptor charges near the junction on the p-side. These regions of fixed charges, distributed inside the crystal, act as dielectric or insulating material.

The conditions existing at the p-n junction in equilibrium are illustrated in figures 6.8(b-e). When equilibrium is attained, the Fermi levels line up and band bending occurs in the depletion regions (width x_0) on both sides of the junction (figure 6.8b). The polarization of the fixed charges, $+$ for n-type and $-$ for p-type (figure 6.8c), creates a potential barrier $V_b = V_c$ (the contact potential), figure 6.8e. The contact potential sets up an electric field across the junction, which tends to discourage further movement of majority charge carriers across the junction. The majority charge carriers now need to acquire extra energy ($=qV_c$) to cross the junction. The electric field however aids the minority charge carriers to flow across the junction.

The resulting minority and majority electron and hole currents are shown in figure 6.9 for zero and external bias conditions. The band bending has been made linear for convenience. With zero external bias, the conduction band electron current I_{en} and I_{ep} are equal, and the valence band hole currents I_{hn} and I_{hp} are also equal. Hence, the net current flow across the junction is zero. When an external potential (V) is applied across the junction, two effects are observed. Depending on the sign of the applied potential,

- (i) A forward bias (figure 6.9b) reduces x_0 to $x_1 < x_0$, and V_b to $(V_c - V)$.
- (ii) A reverse bias (figure 6.9c) increases x_0 to $x_2 > x_0$, and V_b to $(V_c + V)$.

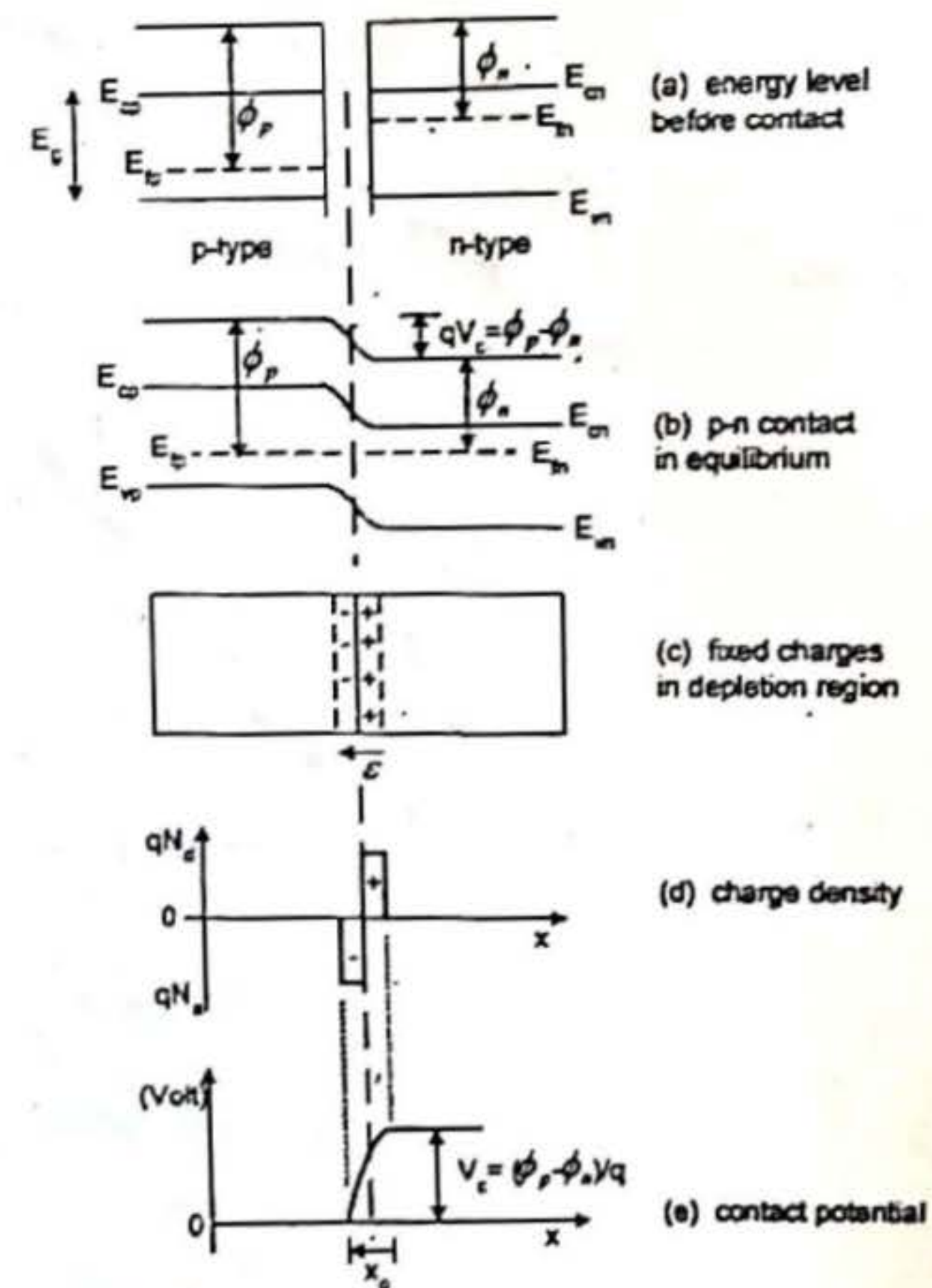


Fig. 6.8 The p-n junction

However, the majority 'diffusion' current I_{en} and I_{hp} are affected by the polarity of the bias. Forward biased (i.e p-type positive with respect to n-type), the barrier potential V_h is reduced to $(V_c - V)$. Thus more electrons and holes can acquire the smaller energy $q(V_c - V)$ to cross the junction. Assuming the extra energy is thermally acquired, then by a similar analysis to equation 6.4, the resulting majority currents due to V becomes

$$I_{fv} = (I_{cn} + I_{hp}) \exp (qV/kT).$$

The total forward current becomes,

$$I_f = I_{fv} - (I_{cp} + I_{hn}) = (I_{eo} + I_{hp}) \exp(qV/kT) - (I_{cp} + I_{hn}).$$

But under zero bias, $(I_{en} + I_{hp}) = (I_{ep} + I_{tn}) = (I_{eo} + I_{to})$, and hence

$$I_f = (I_{\infty} + I_{ho}) [\exp(qV/kT) - 1] \quad \dots\dots\dots (6.11)$$

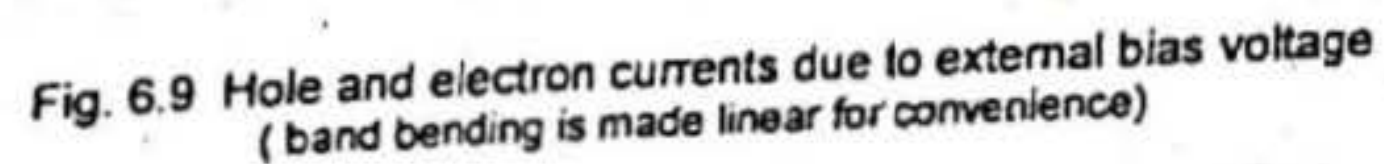
✓ Also, by a similar analysis to equation 6.5, the reverse current is obtained as

$$I_r = (I_{\infty} + I_{ho})[1 - \exp(-qV/kT)]$$

$$= (I_{\infty} + I_{ho}) \quad \text{approximately} \quad \dots\dots\dots (6.12)$$

¶The p-n junction is therefore a rectifying junction. The rectifying characteristic of the p-n junction, at a given temperature, is illustrated in figure 6.10. We note that while the drift current remains essentially constant, diffusion current varies with the applied voltage.

In conclusion, the length of the depletion region is a function of the doping concentration of the material and of the external bias. Where the doping concentration is the same for both materials (i.e. $N_D = N_A$), then the extension of the depletion region into either side of the junction is equal to $x_0/2$. For $N_D \neq N_A$ the dept of ionisation differ. The effect of external bias depends on the polarity of the bias voltage. Forward bias reduces x_0 , while reverse bias increases x_0 .



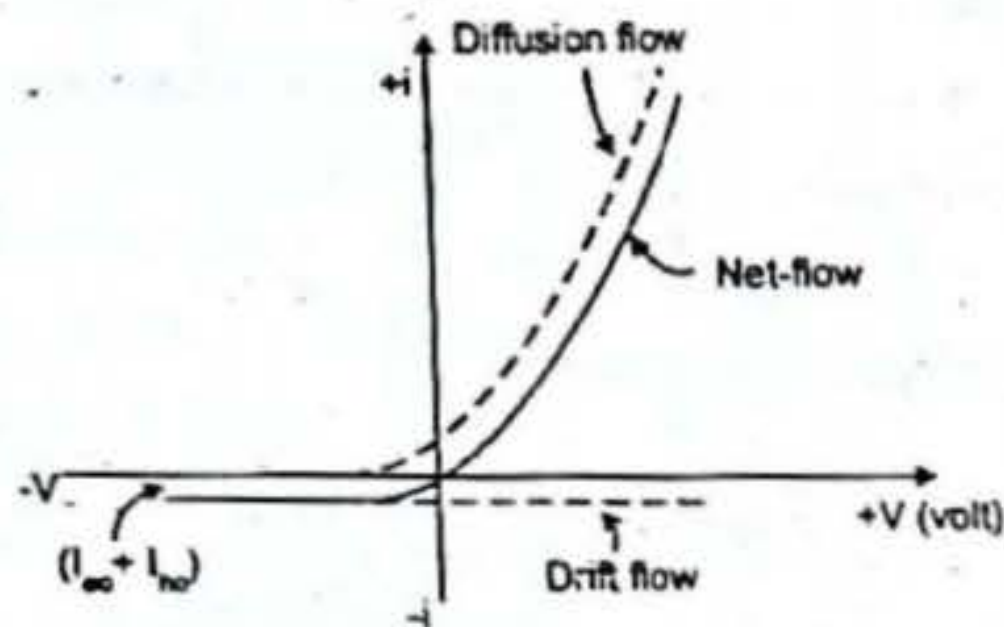


Fig.6.10 Current flow in a p-n junction

The Tunnel Diode

The tunnel diode is a p-n junction diode with very high impurity concentration (of the order of 1-in- 10^4) in the p- and n-type materials. The high doping concentration produces two effects;

- (i) the Fermi level moves into the conduction band in the n-type material and into the valence band in the p-type material. Thus, the materials become degenerate (i.e. near metallic).
- (ii) The depletion region becomes very thin (of the order of 10^{-9} m).

Thus, when the p-n junction is formed and the Fermi levels line up, the bottom of the conduction band (E_{cn}) of the n-type material lies below the top of the valence band (E_{vp}) of the p-type material (figure 6.11). As a result, majority carriers can tunnel through the narrow depletion layer across the junction at zero bias (i.e. with no external voltage applied).

With a forward bias (figure 6.11b), the bottom of the conduction band of the n-type material starts to rise above the top of the valence band of the p-type. Conduction band electrons then tunnel across to fill the lower energy holes in the valence band of the p-material. Thus initially, tunnelling continues with increasing forward bias, but eventually ceases and the current decreases sharply. With further increase in forward bias (figure 6.11c), the normal diffusion current starts to predominate as the energy level diagram returns to that of a forward biased normal p-n junction. With reverse bias, E_{cn} lies below E_{vp} . The overlap between the bottom of the conduction band and top of the valence band increases (figure 6.11d) leading to immediate zener breakdown and consequent high reverse current flow.

The current-voltage characteristic of the tunnel diode is sketched in figure 6.12. Section OB of the characteristic represents the increase in tunnel current as forward bias is increased. Section BC is a negative resistance region, representing current decrease with increase in applied forward voltage. Section CD of the characteristics joins the diffusion current graph of a normal p-n junction in forward bias.

It is important to note that tunnelling takes place at the speed of light, hence the device can respond to fast varying signals. If the device is constrained to operate between the peak point B and the valley point C, it can be used as an electronic switch. The device can switch between points B and C in pico-seconds (i.e. 10^{-12} s) and hence finds useful application in fast switching circuits. Again, its negative resistance region can be utilized in high frequency oscillator circuits working at about 10^{10} Hz (i.e. in the microwave region).

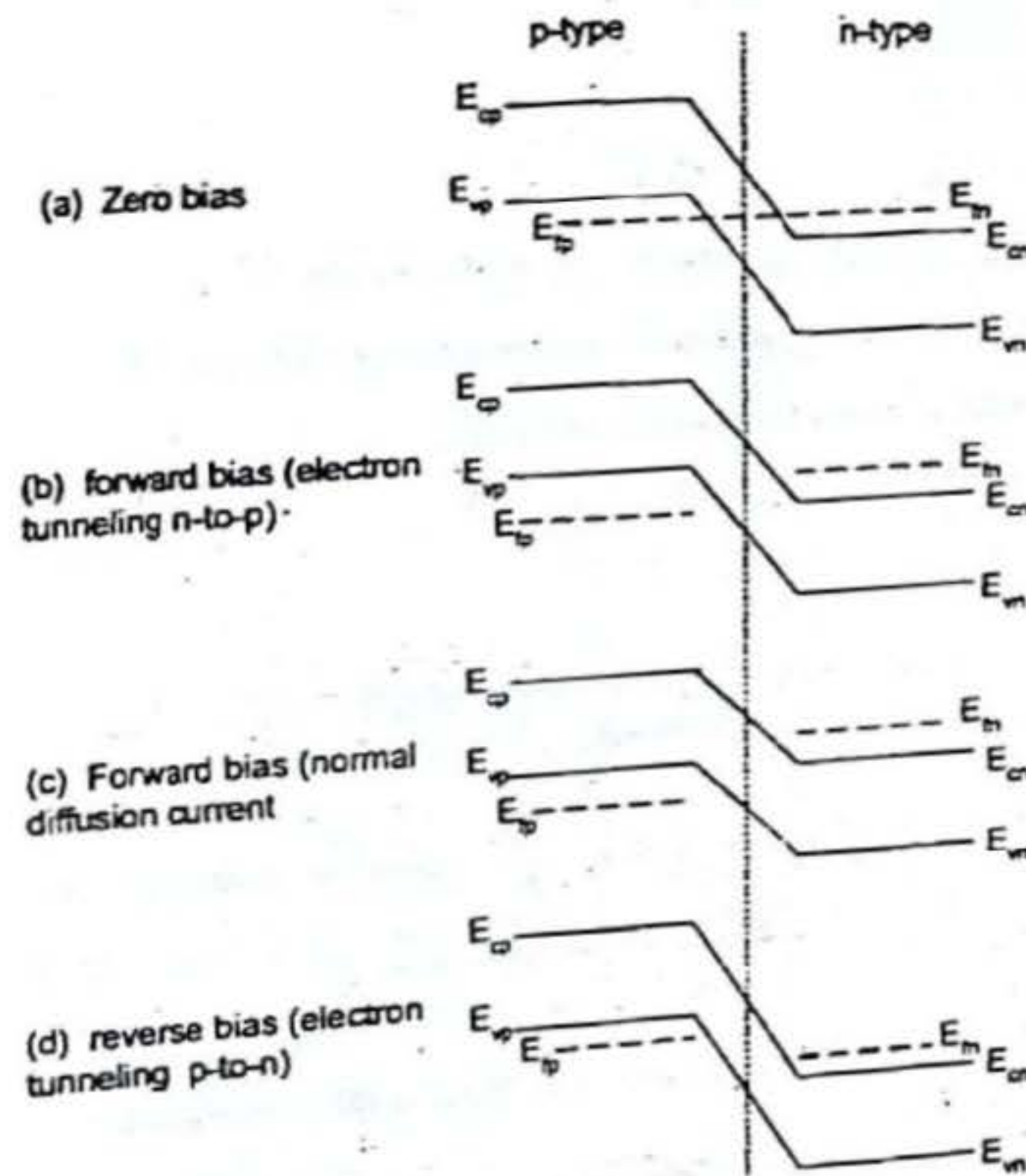


Fig. 6.11 Energy level diagrams for the tunnel diode

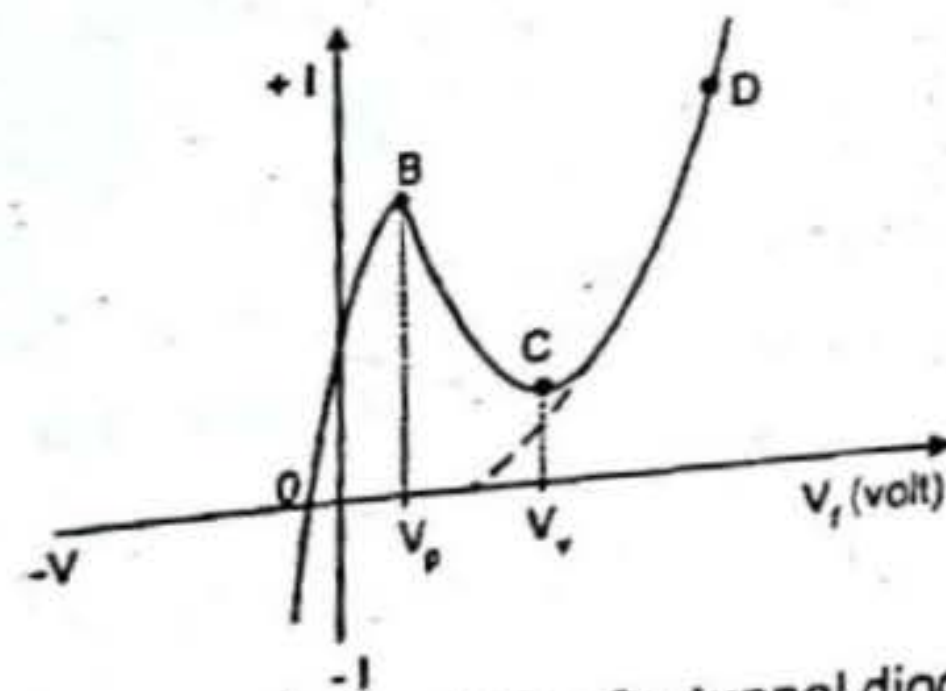


Fig. 6.12 I-V characteristic of a tunnel diode

CHAPTER SEVEN

OPTOELECTRONICS

	Page
7.1 Transition in Direct and Indirect Gap Materials	88
7.2 Opto-electronic Devices	91
Solid-State Radiation Sources	92
Photon Detectors	97
Photoconductive Cell	98
Photo-junctions	99

CHAPTER SEVEN

OPTOELECTRONICS

Opto-electronic devices are transducers, which rely on photoelectric effect to convert light into electrical energy or vice versa. A wide variety of commercially available devices include photo diode, solar cell, photoconductor (or light dependent resistor), light emitting diode (LED), laser diode, etc. Photoelectric effect rely on the principle that when a photon of light is absorbed in a semiconductor, an electron gains its energy. Conversely, when an electron is trapped (e.g. when it recombines with a hole) it loses energy in the form of electromagnetic radiation, thus releasing photon or light. The efficiency of absorption and release of photon energy depends on the type of semiconductor material in which the electron transition takes place.

7.1 Transitions in Direct and Indirect gap Materials

In considering the energy diagrams for semiconductors, we note that free electrons are not totally free but do enjoy a certain mobility, since they can move through the crystal until they recombine with a hole. Holes can also be said to have such mobility. Thus, electrons have momentum as well as energy.

A matter wave, such as an electron, which has energy E_0 at rest (i.e. its potential energy), has total energy E equal to $(E_0 + \frac{1}{2}mu^2)$ when it moves with velocity u . In terms of momentum $p (= mu)$, the total energy E equals $(E_0 + p^2/2m)$.

The energy-momentum plot, for such an electron is shown in figure 7.1. In curve I, the electron has no potential energy. The potential energy E_0 is added to all parts of curve I to obtain curve II.

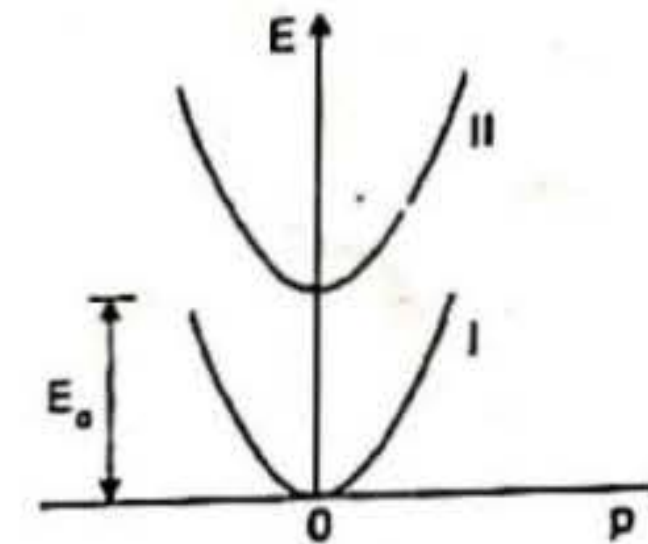


Fig. 7.1 Energy-momentum curve for matter wave

In a semiconductor, the momentum of both electrons and holes must be considered. The illustration in figure 7.2 shows the simplified energy diagram for Gallium Arsenide (GaAs) and Silicon (Si) materials. Normally, the top of the valence band is taken as the reference level. We note that the lowest conduction band minimum in GaAs is located at zero momentum (i.e. directly above the valence band maximum). The band gap equals 1.43 eV. For Silicon, the conduction band minimum is displaced at an angle θ on the energy-momentum plane. Thus, if an electron makes a transition from the valence band (point 0) to conduction band (point A) on the energy-momentum plane, the energy and momentum changes involved are $\Delta E (= \frac{1}{2}mu^2)$ and $\Delta p (=mu)$ respectively as illustrated in figure 7.2b. The angle θ that this transition makes on the diagram is given by

$$\tan \theta = \frac{\frac{1}{2}mu^2}{mu} = \frac{1}{2}u \quad \dots\dots\dots (7.1)$$

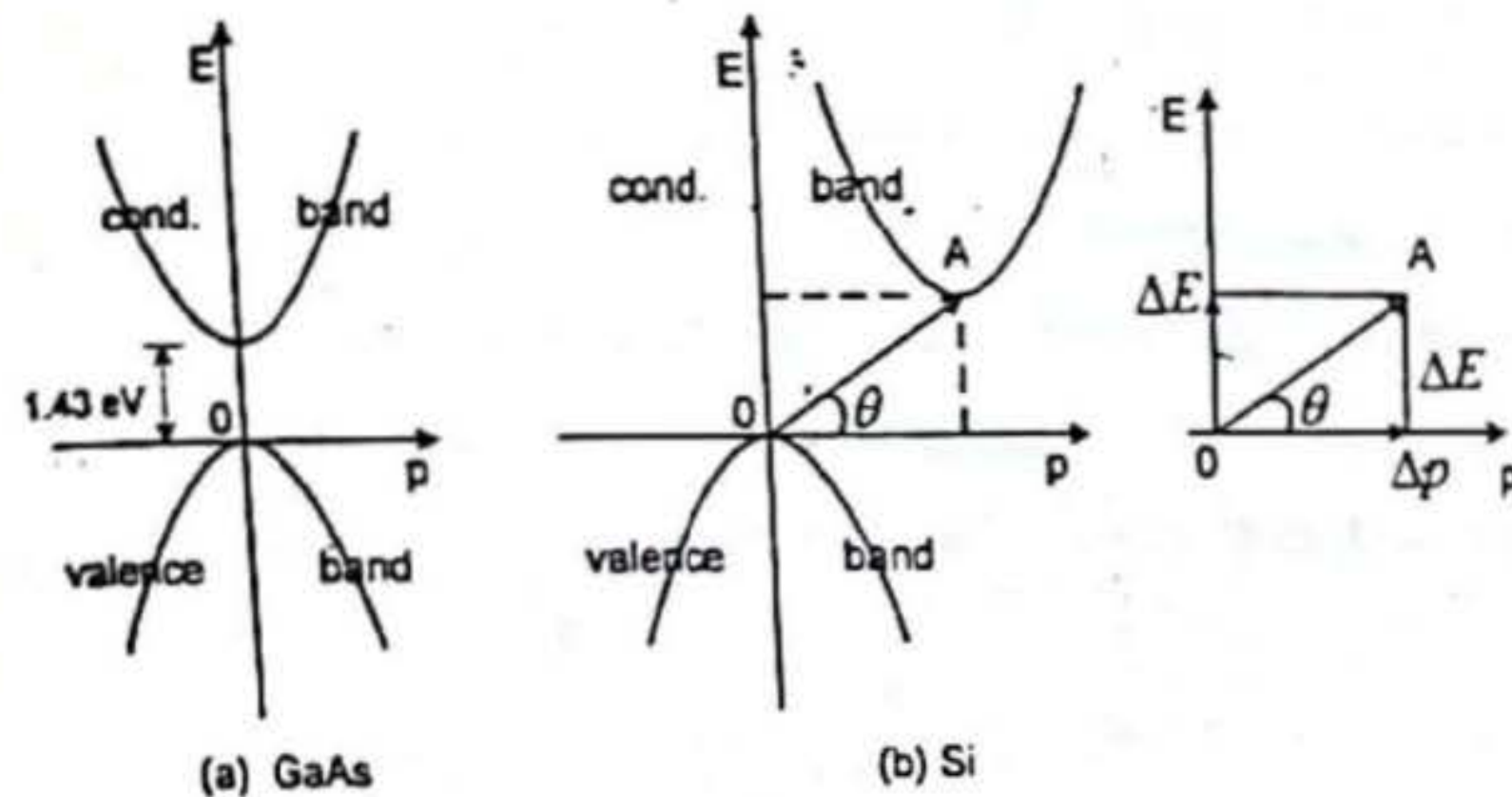


Fig.7.2 Simplified energy-momentum diagram in semiconductor

semiconductor material such as GaAs, where $\theta = 90^\circ$, is called a **direct-gap material**. A transition from valence to conduction band (or vice-versa) in the **direct-gap** material involves energy change only. In a material like Silicon, the transition angle is smaller and both energy and momentum changes involved. Such a material is called an **indirect-gap material**.

The electrons in a semiconductor material interact with **photons** and **phonons**. The photon, as we have seen in Section 5.4, may be described as a quantum of light energy having zero mass, and hence no momentum. Thus, electrons involved in transitions in direct-gap material interact with photons only. We note also that photons travel with the velocity of light (3×10^8 m/s) and hence $\tan \theta (= \frac{1}{2}u)$ is very large indeed, giving an angle θ of almost 90° .

The **phonon** may be described briefly as a quantum of lattice vibration. To understand the concept of phonons, we note that the atoms in a lattice vibrate with thermal energy about their mean position in the lattice. Because the atoms are bound together in the lattice, the vibration of one atom propagates to the other atoms by lattice vibration waves at the speed of sound. If the lattice vibration has a frequency f , its quantized energy $E (=hf)$ is known as phonon. A phonon can therefore be described as a packet of heat energy, always available in a body at any finite temperature, which shows itself as a travelling vibration of the atoms in the lattice. Compared with the velocity of light, phonons have smaller velocities (i.e. the velocity of sound in a solid which typically is 3×10^3 m/s), and hence have much smaller θ value on the energy-momentum plane. Thus phonons have more momentum than energy. Transitions involving both photons and phonons take place in an indirect-gap material.

For photons and phonons, momentum $p = h/\lambda$ and energy $E = hu/\lambda$, where u is velocity. The value of the wavelength λ for phonons and photons are 10^{-6} m and 10^{-10} m respectively. Thus, the phonon has a momentum about 10^4 times that of a photon and an energy about 10^{-1} that of a photon. From the above, it is obvious that transition in direct-gap material is much easier since it does not involve the simultaneous action of photons and phonons.

7.2 Opto-electronic Devices

Light is used as a general title to cover electromagnetic radiation with wavelengths in the range 10 nm to 100 μ m. This optical range covers ultraviolet radiation (10 nm to 400 nm), visible light (400 nm to 700 nm), and infrared radiation (700 nm to 100 μ m).

Opto-electronic devices (or solid-state radiation devices) fall into two categories; i.e. photo-emitters and photo-detectors. Photo-emitters (or radiation sources) convert electrical energy to light energy. They include laser and light emitting diodes. Photo-detectors (or photon detectors) convert light energy (i.e. photon energy) to an electrical output. Photo-detectors include, photo-diode, photo-transistor, photo-thyristor, photo-FET, solar cell, and photo-resistors or photoconductive cell.

Other devices include, liquid-crystal-devices (LCD), solid-state relays, optical shaft encoders, and image intensifiers.

Solid-State Radiation Sources

Solid-state radiation sources emit optical radiation when an electron makes a transition from higher to lower energy level. This transition must of necessity take place in a direct-gap material in order to release photons (or radiation). They emit radiation over a continuous band of wavelengths rather than at a single wavelength. Solid semiconductor optical radiation sources (which include light emitting diode and laser) are made from p-type and n-type semiconductor materials. They operate on the principle that if an electron moves from a higher energy level E_2 to a lower energy level E_1 , it loses energy in the form of optical radiation. The wavelength of the optical radiation is given by

$$\lambda = hc/(E_2 - E_1) \quad \dots\dots\dots (7.2)$$

where h is Plank's constant and c is velocity of light.

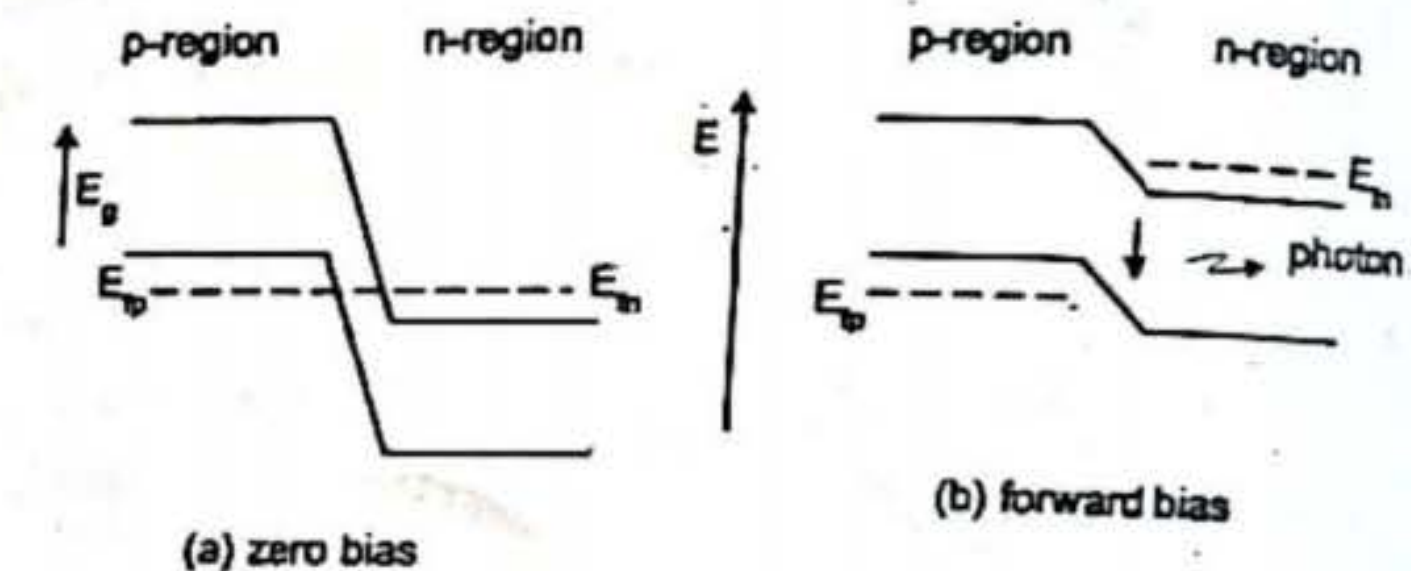


Fig. 7.3 Energy diagram for LED and diode laser

In solid-state radiation sources, the higher energy E_2 is usually the conduction band energy and the lower energy E_1 is the valence band energy. In order to increase radiation intensity, there is the need to enhance the population of electrons in the conduction band. If a p-n junction is formed using direct-gap degenerate semiconductor, there is a high concentration of electrons in the conduction band of the n-region and a high concentration of holes in the valence band of the p-region. With zero external bias (figure 7.3), there is little energy difference between the conduction band of the n-material and valence band of the p-material. The energy difference increases when the junction is forward biased. The higher concentration of electrons, at the edge of the conduction band, occupies energy levels directly above the level of holes at the edge of the valence band. The forward bias of the p-n junction causes the injection of conduction band electrons into the junction region. There the electrons recombine with holes in the p-region and energy is radiated in the form of photons, with one photon emitted for each electron transition.

The radiation therefore relies on current injection into the p-n junction. The wavelength of the radiation is given approximately by $\lambda \text{ (nm)} = 1240/E_g \text{ (eV)}$, where E_g is the energy difference between the conduction band of the n-material and the valence band of the p-material. The energy difference E_g depends on the material and on doping. Thus, by choice of material and doping, different E_g is obtained and hence different wavelengths (or colours of emission) are produced.

In the LED, the electrons randomly return to the lower energy level and spontaneous emission is said to occur. The high carrier concentrations give a useful light output, but not all the light generated at the junction will be emitted. This is due to two main factors;

- (i) light absorption in the anode, which can be reduced by making the anode very thin, and
- (ii) internal reflection at the top of the anode surface.

We note that a ray such as R in figure 7.4 may be part of the light emitted at the anode, which has been reflected back into the diode. At a critical angle θ_c however, all of the light will be totally internally reflected. The angle θ_c is given (from Snell's law) by the expression

$$\sin \theta_c = n_2/n_1 \quad \dots\dots\dots (7.3)$$

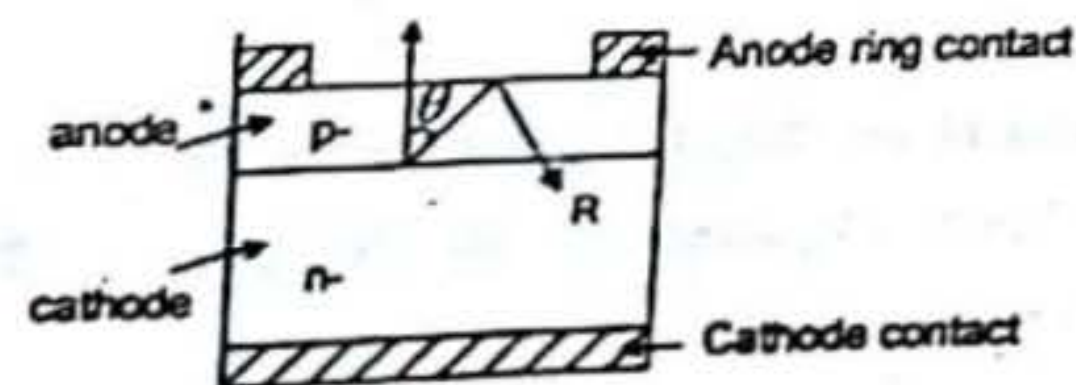


Fig. 7.4 Section through p-n junction showing internal reflection

where n_1 is the refractive index of the medium in contact with the top anode surface (in this case air for which $n_1 = 1$), and n_2 is the refractive index of the anode material. Gallium Arsenide for an example has $n_2 = 3.6$. A transparent epoxy dome acting as a lens (for which $n_1 = 1.5$) can be mounted on the diode to give a critical angle of 24.6° . By suitable choice of lens materials, viewing angles can be varied up to about 60° . Where the LED is used as a radiation source for a fibre optic system, narrow angles between 8° and 11° are provided in order to concentrate the light energy.

The diode **laser** is a development of the LED. The word **laser** stands for 'light amplification by stimulated emission of radiation'. Consider a process whereby a released photon causes another electron to return to the lower energy state, thereby generating a second photon in phase with the first. The two photons then give rise to four, then eight, and so on. This process is called **stimulated emission**, and is the key to laser action. In laser diode, reflecting cavity are formed simply by polishing the ends of the crystal perpendicular to the plane of the p-n junction (Figure 7.5). The polished surfaces, acting as mirrors, reflect photons forwards and backwards through the cavity and stimulated transitions occur. At low current values, spontaneous emission occurs in all directions as in the LED. As forward current increases, photon multiplication, (and hence optical gain), increases due to feedback of the light from the mirrors. The generated photons are eventually allowed to leave the cavity through a small hole in the mirror. The radiation through the hole can be very intense.

In stimulated emission, a photon incident on an atom causes the emission of another photon. As a result, their associated electromagnetic waves are in phase, and reinforce each other (figure 7.6).

Such an emission is said to be **coherent**. In order that the light waves may reinforce each other at a particular wavelength, the distance between the mirrors must be an integral multiple of the $\frac{1}{2}$ -wavelength of the radiation.

The spontaneous emission of photons is a random process and so the phase relationship between their waves and the incident wave is also random. The emission is said to be **incoherent** because their waves tend to cancel one another out.

Materials used in commercial LEDs and lasers include Gallium Arsenide (GaAs), Gallium Phosphide (GaP), Gallium Arsenide Phosphide (GaAsP), Gallium Aluminium Arsenide (GaAlAs), and Indium Gallium Arsenide Phosphide (InGaAsP). Their spectral density covers a narrow band of wavelengths centred on 560 nm (for GaP) and 655 nm (for GaAsP). GaAlAs

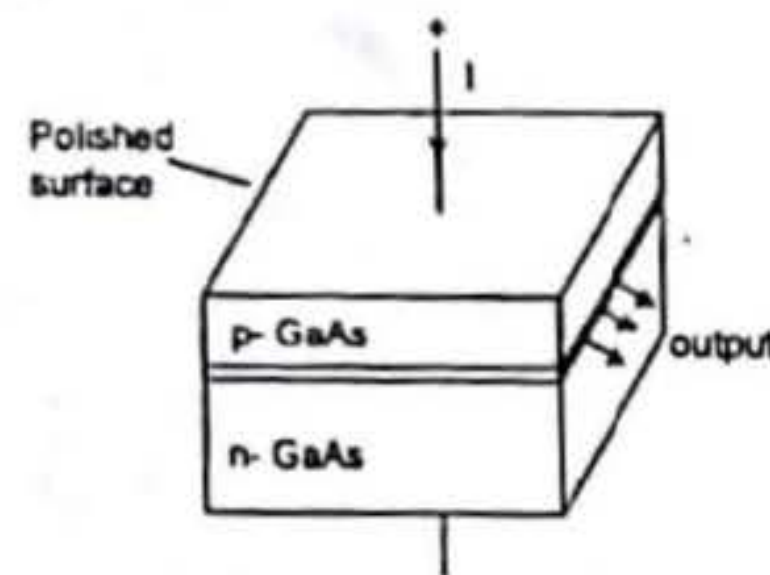


Fig. 7.5 Semiconductor laser diode (construction)

emits radiation wavelengths in the region 800 to 900 nm, while InGaAsP emits wavelengths in the region 1.30 to 1.60 μm . Light emitting diodes find applications as display indicators, while laser applications are many ranging from high intensity radiation sources for cutting, drilling and welding operations, for eye surgery in medicine, in communications, compact discs, optical printers, etc, etc.

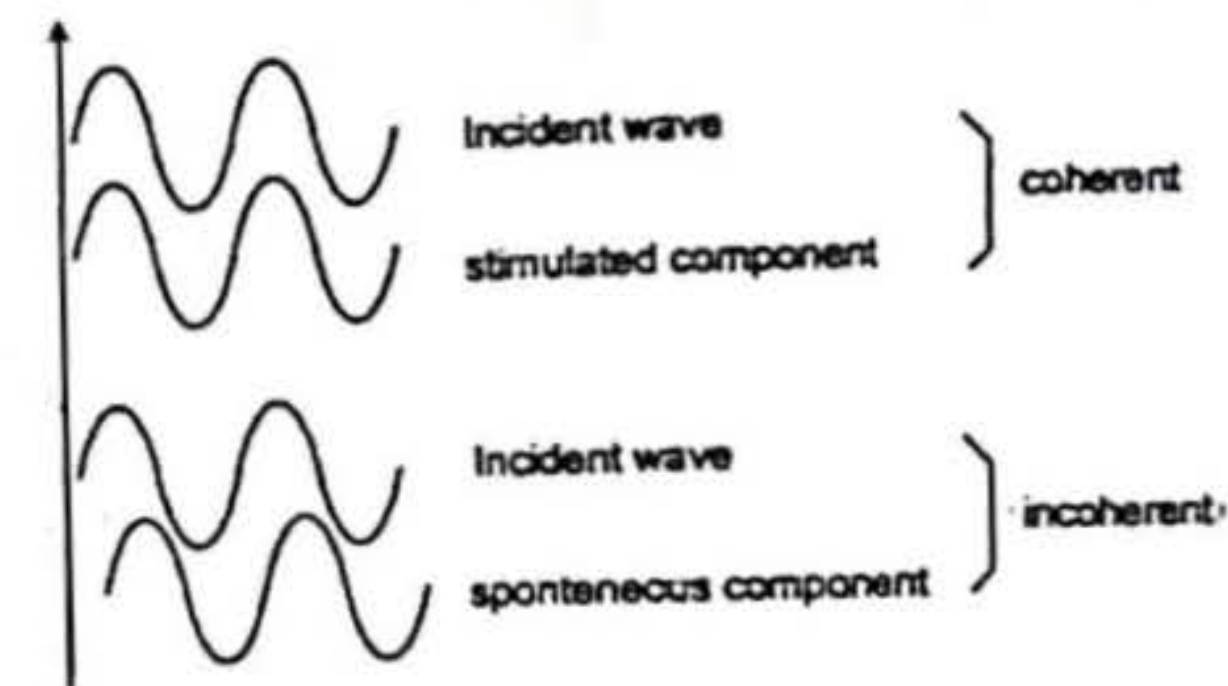


Fig. 7.6 Phase relationships in radiation components

Photon Detectors

A photon detector is basically a semiconductor device that converts photons to electrons. Incident photon of radiation causes electrons to be excited from valence to conduction band thus producing measurable electrical effect. For these devices the energy of the radiated photon (hc/λ) approximately equals the energy gap (E_g) between valence and conduction bands. The response of the detector is therefore centred around the wavelength $\lambda_0 = hc/E_g$.

At room temperature, a semiconductor has partially filled conduction band. If the semiconductor is irradiated by photons (or light) of sufficient energy (hc/λ greater than E_g), electrons move from valence to conduction band, thereby increasing the concentration of free electrons in the conduction band. Since conductivity $\sigma (=qn\mu)$ is proportional to the density of free electrons (n), the conductivity of the irradiated semiconductor increases. The change in conductivity due to light (photons) is the photoconductive effect.

Photoconductive Cell (Or Photoconductor)

The photoconductor is a practical device which relies on the photoconductive effect for its operation. Consider the unit cube of doped semiconductor block in figure 7.7. Let photon flux ϕ (photons per-second per metre cubed) fall on the surface. If each photon successfully releases a free electron, then the rate of electron production is ϕ electrons/second. This leads to a rise Δn in the number of free electrons present in the material. If τ is the life time of an electron, then the recombination (or death) rate will increase by $\Delta n/\tau$. When carrier population attains a new steady state value, then the production and the recombination rates must equal. Thus,

$$\phi = \Delta n/\tau \text{ or } \Delta n = \tau\phi \quad \dots\dots\dots(7.4)$$

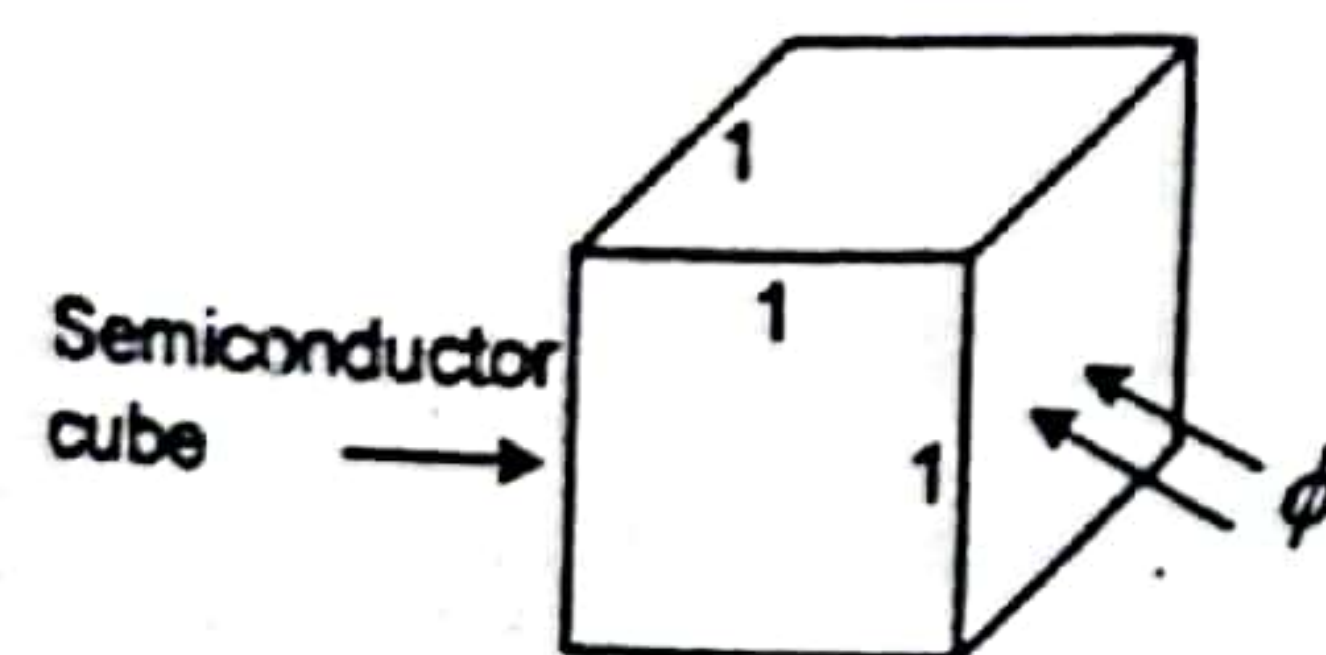


Fig.7.7 Photoconductive effect

From the normal conductivity expression, the increase in conductivity becomes

$$\Delta\sigma = q\Delta n\mu = q\mu\tau\phi \quad \dots\dots\dots(7.5)$$

where q is charge on the electron and μ is electron mobility.

In general, not every photon will be successful in releasing an electron. The fraction η of photons effective in releasing electrons is known as the **quantum efficiency**. Allowing for this factor, the number of electrons produced by the photon flux ϕ equals $\eta\phi$ electrons/sec/m². Equation 7.5 then becomes

$$\Delta\sigma = q\tau\mu\eta\phi \quad \dots\dots\dots(7.6)$$

The fractional change in conductivity in relation to the initial undisturbed electron population, becomes

$$\Delta\sigma/\sigma = q\mu\eta\phi\tau/q\mu n_i = \eta\phi\tau/n_i \quad \dots\dots\dots(7.7)$$

where n_i is the electron density of undisturbed population.

Photo-Junctions

If radiation (light) penetrates to the junction region of a p-n junction, and provided the photon energy is adequate, an electron-hole pair is generated in the depletion region. In a reverse biased p-n junction, the minority carriers so generated are aided across the junction. A minority electron released on the p-side crosses the junction into the n-region, while minority holes generated on the n-side cross to the p-region. The flow of these minority charges give rise to a photo-current (I_p), which adds directly to thermally generated leakage current (I_0).

In a reverse biased photodiode with zero illumination; only the thermally generated leakage current I_0 flows. (It is known as the dark current). When the diode is illuminated, the total reverse current I_r equals $(I_0 + I_p)$. We note that at a given temperature, I_0 is constant and I_p varies with the radiation intensity. For a given radiation wavelength, I_p is proportional to the power of the incident radiation. Let ϕ photons/second be incident on the junction, and n_p be the electron-hole pairs released per second by the radiation. Since photon energy $E = hc/\lambda$, the power P of the incident radiation is given as

$$P = hc\phi/\lambda \quad \dots\dots\dots (a)$$

And current

$$I_p = qn_p \quad \dots\dots\dots (b)$$

Dividing (b) by (a) gives the expression

$$I_p = Pq\lambda n_p/hc\phi = P(q\lambda/hc)(n_p/\phi) \quad \dots\dots\dots (7.8)$$

where n_p/ϕ is called the **quantum yield**. It is a measure of the efficiency of the diode. The symbol and i-v characteristic of a Silicon p-n junction photodiode is shown in figure 7.8. We note that the reverse current $I_r (= I_0 + I_p)$ is almost independent of the reverse voltage, and is proportional to the light intensity ϕ .

A variation of the photodiode is the PIN-photodiode. 'PIN' stands for positive-intrinsic-negative. Its construction is shown in figure 7.9. The incident photons release additional electron-hole pairs in the layer of the intrinsic material (sandwiched between the p-layer and n-layer), thus increasing the photocurrent. A typical silicon PIN-photodiode has an active area of 1 mm^2 and centre wavelength λ_0 of $0.85 \mu\text{m}$ (located in the infrared region).

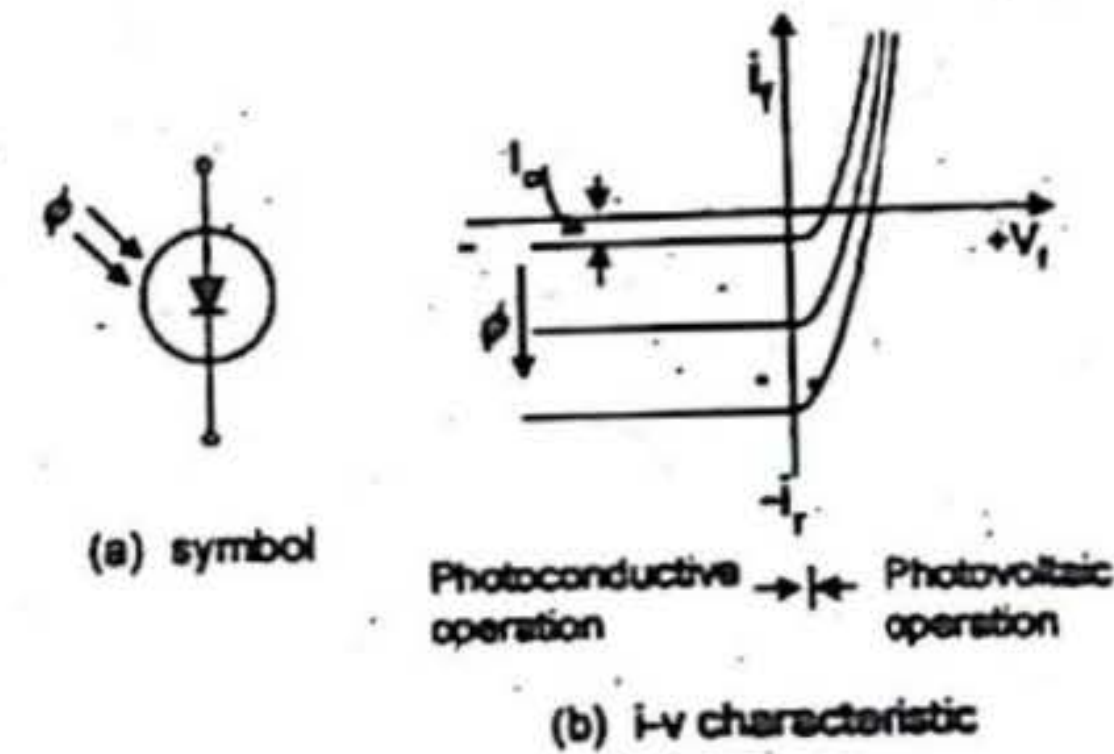


Fig.7.8 The Silicon p-n junction photodiode

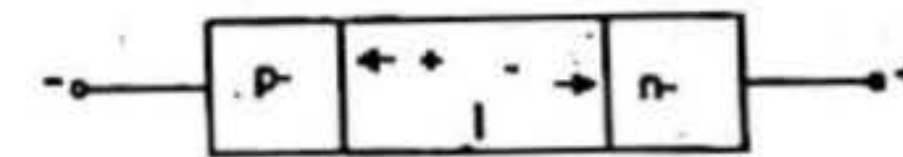


Fig.7.9 PIN-photodiode construction

The forward characteristic of the p-n junction photodiode shows that the characteristic passes through a forward voltage, at zero current. This forward voltage drop (approximately 0.4V) is almost independent of light intensity. This can be interpreted to mean that an open circuited photodiode generates an emf when the junction is illuminated. This is the **photovoltaic effect**, on which the operation of the photovoltaic cell (or solar cell) depends. Thus, a p-n junction photodiode can function as a photovoltaic cell.

As previously noted, the illumination falling on a p-n junction releases electron-hole pairs near the junction. The holes move into the p-region while the electrons move into the n-region (figure 7.10). This movement of minority charges are due to the influence of internal electric field across the junction. If a low resistance circuit is connected externally across the diode, a current flow will be indicated. This means that the junction is converting light energy directly into electrical energy. The magnitude of the current will equal the rate of excitation of the free carriers. Hence,

$$\text{Current } I_p = q\eta\phi \quad (\text{Amps}) \quad \dots\dots\dots (7.9)$$

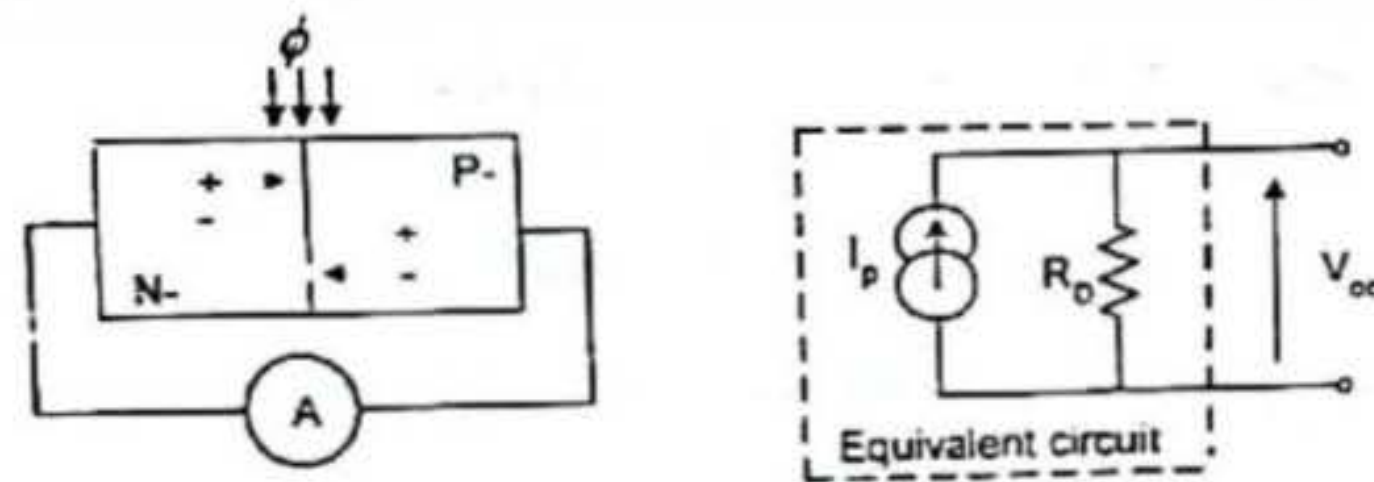


Fig. 7.10 The photovoltaic cell

The junction may therefore be represented by an equivalent current generator, as shown in figure 7.10b. The resistance R_0 is the resistance of a normal p-n junction diode, which is a function of current I_p . Substituting the current value of equation 7.9 into the diode current equation, gives (for the open-circuit condition)

$$I_p = I_d = q\eta\phi = I_0[e^{qV/kT} - 1] \quad \dots\dots\dots (7.10)$$

Solving for V and noting that $I_d = I_p$, the open circuit photovoltaic voltage V_{oc} is given as

$$\begin{aligned} V_{oc} &= (kT/q) \log_e [q\eta\phi/I_0 + 1] \\ &= (kT/q) \log_e [I_p/I_0 + 1] \quad \dots\dots\dots (7.11) \end{aligned}$$

CHAPTER EIGHT

DIELECTRIC MATERIALS

	Page
8.1 Introduction	104
Polarization	106
Capacitance	107
Ferroelectrics	108
8.2 Applications of Dielectrics	109
Energy storage	109
Pyroelectric Materials	110
Piezoelectric Materials	110
Dielectrics as Transmission Medium	112
8.3 Optical fibre transmission Medium	112

CHAPTER EIGHT

DIELECTRIC MATERIALS

8.1 Introduction

The term dielectric is generally used to describe materials which are insulators, with resistivity as high as $10^{20} \Omega\text{-m}$. Their properties determine the performance of components such as capacitors, cables, etc. Insulators, whether solid or liquid, are generally compounds and hence are composed of molecules which may be regarded as collections of positive and negative charges. There are no free electrons to conduct electricity under the influence of an electric field.

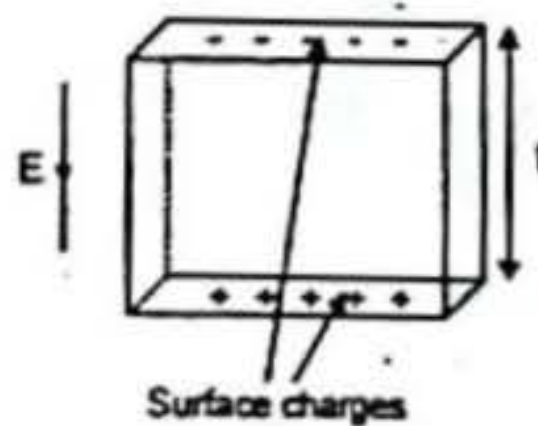


Fig. 8.1 Bound charges in polarized dielectric

Although dielectric materials do not conduct electricity, they respond to the application of an electric field through the displacement of electric charges in the material. This charge displacement causes a small separation of the effective positive and negative charges. Hence electric dipoles are induced.

For two isolated charges $+q$ and $-q$ separated by a distance r , an electric dipole moment (defined as $\mu = qr$), is formed. When the charges separate under the influence of an electric field, they appear on the surfaces of the material normal to the direction of the field, as shown in figure 8.1. These bound charges, as they are called, disappear when the electric field is removed. All materials show (to a limited extent) this dielectric response to electric field. In the case of conductors and semiconductors, it is swamped by the movement of conduction electrons.

In some other dielectric materials, such as ionic crystals or materials with asymmetric molecules (where the distribution of positive and negative ions are asymmetrical about a point), permanently separated charges do exist even without an electric field. The permanent electric dipoles associated with such a material are considered as a mass of crystallites, randomly oriented with respect to each other (figure 8.2a). The random tendency is caused by thermal effect. In such a material, the effect of an electric field is to align the dipoles in the direction of the field (figure 8.2b) and in the process produce charges on the surface of the material normal to the direction of the electric field, as in figure 8.1.

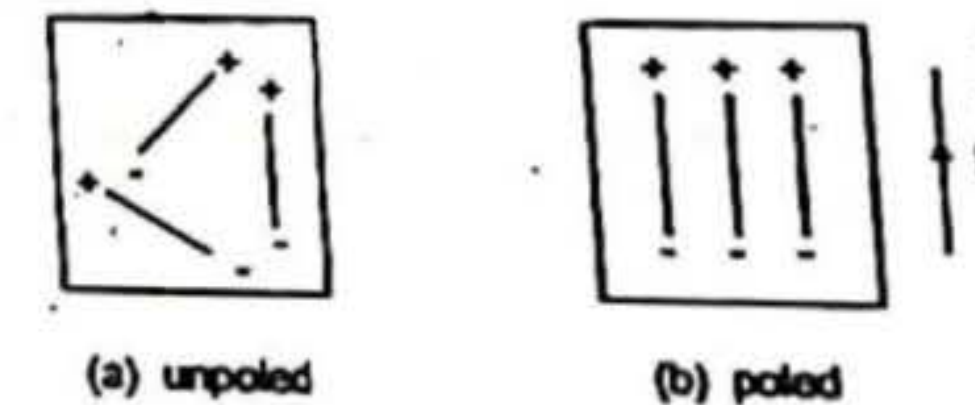


Fig. 8.2 Dipole orientation

Polarization

Whether the dipoles are induced or permanent, there is polarization P , defined as charges per unit area of dielectric surface (or surface charge density). It has the dimension C/m^2 , same as for electric flux density $D (= \epsilon E)$, where ϵ is permittivity and E is electric field.

Polarization can be classified as electronic (optical) or ionic (molecular). In electronic polarization, the atom may simply be considered to consist of a positive nucleus and a negative electron cloud of equal charge magnitude. In the absence of an electric field, the charge centres coincide. When electric field is applied, the electrons move relative to the nucleus, thus creating an induced dipole. The electron motion can respond to rapidly alternating field (of the order of 10^{15} Hz) because of the low mass of the electron.

In ionic (or molecular) polarization, the ionic crystals contain permanent dipole moments, due to positive and negative ions held in the lattice (e.g. sodium chloride which might be considered as Na^+Cl^-). Due to their larger mass, the upper frequency response of ions to alternating field is lower than for electrons.

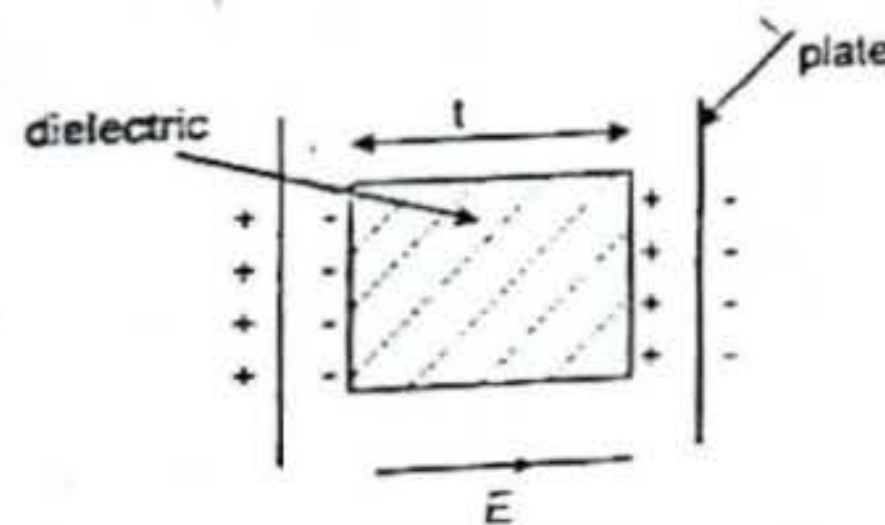


Fig. 8.3 Effect of dielectric on electric flux density (D)

The reverse effect is also true. When an electric potential (field) is applied across a pair of surfaces, the crystal deforms mechanically. Piezoelectric materials are therefore used as electromechanical transducers (or sensors) which allow an exchange between electrical and mechanical energy.

Piezoelectric materials can be divided into three groups; monocrystalline materials (e.g. naturally occurring quartz), polycrystalline aggregates (e.g. ferroelectric ceramics), and polymeric substances (e.g. polyvinylidene fluoride). For measurement applications, the quartz and man-made polycrystalline ceramics (such as lead-zirconate-titanate, barium titanate, and lead metaniobate) are most used materials. Polymeric substances are normally considered to be too unstable for measurement purposes. Piezoelectric sensors are used in the measurement of force, pressure and acceleration.

The properties of piezoelectric crystals can be described by two equations

$$T = yS - pE \quad \dots\dots\dots (a)$$

and

$$D = \epsilon E + pS \quad \dots\dots\dots (b)$$

where T is stress ($force/m^2$), S is strain, y is Young's modulus of elasticity, p is piezoelectric constant, $\epsilon (= \epsilon_0 \epsilon_r)$ is permittivity, E is electric field (V/m), and D is electric flux density (C/m^2).

If stress is applied with electric field E set zero, then,

$$D = pS = T(p/y) \propto T \quad \dots\dots\dots (c)$$

Equation (c) shows that the charge produced per unit area is proportional to force per unit area acting on the crystal.

On the other hand, if an electric field E is applied when $T = 0$, we have

$$S = E(p/y) \propto E \quad \dots\dots\dots (d)$$

This result shows that the crystal is strained when an electric field is developed across it. The quantity p/y , called the piezoelectric coefficient, is an important property of the transducer. From the above, it is deduced that the piezoelectric material acts as electro-mechanical transducer which allows an exchange between electrical and mechanical energy.

Dielectric As Transmission Medium

Dielectric materials find applications as transmission medium, for example, polythene (in coaxial cables), and optical fibre (in optical measurement systems and optical communication links). The velocity (c) of an electromagnetic wave in a medium is equal to $(\epsilon\mu)^{-1/2}$. For dielectric materials of interest, $\mu (= \mu_0)$ is a constant. Thus, $c \propto \epsilon^{-1/2}$. Since the refractive index (n) of a medium is inversely proportional to velocity, then $n \propto \sqrt{\epsilon}$. In fact, the relationship between refractive index and permittivity is given as $n^2 = \epsilon_r$.

In practice, the dimensions of a dielectric material used as a transmission medium is determined by the wavelength (λ) of the electromagnetic wave. Since wavelength is proportional to velocity, it is inversely proportional to refractive index. Thus, materials of high permittivity are needed in order to reduce the dimensions of the transmission medium.

Optical Fibre Transmission Medium

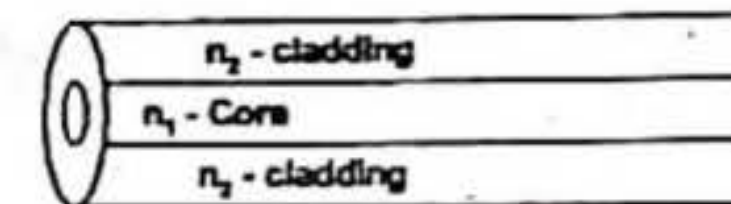
The optical fibre is widely used as transmission medium in optical measurement systems and in communication links. An optical fibre typically consists of two concentric cylinders (figure 8.5a), with different refractive index.

The inner cylinder is called the **core** and has a refractive index n_1 . The outer cylinder called the **cladding** has refractive index n_2 , where $n_2 < n_1$. When a ray is incident at the boundary between the two media (figure 8.5b) at an angle θ_1 , part of the ray is reflected back into the first medium, while the remainder is refracted into the second medium. From Snell's law,

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

where θ_2 is the angle of refraction. This can be written as

$$n_1/n_2 = \sin \theta_2 / \sin \theta_1 \quad \dots\dots\dots (8.5)$$



(a) optical fibre construction

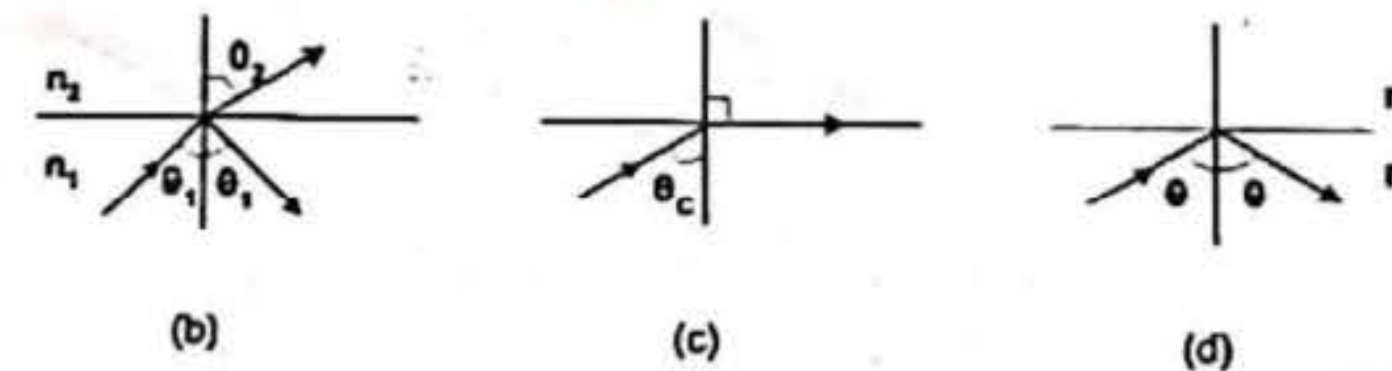


Fig. 8.5 Optical fibre principle

When $\theta_2 = 90^\circ$, the reflected ray travels along the boundary (figure 8.5c). The corresponding value of θ_1 at which $\theta_2 = 90^\circ$ is known as the critical angle θ_c . Thus, θ_c is given by $n_1 \sin \theta_c = n_2$, or $\sin \theta_c = n_2 / n_1$. From which

$$\theta_c = \sin^{-1}(n_2/n_1) \quad \dots\dots\dots (8.6)$$

The critical angle can therefore be calculated for a given n_1 and n_2 . When $\theta_i > \theta_c$, total internal reflection occurs (figure 8.5d) and the ray is reflected back into the first medium (core). Subsequently, the ray is internally reflected and will be transmitted along the fibre. By means of many internal reflections, all rays with angles of incidence greater than the critical angle will be transmitted along the fibre.

In practice, the core/cladding fibre is surrounded by an external medium (e.g air) with refractive index n_0 . When a ray is launched from outside the fibre, it is refracted at the air/core interface into the core and then reflected at the core/cladding interface, as illustrated in figure 8.6. If θ_0 is the angle of incidence of the ray to the fibre, the ray enters into the fibre core at an angle θ given by

$$n_0 \sin \theta_0 = n_1 \sin \theta.$$

If the angle of incidence at the core/cladding interface is ϕ , then

$$n_1 \sin \theta = n_1 \cos \phi = n_0 \sin \theta_0$$

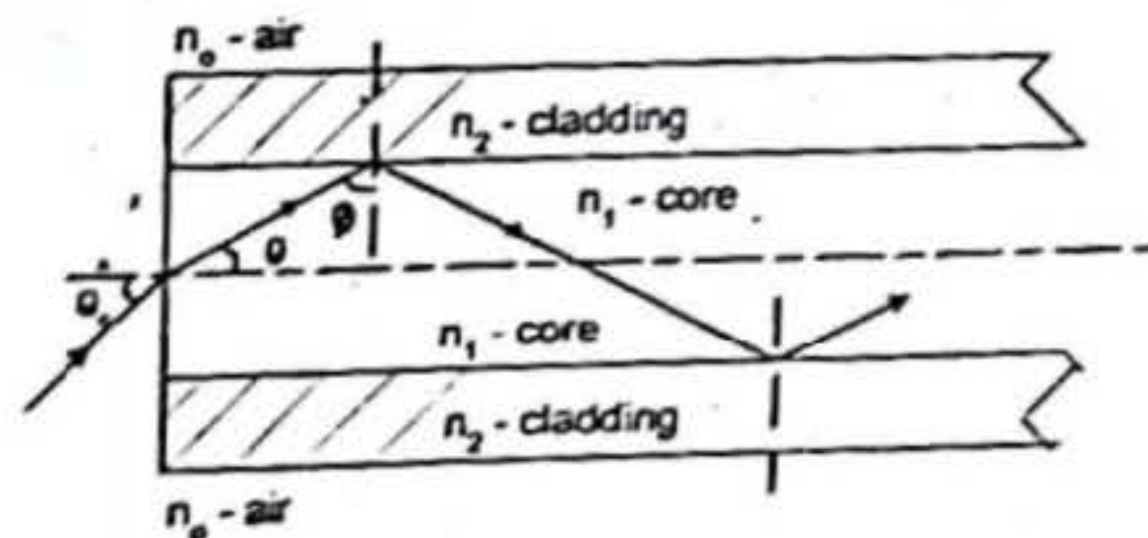


Fig. 8.6 Reflection and refraction of externally launched ray

For two plane parallel plates forming a capacitor in vacuum, the electric flux density $D = \epsilon_0 E$, where ϵ_0 ($= 8.854 \times 10^{-12} \text{ F/m}$) is permittivity of free space and E (Volt/m) is the electric field applied between the plates. When a dielectric material of permittivity ϵ is now introduced into the gap between the plates (figure 8.3), the internal polarization induces extra charges of opposite sign on the plates of the capacitor. Put differently, the internal charges on the dielectric material, induce extra external charges of opposite polarity on the external surfaces of the plates of the capacitor. The increase in electric flux density is given as

$$D = \epsilon E = \epsilon_0 E + P \quad \dots\dots\dots (8.1)$$

The relative permittivity ϵ_r of a material, often referred to as its dielectric constant, is given as $\epsilon_r = \epsilon / \epsilon_0$. Thus $\epsilon = \epsilon_0 \epsilon_r$. Substituting into equation 8.1, yields $D = \epsilon_0 \epsilon_r E = \epsilon_0 E + P$, from which

$$P = (\epsilon_0 \epsilon_r E - \epsilon_0 E) = \epsilon_0 (\epsilon_r - 1) E \quad \dots\dots\dots (8.2)$$

Since ϵ_r for a given material is constant, polarization (P) varies with variation in average electric field between the plates.

Capacitance

The capacitance of a parallel plate capacitor is given as $C = \epsilon A/t = \epsilon_0 \epsilon_r A/t$, where A is the overlapping surface area of the plates and t is the separation between the plates (i.e the thickness of the dielectric material). This can be written as

$$C = \epsilon_r C_0 \quad \dots\dots\dots (8.3)$$

where $C_0 = \epsilon_0 A/t$ is the capacitance of the parallel plate in vacuum. Thus for given dimensions, C is proportional to ϵ_r . Relative permittivity therefore affects capacitance. It is an important property of dielectric materials.

Example 8.1

The plates of a capacitor have an effective area $A = 1 \text{ m}^2$ and a separation $t = 0.5 \text{ mm}$. A potential of 1.5 Volt is applied across the plates. The permittivity of free space is given as $\epsilon_0 = (1/36\pi) \times 10^{-9} \text{ Farad /m}$.

- (a) Calculate the capacitance of the capacitor (i) in vacuum, and (ii) if a dielectric material of relative permittivity $\epsilon_r = 5$ is introduced in the space between the plates.
- (b) Calculate the total charge which is accumulated on the plates.

Solutions

Capacitance expression is given as $C = \epsilon_0 \epsilon_r A/t = \epsilon_r C_0$.

- (ai) In vacuum, $C_0 = \epsilon_0 A/t = (1/36\pi) \times 10^{-9} \times 1/0.5 \times 10^{-3} = 0.0176 \mu\text{F}$.
- (aii) $C = \epsilon_r C_0 = 5 \times C_0 = 5 \times 0.0176 = 0.088 \mu\text{F}$.
- (b) For total charge we note that $q = CV = 0.088 \times 1.5 = 0.133 \mu\text{C}$.

Ferroelectrics

In certain polycrystalline dielectric ceramics, if displacement of electric charges is plotted as a function of electric field, it exhibits hysteresis very similar to those found in ferromagnetic B-H curves. The spontaneous polarization disappear when the dielectric is heated to above a critical temperature just as does the spontaneous magnetization in a ferromagnet. These materials are said to possess ferroelectric properties, and the critical temperature at which polarization disappears is called its ferroelectric Curie temperature.

The ferroelectric property is obtained by slowly cooling the material to below its Curie temperature while maintaining a strong electric field. Under this condition, the orientation of the dipoles along the direction of the electric field is frozen in. The polarization of such a material can vary with temperature (pyroelectric effect) and also with pressure (piezoelectric effect).

8.2 Applications Of Dielectrics

The properties of ferroelectrics have been used in the manufacture of electrostatic microphones, and capacitors of very small size. In addition to their general use as electrical insulators, dielectric materials are also used for energy storage (in capacitors), as sensors (for temperature and pressure measurements), as transmission medium (in cables and optical fibres), etc.

Energy Storage

One use of dielectrics is in the storage of electrical energy. When an electric field is applied to a dielectric, the displacement of the electric charges will represent a storage of energy. If the field is allowed to vary, the stored energy will vary.

For our parallel plate capacitor, $C = \epsilon A/t$. If a potential voltage $V (= Et)$ is applied to the capacitor, the energy stored is

$$W = \frac{1}{2} CV^2 = \left(\frac{1}{2} \epsilon A/t\right) E^2 t^2 = \left(\frac{1}{2} \epsilon E^2\right) v \quad \dots\dots\dots (8.4)$$

where $v = tA$ is the volume of the dielectric material. Thus, the energy stored per unit volume of the dielectric is $\frac{1}{2} \epsilon E^2$.

Pyroelectric materials

Variation of polarization (and hence capture surface charges) with temperature is called pyroelectric effect. Increase in temperature slightly disorientates the dipoles and polarization decreases. The reduced polarization causes a reduction in capture surface charges in the material. Thus the effect of temperature increase is to reduce the capture surface charges in the material. The decrease in polarization (P) with temperature (T) is non-linear, as illustrated in the graph of figure 8.4. It can be observed that polarization vanishes at the Curie temperature (T_c). Pyroelectric ceramics are used as heat sensors for the measurement of temperature.

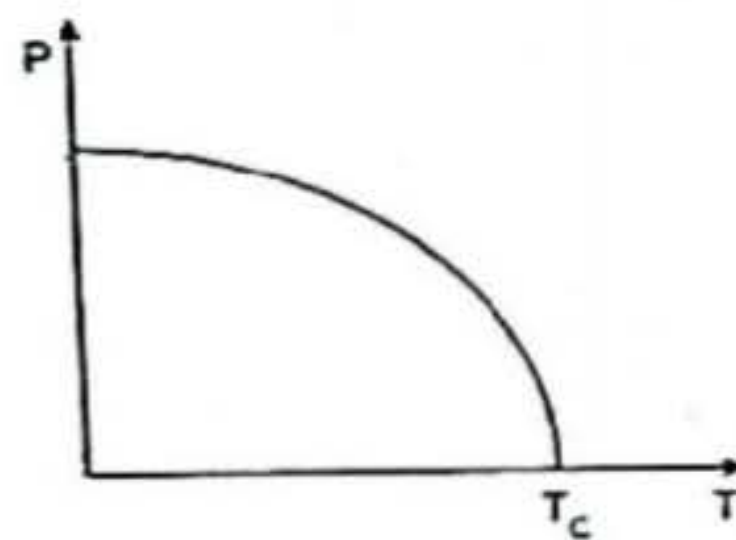


Fig.8.4 Variation of polarization with temperature

Piezoelectric Materials

Piezoelectric effect refers to the appearance of electric charges on the surface of a dielectric crystal under pressure. (piezo- is Greek word meaning to press). When these crystals are deformed mechanically, electric charges appear on opposite surfaces normal to the line of pressure, thus producing an electric field.

Thus,

$$\cos \phi = (n_2/n_1) \sin \theta_0 \quad \dots\dots\dots (8.7)$$

Total internal reflection occurs in the fibre if ϕ is equal to or greater than the critical angle θ_c . That is if

$$\sin \phi \geq n_2/n_1, \text{ or } \cos \phi \leq \sqrt{1 - (n_2/n_1)^2} \quad \dots\dots\dots (8.8)$$

Substituting into 8.7 yields

$$\sin \theta_0 \leq (n_1/n_2) \cos \phi \leq (n_1/n_2) [1 - (n_2/n_1)^2]^{1/2} \leq [(n_1/n_2)^2 - (n_2/n_1)^2]^{1/2}$$

For air, $n_2 = 1$. Thus, the maximum angle of acceptance of the fibre is given by

$$\sin \theta_{\alpha(\max)} = [n_1^2 - n_2^2]^{1/2} \quad \dots\dots\dots (8.9)$$

The quantity $\sin \theta_{\alpha(\max)}$ is called the numerical aperture of the fibre. From the above, rays with $\theta_0 \leq \theta_{\alpha(\max)}$ have $\phi \geq \theta_c$ and are internally reflected continuously.

INDEX

Acceptor, (p-type), 19
 Atom,
 hydrogen atom, 5
 quantum atom, 2

 Band bending, 73
 Band structure, 8
 conduction band, 12
 density of states, 12
 energy gap, 15
 Fermi energy, 27, 65
 valence band, 12
 Bonding,
 covalent bond, 9
 metallic bond, 9
 Bound charges, 104

 Capacitance, 107
 Carrier,
 lifetime, 41
 population, 26
 Charges (bound), 104
 Coherent emission, 96
 Conduction band, 12
 Conductivity(σ), 38
 Conductors, 13
 Contact,
 contact potential, 67
 ohmic contact, 67
 rectifying contact, 68
 Coulomb's force, 2.5.50
 Current density, 38
 Current-voltage relationship
 (rectifying contact), 68
 Cuts temperature, 102

 Dark current, 100

Density of energy states, 12
 Depletion region (width), 80
 Diffusion flow, 36
 Diode (p-n junction), 79
 photodiode, 100
 tunnel diode, 84
 Dielectric materials, 104
 Direct gap materials, 18
 Dipoles (dipole moment), 105
 Distribution function (Fermi), 25
 Donor impurity, 20
 Doping (impurity), 18
 Drift current flow, 36

 Electromagnetic radiation, 3
 Electron volt, 52
 Emission current, 54
 field effect, 57
 photoelectric effect, 61
 Energy band structure, 10
 Energy gap, 15
 Energy-momentum curves, 89
 Exclusion principle, 12.23
 Extrinsic materials, 17

 Ferroelectric materials, 102
 Fermi-Dirac statistics
 (distribution function), 26
 Fermi energy (level), 27.65
 Field effect
 (on emission current), 57
 Field emission, 55.59
 Flat band (MOS), 79
 Forbidden energy band
 (energy gap), 12

Hall angle, 45
 Hall effect, 43
 Hall coefficient, 44
 Hole (positive carrier), 17
 conduction, 18

 Image force, 49
 Impurities (semiconductor)
 acceptor impurity, 19
 donor impurity, 20
 Indirect gap materials, 18
 Insulators, 13
 Inversion (MOS), 79

 Junctions, 65-86
 metal-metal, 66
 metal-semiconductor, 71-78
 p-n junction, 79-86
 Laser diode, 82.91
 Lifetime, 41.58
 Light-emitting diode, 82.91
 Lorentz force, 43

 Magnetic effects, 43
 Hall effect, 43
 Lorentz force, 43
 Mean free time (path), 38
 Majority-minority carriers, 20
 Metallic (degenerate), 75
 Mobility (μ), 38
 electron, 38
 hole, 38

 n-type semiconductor, 19
 Negative resistance, 85

 Ohmic contact, 65

Optical effects,
 photoconductive, 18
 photoelectric, 61.88
 quantum efficiency, 99
 Optical fibre, 112

 Pauli exclusion principle, 12.23
 Phonon, 91
 Photoconductive cell, 98
 Photodiode (PD), 100
 photo junction, 99
 Photons, 3.90
 Flux, 98
 Photovoltaic, 101
 Piezoelectric materials, 110
 Planck's constant, 3
 p-n junctions, 79
 current-voltage relationship, 83
 reverse current, 83
 Polarization, 106
 Positive hole, 17
 Principle of indeterminacy,
 4.23
 p-type semiconductor, 19
 Pyroelectric materials, 110

 Quantum atom model, 2
 Quantum efficiency, 99
 Quantum mechanics, 4
 Quantum number, 5

 Recombination centres, 41
 Rectifying contact, 68
 Rectification (at junctions), 68-
 83
 Resistivity (ρ), 38
 Reverse current, 70.83
 Richardson equation, 68